Template Heaps for Full-Ground Local State

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1 A theory of template heaps

In this section we introduce *template heaps*, a notion that describes multiple heaps with a common segment. We develop a small theory around template heaps in order to achieve our main objective: to give a finitary submonad of the Full-Ground Local State monad [Kam+17].

We begin by revisiting the categories $\mathbb{W}$ (worlds and injections) and $\mathbb{WI}$ (world and injections with instantiation data) where we mainly focus on their local independence structure [Sim18]. We introduce the category of templates $\mathbb{T}$ that introduces a notion of location variables. We extend the local independence structure to $\mathbb{T}$ in order to define a suitable generalisation of the hiding monad on template heaps. We give the category of template instantiations $\mathbb{TI}$, which introduces instantiation data to $\mathbb{T}$.

1.1 The category of worlds and injections $\mathbb{W}$

Given a set of sorts $C \in \mathsf{Sort}$, the category $\mathbb{W}$ is defined as

- Objects: worlds $w = \{l_1 : C_1, \dots, l_n : C_n\}$.
- Morphisms: Injective, sort-preserving $\sigma : w \to w'$.

Notation. We use $\mathsf{Img}(\sigma)$ to refer to its image and $w' \ominus \sigma$ for its complement, that is, $w' \setminus \mathsf{Img}(\sigma)$, which induces an inclusion $\sigma^{\mathsf{c}} : w' \ominus \sigma \to w'$.
We denote morphisms from the empty world as $?_w : \emptyset \to w$ and inclusions as $i_i^{w_1,w_2} : w_i \to w_1 \oplus w_2$, where $\oplus : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ is the independent coproduct functor (see [Kam+17] for definition). We also denote by $\cong^{\sigma : w \to w'} : w' \cong w \oplus (w' \ominus \sigma)$ the canonical isomorphism.

Lemma 1.1.1 (Monomorphisms in $\mathbb{W}$)**.** Every morphism in $\mathbb{W}$ is mono.

Proof. Immediate from their injectivity. □

Lemma 1.1.2 (Initial Object in $\mathbb{W}$)**.** The category $\mathbb{W}$ has an initial object.

Proof. Consider the empty world. □

Lemma 1.1.3 (Equalizers in $\mathbb{W}$)**.** The category $\mathbb{W}$ has all equalizers.

Proof. Define the equalizer in $\mathsf{eqz}(\sigma_1, \sigma_2) \xhookrightarrow{e} w_1 \underset{\sigma_2}{\overset{\sigma_1}{\rightrightarrows}} w_2$ to be

$$\mathsf{eqz}(\sigma_1, \sigma_2) := \{a : C \in w_1 \mid \sigma_1(a) = \sigma_2(a)\}$$

with e being the inclusion into w_1.

Given a $\sigma_1 \circ \hat{\sigma} = \sigma_2 \circ \hat{\sigma}$, let $h : \hat{w} \to \mathsf{eqz}(\sigma_1, \sigma_2)$ be given by $\hat{\sigma}$. It is clear that it is well-defined, and uniqueness follows from e being mono. □

Example 1.1.4. This is an equalizer diagram

$$b : C' \xrightarrow{i_2} a : C \oplus b : C' \underset{i_2 \oplus 1}{\overset{i_1 \oplus 1}{\rightrightarrows}} a_1 : C \oplus a_2 : C \oplus b : C'$$

Notation. We sometimes overload $\mathsf{eqz}(\sigma_1, \sigma_2)$ to refer both to the object and the morphism when it is clear from context, e.g.,

$$\mathsf{eqz}(\sigma_1, \sigma_2) \xrightarrow{\mathsf{eqz}(\sigma_1, \sigma_2)} w_1 \underset{\sigma_2}{\overset{\sigma_1}{\rightrightarrows}} w_2$$

Lemma 1.1.5 (Pullbacks in $\mathbb{W}$)**.** The category $\mathbb{W}$ has all pullbacks.

Proof. Define the pullback of injections $w_1 \xrightarrow{\sigma_1} \hat{w} \xleftarrow{\sigma_2} w_2$ to be

$$\begin{aligned} w_1 \otimes_{\hat{w}} w_2 &:= \mathsf{Img}(\sigma_1) \cap \mathsf{Img}(\sigma_2) \\ \sigma_{1*}\sigma_2 &:= \sigma_1^{-1} \\ \sigma_{2*}\sigma_1 &:= \sigma_2^{-1} \end{aligned}$$

It is clear that the square $\begin{array}{ccc} w_1 \otimes_{\hat{w}} w_2 & \xrightarrow{\sigma_{2*}\sigma_1} & w_2 \\ {\scriptstyle \sigma_{1*}\sigma_2}\downarrow & & \downarrow{\scriptstyle \sigma_2} \\ w_1 & \xrightarrow[\sigma_1]{} & \hat{w} \end{array}$ commutes. Further assume $w' \xrightarrow{\sigma_1'} w_1$, $w' \xrightarrow{\sigma_2'} w_2$. Define the induced morphism $h : w' \to w_1 \otimes_{\hat{w}} w_2$ as $\sigma_1 \circ \sigma_1'$. It is well-defined by $\sigma_1 \circ \sigma_1' = \sigma_2 \circ \sigma_2'$. Uniqueness follows from injectivity of σ_1^{-1}, σ_2^{-1}. □

Example 1.1.6. This is a pullback diagram $\begin{array}{ccc} b : C' & \xhookrightarrow{j_* i} & b : C', c : C'' \\ {\scriptstyle i_* j}\downarrow & \lrcorner & \downarrow{\scriptstyle j} \\ a : C, b : C' & \xhookrightarrow{i} & a : C, b : C', c : C'' \end{array}$

1.1.1 Local Independence Structure

Pushouts in $\mathbb{W}$ do not exist in general as witnessed by $\mathbb{W} \hookrightarrow \mathsf{Fin} \Downarrow \mathsf{Sort}$ not reflecting pushouts. The reason for this stems from the inherent injectivity of morphisms in $\mathbb{W}$ certain induced morphisms coming out of the pushout in $\mathsf{Fin} \Downarrow \mathsf{Sort}$ are not injective. We instead consider a local independence structure on $\mathbb{W}$ which recovers the initia(L.I.C.) lity of the desired construct. Instead of considering any square as a solution to the problem of closing two morphisms $w_1 \xleftarrow{\sigma_1} w \xrightarrow{\sigma_2} w_2$, we consider independent squares that only allow locations from w to be conflated. The pushout in $\mathsf{Fin} \Downarrow \mathsf{Sort}$ becomes their local independent coproduct which takes the disjoint union of their coimages. We recall here the local independent structure on $\mathbb{W}$ [Sim18].

Definition 1.1.7 (Local Independent Squares in $\mathbb{W}$)**.** $\begin{array}{ccccc} & & w & & \\ & \overset{\sigma_1}{\swarrow} & & \overset{\sigma_2}{\searrow} & \\ w_1 & & & & w_2 \\ & \underset{\sigma_1'}{\searrow} & & \underset{\sigma_2'}{\swarrow} & \\ & & \hat{w} & & \end{array}$ is a local independent square (L.I.S.) when it is a pullback

$$\begin{array}{ccc} w & \xrightarrow{\sigma_2} & w_2 \\ {\scriptstyle \sigma_1}\downarrow & \lrcorner & \downarrow{\scriptstyle \sigma_1'} \\ w_1 & \xrightarrow[\sigma_1']{} & \hat{w} \end{array}$$

Lemma 1.1.8 (Local Independence Structure on $\mathbb{W}$)**.** We can equip $\mathbb{W}$ with a local independent structure [Sim18] by defining the collection of local independent multicospans $\mathcal{I} = \{\sigma_i \xrightarrow{\sigma_i'} \hat{\sigma}\}_{i\in I}$ as multicospans that satisfy (note that I is finite):

$$\forall i, j \in I.\, i \neq j \implies \begin{array}{ccccc} & & w & & \\ & \overset{\sigma_i}{\swarrow} & \downarrow & \overset{\sigma_j}{\searrow} & \\ w_i & & {\scriptstyle \hat{\sigma}}\downarrow & & w_j \\ & \underset{\sigma_i'}{\searrow} & & \underset{\sigma_j'}{\swarrow} & \\ & & \hat{w} & & \end{array} \text{ is a local independent square.}$$

Proof.
We only need to show that the collection $\mathcal{I}$ is closed under composition, all other axioms follow immediately. Consider multicospan $\{\sigma_i \xrightarrow{\sigma_i'} \hat{\sigma}\}_{i\in I} \in \mathcal{I}$ and a family of multicospans $\{\{\tau_{ij} \xrightarrow{\tau_{ij}'} \sigma_i\}_{j\in J_i} \in \mathcal{I}\}_{i\in I}$. We need $\{\tau_{ij} \xrightarrow{\sigma_i'\circ\tau_{ij}'} \hat{\sigma}\}_{j\in J_i, i\in I} \in \mathcal{I}$, which follows from

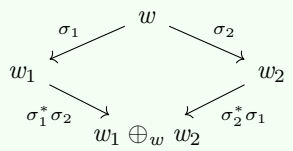
1. For all $i \in I, k \neq l \in J_i$, $\tau_{ik} \xrightarrow{\sigma_i'\circ\tau_{ik}'} \hat{\sigma} \xleftarrow{\sigma_i'\circ\tau_{il}'} \tau_{il}$ is a local independent square. $\begin{array}{ccc} & \overset{\tau_{il}}{\longrightarrow} & \\ {\scriptstyle \tau_{ik}}\downarrow & \lrcorner & \downarrow{\scriptstyle \tau_{il}'} \\ & & {\scriptstyle \sigma_i'}\downarrow \\ & \underset{\tau_{ik}'}{\longrightarrow} \quad \underset{\sigma_i'}{\longrightarrow} & \end{array}$

   follows from $\begin{array}{ccc} & \overset{\tau_{il}}{\longrightarrow} & \\ {\scriptstyle \tau_{ik}}\downarrow & \lrcorner & \downarrow{\scriptstyle \tau_{il}'} \\ & \underset{\tau_{ik}'}{\longrightarrow} & \end{array}$ and injectivity of σ_i' (pullbacks are preserved under post-composition with monos, Lemma A.1.1).

2. For all $i \neq j \in I, k \in J_i, l \in J_j$, $\tau_{ik} \xrightarrow{\sigma_i'\circ\tau_{ik}'} \hat{\sigma} \xleftarrow{\sigma_j'\circ\tau_{jl}'} \tau_{jl}$ is a local independent square. We need $\begin{array}{ccc} & \overset{\tau_{jl}}{\longrightarrow} & \\ {\scriptstyle \tau_{ik}}\downarrow & \lrcorner & \downarrow{\scriptstyle \tau_{jl}'} \\ & & {\scriptstyle \sigma_j'}\downarrow \\ & \underset{\tau_{ik}'}{\longrightarrow} \quad \underset{\sigma_i'}{\longrightarrow} & \end{array}$ which follows from pullback pasting $\begin{array}{ccccc} & & \overset{\tau_{jl}}{\longrightarrow} & & \\ {\scriptstyle 1}\downarrow & \lrcorner & & \downarrow{\scriptstyle \tau_{jl}'} & \\ & \overset{1}{\longrightarrow} & & \overset{\sigma_j}{\longrightarrow} & \\ {\scriptstyle \tau_{ik}}\downarrow & \lrcorner & & \downarrow{\scriptstyle \sigma_j'} & \\ & & \underset{\sigma_i}{\longrightarrow} & & \\ & \underset{\tau_{ik}'}{\longrightarrow} & \quad \underset{\sigma_i'}{\longrightarrow} & & \end{array}$ (see also Lemma A.1.2)

□

Definition 1.1.9 (Local Independent Coproducts from [Kam+17])**.** Recall that a local independent coproduct of $w_1 \xleftarrow{\sigma_1} w \xrightarrow{\sigma_2} w_2$ from Kammar et al. [Kam+17] is given by the following square



defined by

$$w_1 \oplus_w w_2 := w \oplus (w_1 \ominus \sigma_1) \oplus (w_2 \ominus \sigma_2)$$
$$\sigma_1^* \sigma_2 := 1 \oplus i_1 \circ \cong^{\sigma_1}$$
$$\sigma_2^* \sigma_1 := 1 \oplus i_2 \circ \cong^{\sigma_2}$$

Lemma 1.1.10 (Local Independent Coproducts on $\mathbb{W}$)**.** The local independent coproducts (L.I.C.) from Kammar et al. [Kam+17] give the structure for the L.I.C. in Simpson [Sim18] (considering the independent structure above).

Proof. One needs to show:

- $\{\sigma_1^* \sigma_2, \sigma_2^* \sigma_1\}$ is $\mathcal{I}$-neutral and independent (Definition 3.5).
- The unique morphism exists (Definition 3.5).
- Independent squares compose (Definition 5.2).
- The pasting-like law for L.I.C. (Definition 6.2).

□

Remark 1.1.11. In a L.I.C. square of span $w_1 \xleftarrow{\sigma_1} w \xrightarrow{\sigma_2} w_2$, if σ_1 is equipped with instantiation data $\eta : \underline{\mathbb{H}}(w_1 \ominus \sigma_1, w_1)$, then $\sigma_2^* \sigma_1$ can be equipped with instantiation data as well (See [Kam+17] for definition of the heaplet bifunctor $\underline{\mathbb{H}} : \mathbb{W}^{\mathsf{op}} \times \mathbb{W} \to \mathsf{Set}$)

$$\underline{\mathbb{H}}(\cong, \sigma_1^* \sigma_2)(\eta) : \underline{\mathbb{H}}((w_1 \oplus_w w_2) \ominus \sigma_2^* \sigma_1, w_1 \oplus_w w_2)$$

using the canonical iso $(w_1 \oplus_w w_2) \ominus \sigma_2^* \sigma_1 \cong w_1 \ominus \sigma_1$.

1.2 The category of instantiations $\mathbb{WI}$

The category $\mathbb{WI}$ is defined as

- Objects: worlds $w = \{l_1 : C_1, \ldots, l_n : C_n\}$.
- Morphisms: Injective, sort-preserving $\rho : w \to w'$ and instantiation data $\eta : \underline{\mathbb{H}}(w' \ominus \rho, w')$.

We have a forgetful functor $u : \mathbb{WI} \to \mathbb{W}$ that forgets the instantiation data.

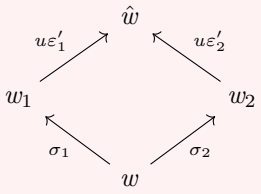
Definition 1.2.1 (Inhabited Sorts)**.** A sort $C \in \mathsf{Sort}$ is said to be *inhabited* if there exists a world w, renaming $a : C \to w$, and $\eta : \mathbb{H}(w)$.

Similarly we say that a set of sorts is inhabited if every sort in the set is inhabited.

Lemma 1.2.2. The local independent structure on $\mathbb{W}$ can be directly extended to $\mathbb{WI}$ such that u reflects local independent squares.

Recall that the hiding monad $\mathbb{P}$ on $\mathbb{WI}$ is given by the colimit $\mathbb{P}Aw = \mathit{colimit}\,(w/u \xrightarrow{U} \mathbb{WI} \xrightarrow{A} \mathsf{Set})$. The following lemma describes the quotienting that happens by this colimit.

Lemma 1.2.3 (Quotienting in the Hiding Monad)**.** For $\sigma_1 : w \to w_1$, $a_1 \in A(w_1)$, and $\sigma_2 : w \to w_2$, $a_2 \in A(w_2)$, we have that $q_{\sigma_1} a_1 = q_{\sigma_2} a_2 \in \mathbb{P}Aw$ iff there exist

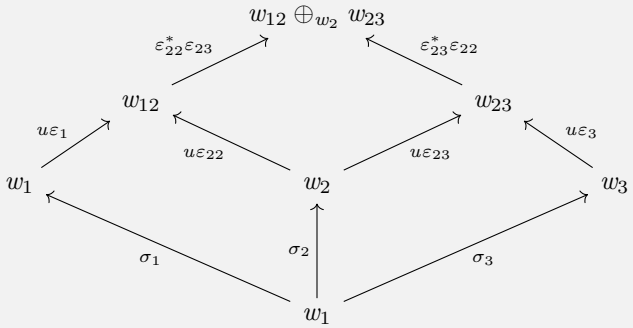


s.t. $A(\varepsilon_1')(a_1) = A(\varepsilon_2')(a_2)$.

Proof. We first show that the relation

$$\mathit{inj}_{\sigma_1}(a_1) \sim \mathit{inj}_{\sigma_2}(a_2) \iff \exists \varepsilon_1', \varepsilon_2'.\, u\varepsilon_1' \circ \sigma_1 = u\varepsilon_2' \circ \sigma_2 \wedge A(\varepsilon_1')(a_1) = A(\varepsilon_2')(a_2)$$

is an equivalence. Reflexivity and symmetry is immediate. For transitivity, we have



Notice that the top square commutes by Lemma 1.2.2. which implies that $\mathit{inj}_{\sigma_1}(a_1) \sim \mathit{inj}_{\sigma_3}(a_3)$. To prove that the generated relation by the colimit is exactly $\sim$, one shows direction $\to$ by inducting on the list of related elements, and for $\leftarrow$, notice that if $\mathit{inj}_{\sigma_1}(a_1) \sim \mathit{inj}_{\sigma_3}(a_3)$, then

$$\mathit{inj}_{\sigma_1} a_1 \leq \mathit{inj}_{u\varepsilon_1 \circ \sigma_1} A(\varepsilon_1)(a_1) \geq \mathit{inj}_{\sigma_2} a_2$$

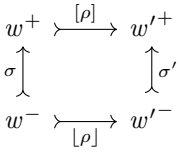
□

1.3 The category of templates $\mathbb{T}$

The first step towards introducing template heaps requires us to distinguish between two different kinds of locations. We have (i) concrete locations that are associated with some cell contents and (ii) location variables that stand for *any arbitrary cell contents*.

To model these two kinds of locations we make use of the category of templates defined as the arrow category $\mathsf{Arr}(\mathbb{W})$

- Objects: Sort-preserving injective renamings between worlds $w^- \xrightarrow{\sigma} w^+$.
- Morphisms: Injective, sort-preserving $\lfloor\rho\rfloor : w^- \to w'^+$ and $[\rho] : w^- \to w'^+$, such that:



We have two forgetful functors $\lfloor - \rfloor, [-] : \mathbb{T} \to \mathbb{W}$, projecting the $-$ and $+$ components respectively.

Notation.

- We refer to the elements of $\lfloor\sigma\rfloor$ as the concrete locations of $\sigma \in \mathsf{Obj}(\mathbb{T})$.
- We refer to the elements of $w^+ \ominus \sigma$ as the location variables of $\sigma \in \mathsf{Obj}(\mathbb{T})$, notation $\lceil\sigma\rceil$.
- We refer to elements of $[\sigma]$ as *all* the locations of $\sigma \in \mathsf{Obj}(\mathbb{T})$.
- If it is clear from the context we overload the notations $\lfloor\sigma\rfloor = \lfloor\sigma\rfloor \xrightarrow{\sigma} [\sigma]$ and $\lceil\sigma\rceil = \lceil\sigma\rceil \xrightarrow{\sigma^c} [\sigma]$ for the inclusions into $[\sigma]$.
- We have an induced morphism $\lceil\rho\rceil : \lceil\sigma\rceil \xrightarrow{\lceil\sigma\rceil} [\sigma] \xrightarrow{[\rho]} [\sigma']$.
- We refer to elements $[\sigma'] \ominus [\rho]$ as the new locations introduced by ρ, and $\lfloor\sigma'\rfloor \ominus \lfloor\rho\rfloor$ as the concrete locations introduced by ρ.
- We refer to the variables $\lceil\sigma\rceil$ that are instantiated by ρ as $\lfloor\sigma'\rfloor \otimes_{[\sigma']} \lceil\sigma\rceil$:

$$\begin{array}{ccc} \lfloor\sigma'\rfloor \otimes_{[\sigma']} \lceil\sigma\rceil & \longrightarrow & \lceil\sigma\rceil \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \lceil\rho\rceil} \\ \lfloor\sigma'\rfloor & \xrightarrow[\lfloor\sigma'\rfloor]{} & [\sigma'] \end{array}$$

  which are exactly the $\lceil\sigma\rceil$ that are mapped to concrete locations $\lfloor\sigma'\rfloor$ by ρ.
- We refer to the variables $\lceil\sigma'\rceil$ that are fresh w.r.t. ρ as $\lceil\sigma'\rceil \otimes_{[\sigma']} ([\sigma'] \ominus [\rho])$:

$$\begin{array}{ccc} \lceil\sigma'\rceil \otimes_{[\sigma']} ([\sigma'] \ominus [\rho]) & \longrightarrow & [\sigma'] \ominus [\rho] \\ \downarrow & \lrcorner & \downarrow{\scriptstyle [\rho]^c} \\ \lceil\sigma'\rceil & \xrightarrow[\lceil\sigma'\rceil]{} & [\sigma'] \end{array}$$

  which are exactly the $\lceil\sigma'\rceil$ that are not mapped from any location $[\sigma]$ by ρ.
- Similarly, $\lceil\sigma'\rceil \otimes_{[\sigma']} \lceil\sigma\rceil$ refers to the variables $\lceil\sigma'\rceil$ that are mapped from variables in $\lceil\sigma\rceil$ by ρ.

$$\begin{array}{ccc} \lceil\sigma'\rceil \otimes_{[\sigma']} \lceil\sigma\rceil & \longrightarrow & \lceil\sigma\rceil \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \lceil\rho\rceil} \\ \lceil\sigma'\rceil & \xrightarrow[\lceil\sigma'\rceil]{} & [\sigma'] \end{array}$$

- Notice that from $\lfloor\sigma'\rfloor$ and $\lceil\sigma'\rceil$ being disjoint, we have that concrete locations in $\lfloor\sigma\rfloor$ cannot become uninstantiated

$$\begin{array}{ccccc} \emptyset & & \longrightarrow & & \lceil\sigma'\rceil \\ \downarrow & \lrcorner & & & \downarrow{\scriptstyle \lceil\sigma'\rceil} \\ \lfloor\sigma\rfloor & \xrightarrow[\lfloor\rho\rfloor]{} & \lfloor\sigma'\rfloor & \xrightarrow[\lfloor\sigma'\rfloor]{} & [\sigma'] \end{array}$$

Example 1.3.1. Consider the following morphism in $\mathbb{T}$

$$\begin{array}{ccc} a:C, b:C' & \xrightarrow{[\rho]=i^{a,b\hookrightarrow a,b,c}} & a:C, b:C', c:C'' \\ {\scriptstyle i^{a\hookrightarrow a,b}}\uparrow & & \uparrow{\scriptstyle i^{a,b\hookrightarrow a,b,c}} \\ a:C & \xrightarrow[\lfloor\rho\rfloor]{} & a:C, b:C' \end{array}$$

- We have variables $\lceil i^{a\hookrightarrow a,b}\rceil = \{b : C'\}$, and $\lceil i^{a,b\hookrightarrow a,b,c}\rceil = \{c : C''\}$.
- Variables instantiated by ρ:

$$\lfloor i^{a,b\hookrightarrow a,b,c}\rfloor \otimes_{\{a,b,c\}} \lceil i^{a\hookrightarrow a,b}\rceil = \{b : C'\}$$

- Variables $\lceil i^{a,b\hookrightarrow a,b,c}\rceil$ that are fresh w.r.t. to ρ:

$$\lceil i^{a,b\hookrightarrow a,b,c}\rceil \otimes_{\{a,b,c\}} (\lceil i^{a,b\hookrightarrow a,b,c}\rceil \ominus [\rho]) = \{c : C''\}$$

- Variables $\lceil i^{a,b\hookrightarrow a,b,c}\rceil$ mapped from variables in $\lceil i^{a\hookrightarrow a,b}\rceil$ by ρ:

$$\lceil i^{a,b\hookrightarrow a,b,c}\rceil \otimes_{\{a,b,c\}} \lceil i^{a\hookrightarrow a,b}\rceil = \emptyset$$

Definition 1.3.2 (Variable Extension Functor)**.** We define the variable extension functor as a means to add w-many variables.

$$\begin{array}{rl} & (- \oplus w) : \mathbb{T} \to \mathbb{T} \\ & [\sigma] \oplus w \\ ((\lfloor\sigma\rfloor \xrightarrow{\lfloor\sigma\rfloor} [\sigma]) \oplus w) = & \uparrow{\scriptstyle i_1 \circ \lfloor\sigma\rfloor} \\ & \lfloor\sigma\rfloor \\ & (\rho \oplus w) = \langle \lfloor\varepsilon\rfloor, [\varepsilon] \oplus 1_w \rangle \end{array}$$

Further, for every functor $G : \mathbb{T} \to \mathsf{Set}$ and w, we have a $\mathsf{ext}^w_G : G \Rightarrow G(- \oplus w')$.

Example 1.3.3. The action of $(- \oplus \{d : C'''\})$ on morphism ρ from Example 1.3.1 is $(\rho \oplus \{d : C'''\})$ defined as

$$\begin{array}{ccc} \{a:C, b:C'\} \oplus \{d:C'''\} & \xrightarrow{[\rho]\oplus 1_d} & \{a:C, b:C', c:C''\} \oplus \{d:C'''\} \\ {\scriptstyle i_1 \circ i^{a\hookrightarrow a,b}}\uparrow & & \uparrow{\scriptstyle i_1 \circ i^{a,b\hookrightarrow a,b,c}} \\ a:C & \xrightarrow[\lfloor\rho\rfloor]{} & a:C, b:C' \end{array}$$

Lemma 1.3.4 (Monomorphisms in $\mathbb{T}$)**.** Every morphism in $\mathbb{T}$ is mono.

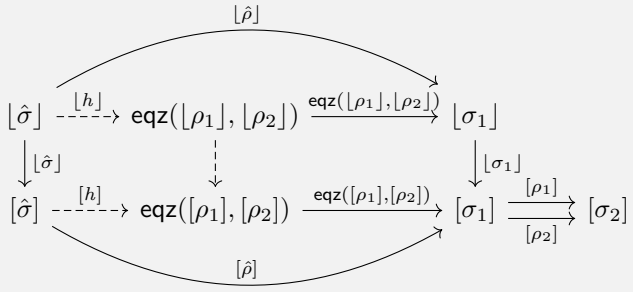
Proof. Follows directly from injectivity of underlying morphisms. □

Lemma 1.3.5 (Initial Object in $\mathbb{T}$)**.** The category $\mathbb{T}$ has an initial object.

Proof. Consider template $?_\emptyset : \emptyset \to \emptyset$. □

Lemma 1.3.6 (Equalizers in $\mathbb{T}$)**.** The category $\mathbb{T}$ has all equalizers, and $\lfloor - \rfloor$, $[-]$ preserve equalizers.

Proof. The equalizer $\mathsf{eqz}(\rho_1, \rho_2) \longrightarrow \sigma_1 \underset{\rho_2}{\overset{\rho_1}{\rightrightarrows}} \sigma_2$ is given componentwise



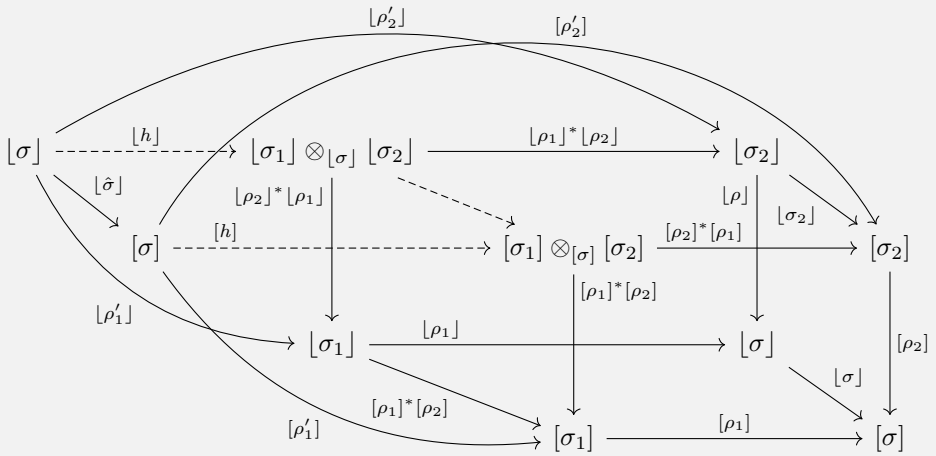
□

Example 1.3.7. This is an equalizer diagram

$$?_b \xrightarrow{\langle ?_a, i_2 \rangle} i_1^{a \to a \oplus b} \underset{\langle i_2, i_2 \oplus 1_b \rangle}{\overset{\langle i_1, i_1 \oplus 1_b \rangle}{\rightrightarrows}} i_1^{a_1 \oplus a_2 \to a_1 \oplus a_2 \oplus b}$$

Lemma 1.3.8 (Pullbacks in $\mathbb{T}$)**.** The category $\mathbb{T}$ has all pullbacks and $\lfloor - \rfloor$, $[-]$ preserve pullbacks.

Proof. The pullback of $\sigma_1 \xrightarrow{\rho_1} \hat{\sigma} \xleftarrow{\rho_2} \sigma_2$ is given componentwise



□

Example 1.3.9. This is a pullback square

$$\begin{array}{ccc} ?_b & \xrightarrow{\langle ?_{a_1}, i_2 \rangle} & i_1^{a_1 \to a_1 \oplus b} \\ {\scriptstyle \langle ?_{a_2}, i_2 \rangle}\downarrow & \lrcorner & \downarrow{\scriptstyle \langle i_1, 1_{a_1} \oplus i_2 \rangle} \\ 1_{a_2 \oplus b} & \xrightarrow[\langle i_2, i_2 \rangle]{} & 1_{a_1 \oplus (a_2 \oplus b)} \end{array}$$

1.3.1 Generated Cospans

Lemma 1.3.10 (Generated Cospans in $\mathbb{T}$)**.** Given cospan $[\sigma_1] \xrightarrow{\tau_1} \hat{w} \xleftarrow{\tau_2} [\sigma_2]$ in $\mathbb{W}$ we generate a cospan $\sigma_1 \xrightarrow{\rho_1^\tau} \sigma^\tau \xleftarrow{\rho_2^\tau} \sigma_2$ in the least possible way, that is, it satisfies

- Satisfy $[\sigma^\tau] = \hat{w}$, $[\rho_1^\tau] = \tau_1$ and $[\rho_2^\tau] = \tau_2$.
- For any other cospan $\sigma_1 \xrightarrow{\rho_1'} \hat{\sigma} \xleftarrow{\rho_2'} \sigma_2$ that factors through $\hat{w}$ $\begin{array}{ccccc} [\sigma_1] & \xrightarrow{\tau_1} & \hat{w} & \xleftarrow{\tau_2} & [\sigma_2] \\ & \searrow^{[\rho_1']} & \downarrow{\scriptstyle \exists \tau'} & \swarrow_{[\rho_2']} & \\ & & [\hat{\sigma}] & & \end{array}$, there exists a unique $h : \sigma^\tau \to \hat{\sigma}$ that satisfies $\rho_1' = h \circ \rho_1^\tau$ and $\rho_2' = h \circ \rho_2^\tau$.

We say that ρ_1^τ, ρ_2^τ are generated by τ_1, τ_2. Intuitively, a generated cospan selects the smallest $\lfloor \sigma^\tau \rfloor$ possible, that is, it selects the least concrete locations possible.

Proof. We define $[\sigma^\tau] = \hat{w}$, $[\rho_1^\tau] = \tau_1$, $[\rho_2^\tau] = \tau_2$. To define $\lfloor \sigma^\tau \rfloor$ we first consider their common concrete locations through the pullback

$$\begin{array}{ccccc} \lfloor \sigma_1 \rfloor \otimes_{\hat{w}} \lfloor \sigma_2 \rfloor & & \xrightarrow{\lfloor \sigma_2 \rfloor ; \tau_{2*} \lfloor \sigma_1 \rfloor ; \tau_1} & & \lfloor \sigma_2 \rfloor \\ \downarrow{\scriptstyle \lfloor \sigma_1 \rfloor ; \tau_{1*} \lfloor \sigma_2 \rfloor ; \tau_2} & & \lrcorner & & \downarrow{\scriptstyle \lfloor \sigma_2 \rfloor} \\ & & & & [\sigma_2] \\ & & & & \downarrow{\scriptstyle \tau_2} \\ \lfloor \sigma_1 \rfloor & \xrightarrow[\lfloor \sigma_1 \rfloor]{} & [\sigma_1] & \xrightarrow[\tau_1]{} & \hat{w} \end{array}$$

and further take the L.I.C. in $\mathbb{W}$ of the pullback morphisms

$$\begin{array}{ccc} \lfloor \sigma_1 \rfloor \otimes_{\hat{w}} \lfloor \sigma_2 \rfloor & \xrightarrow{\lfloor \sigma_2 \rfloor ; \tau_{2*} \lfloor \sigma_1 \rfloor ; \tau_1} & \lfloor \sigma_2 \rfloor \\ \downarrow{\scriptstyle \lfloor \sigma_1 \rfloor ; \tau_{1*} \lfloor \sigma_2 \rfloor ; \tau_2} & & \downarrow{\scriptstyle \lfloor \rho_2^\tau \rfloor} \\ \lfloor \sigma_1 \rfloor & \xrightarrow[\lfloor \rho_1^\tau \rfloor]{} & \lfloor \sigma^\tau \rfloor := \lfloor \sigma_1 \rfloor \oplus_{\lfloor \sigma_1 \rfloor \otimes_{\hat{w}} \lfloor \sigma_2 \rfloor} \lfloor \sigma_2 \rfloor \end{array}$$

The morphism $\lfloor \sigma^\tau \rfloor \xrightarrow{\lfloor \sigma^\tau \rfloor} [\sigma^\tau]$ is the unique morphism induced by the L.I.C. towards the pullback. The UP gives that the morphisms ρ_1^τ, ρ_2^τ are commuting squares.

Least Property Consider a compatible cospan ρ_1', ρ_2'. We only need to extend $[h] = \tau'$ to a morphism in $\mathbb{T}$. Let $\lfloor h \rfloor$ be the induced morphism by the L.I.C.

$$\begin{array}{ccccc} \lfloor \sigma_1 \rfloor \otimes_{\hat{w}} \lfloor \sigma_2 \rfloor & \xrightarrow{\lfloor \sigma_2 \rfloor ; \tau_{2*} \lfloor \sigma_1 \rfloor ; \tau_1} & \lfloor \sigma_2 \rfloor & & \\ \downarrow{\scriptstyle \lfloor \sigma_1 \rfloor ; \tau_{1*} \lfloor \sigma_2 \rfloor ; \tau_2} & & \downarrow{\scriptstyle \lfloor \rho_2^\tau \rfloor} & \searrow^{\lfloor \rho_2' \rfloor} & \\ \lfloor \sigma_1 \rfloor & \xrightarrow[\lfloor \rho_1^\tau \rfloor]{} & \lfloor \sigma^\tau \rfloor & \xrightarrow{\lfloor h \rfloor} & \lfloor \hat{\sigma} \rfloor \end{array} \qquad \lfloor \sigma_1 \rfloor \xrightarrow{\lfloor \rho_1' \rfloor} \lfloor \hat{\sigma} \rfloor$$

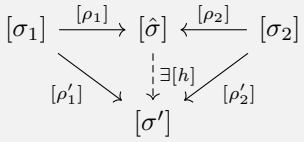
The outer square commutes by

$$\begin{aligned} \lfloor \rho_2' \rfloor \circ (\lfloor \sigma_2 \rfloor ; \tau_{2*} \lfloor \sigma_1 \rfloor ; \tau_1) = \lfloor \rho_1' \rfloor \circ (\lfloor \sigma_1 \rfloor ; \tau_{1*} \lfloor \sigma_2 \rfloor ; \tau_2) \iff \\ \lfloor \hat{\sigma} \rfloor \circ \lfloor \rho_2' \rfloor \circ (\lfloor \sigma_2 \rfloor ; \tau_{2*} \lfloor \sigma_1 \rfloor ; \tau_1) = \lfloor \hat{\sigma} \rfloor \circ \lfloor \rho_1' \rfloor \circ (\lfloor \sigma_1 \rfloor ; \tau_{1*} \lfloor \sigma_2 \rfloor ; \tau_2) = \\ [\rho_2'] \circ \lfloor \sigma_2 \rfloor \circ (\lfloor \sigma_2 \rfloor ; \tau_{2*} \lfloor \sigma_1 \rfloor ; \tau_1) = [\rho_2'] \circ \lfloor \sigma_1 \rfloor \circ (\lfloor \sigma_1 \rfloor ; \tau_{1*} \lfloor \sigma_2 \rfloor ; \tau_2) = \\ \tau' \circ \tau_2 \circ \lfloor \sigma_2 \rfloor \circ (\lfloor \sigma_2 \rfloor ; \tau_{2*} \lfloor \sigma_1 \rfloor ; \tau_1) = \tau' \circ \tau_1 \circ \lfloor \sigma_1 \rfloor \circ (\lfloor \sigma_1 \rfloor ; \tau_{1*} \lfloor \sigma_2 \rfloor ; \tau_2) \end{aligned}$$

and satisfies independence by Lemma A.1.1. Further h is a commuting square, that is $\lfloor\hat{\sigma}\rfloor \circ \lfloor h\rfloor = [h] \circ \lfloor\sigma^{\tau}\rfloor$, by the jointly epi property of $\lfloor\rho_1^{\tau}\rfloor$ and $\lfloor\rho_2^{\tau}\rfloor$. □

Lemma 1.3.11 (Generated Cospans and Jointly Epi Property)**.** A cospan $\sigma_1 \xrightarrow{\rho_1} \hat{\sigma} \xleftarrow{\rho_2} \sigma_2$ is generated by $[\sigma_1] \xrightarrow{[\rho_1]} [\hat{\sigma}] \xleftarrow{[\rho_2]} [\sigma_2]$ iff $\lfloor\rho_1\rfloor$ and $\lfloor\rho_2\rfloor$ are jointly surjective.

Proof.
$\rightarrow$) Generated cospans are unique up to isomorphism, and so ρ_1, ρ_2 is iso to the cospan induced by Lemma 1.3.10 at $[\rho_1]$, $[\rho_2]$. Since the cospan of Lemma 1.3.10 at $\lfloor-\rfloor$ is a L.I.C., we have that $\lfloor\rho_1\rfloor$, $\lfloor\rho_2\rfloor$ are jointly epi.
$\leftarrow$) We only need to show that the cospan ρ_1, ρ_2 satisfies the UP of generated cospans, so consider



Define $\lfloor h\rfloor$ as

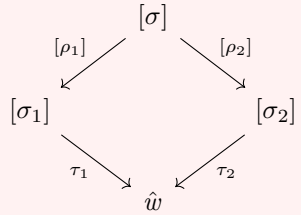
$$\lfloor h\rfloor : \lfloor\hat{\sigma}\rfloor \to \lfloor\sigma'\rfloor$$
$$\lfloor h\rfloor(a : C) := \begin{cases} \lfloor\rho_1'\rfloor(a_1) & \text{if } \lfloor\rho_1\rfloor(a_1) = a \\ \lfloor\rho_2'\rfloor(a_2) & \text{if } \lfloor\rho_2\rfloor(a_2) = a \end{cases}$$

It is well-defined for if $\lfloor\rho_1\rfloor(a_1) = a = \lfloor\rho_1\rfloor(a_2)$, we have

$$\begin{aligned} \lfloor\hat{\sigma}\rfloor\lfloor\rho_1\rfloor(a_1) = \lfloor\hat{\sigma}\rfloor\lfloor\rho_2\rfloor(a_2) &\iff \\ [\rho_1]\lfloor\sigma_1\rfloor(a_1) = [\rho_2]\lfloor\sigma_2\rfloor(a_2) &\iff \\ [h][\rho_1]\lfloor\sigma_1\rfloor(a_1) = [h][\rho_2]\lfloor\sigma_2\rfloor(a_2) &\iff \\ [\rho_1']\lfloor\sigma_1\rfloor(a_1) = [\rho_2']\lfloor\sigma_2\rfloor(a_2) &\iff \\ \lfloor\sigma'\rfloor\lfloor\rho_1'\rfloor(a_1) = \lfloor\sigma'\rfloor\lfloor\rho_1'\rfloor(a_2) &\iff \\ \lfloor\rho_1'\rfloor(a_1) = \lfloor\rho_1'\rfloor(a_2) & \end{aligned}$$

It is easy to show that h is a morphism in $\mathbb{T}$ and uniqueness of h follows from the jointly epi property. □

Lemma 1.3.12 (Instantiation data for Generated Cospans)**.** Given a diagram of the form

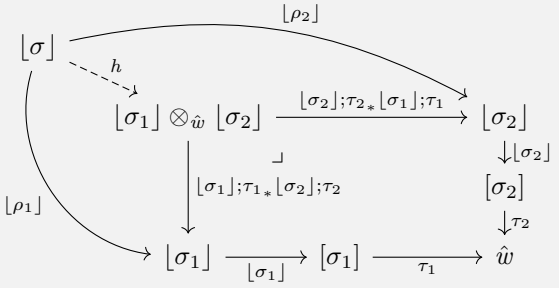


where ρ_2 has data $\eta : \underline{\mathbb{H}}(\lfloor\sigma_2\rfloor \ominus \lfloor\rho_2\rfloor, [\sigma_2])$, we can equip ρ_1^{τ} with data $\eta_{\rho_1^{\tau}} : \underline{\mathbb{H}}(\lfloor\hat{\sigma}\rfloor \ominus \lfloor\rho_1^{\tau}\rfloor, [\hat{\sigma}])$.

Proof. We have a canonical isomorphism

$$\lfloor\hat{\sigma}\rfloor \ominus \lfloor\rho_1^{\tau}\rfloor \cong \lfloor\sigma_2\rfloor \ominus (\lfloor\sigma_2\rfloor; \tau_2)_* \lfloor\sigma_1\rfloor; \tau_1$$

and further from the pullback property



we have

$$\lfloor\sigma_2\rfloor \ominus \lfloor\rho_2\rfloor \cong (\lfloor\sigma_2\rfloor \ominus (\lfloor\sigma_2\rfloor;\tau_2)_*\lfloor\sigma_1\rfloor;\tau_1) \oplus (\lfloor\sigma_1\rfloor \otimes_{\hat{w}} \lfloor\sigma_2\rfloor \ominus h)$$

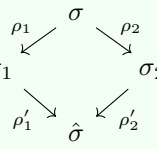
Putting these together gives a morphism

$$\lfloor\hat{\sigma}\rfloor \ominus \lfloor\rho_1^\tau\rfloor \xrightarrow{i_1} \lfloor\sigma_2\rfloor \ominus \lfloor\rho_2\rfloor$$

and so we define $\eta_{\rho_1^\tau} = \underline{\mathbb{H}}(i_1, \tau_2)(\eta)$. □

1.3.2 Local Independence Structure

We equip $\mathbb{T}$ with a local independence structure that takes into consideration the instantiations of location variables.

Definition 1.3.13 (Local Independent Squares in $\mathbb{T}$)**.**  is a local independent square when the pullback of $[\sigma_1] \otimes_{[\hat{\sigma}]} [\sigma_2]$ is $[\sigma]$

$$\begin{array}{ccc} [\sigma] & \xrightarrow{[\rho_2]} & [\sigma_2] \\ {\scriptstyle [\rho_1]}\downarrow & \lrcorner & \downarrow{\scriptstyle [\rho_2']} \\ [\sigma_1] & \xrightarrow[[\rho_1']]{} & [\hat{\sigma}] \end{array}$$

Thus an independent square conflates two locations $a \in [\sigma_1]$ and $b \in [\sigma_2]$ only if they come from $[\sigma]$.

Lemma 1.3.14 (Local Independent Coproducts in $\mathbb{T}$)**.** For every span $\sigma_1 \xleftarrow{\rho_1} \sigma \xrightarrow{\rho_2} \sigma_2$ the local independent coproduct (L.I.C.) exists.

Proof. Consider the L.I.C. in $\mathbb{W}$

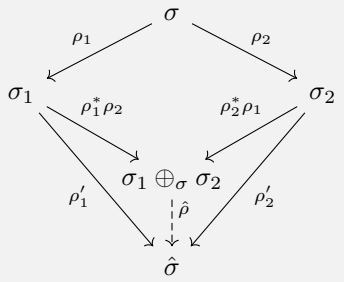
$$\begin{array}{ccc} [\sigma] & \xrightarrow{[\rho_1]} & [\sigma_1] \\ {\scriptstyle [\rho_2]}\downarrow & & \downarrow{\scriptstyle [\rho_1]^*[\rho_2]} \\ [\sigma_2] & \xrightarrow[[\rho_2]^*[\rho_1]]{} & [\sigma_1] \oplus_{[\sigma]} [\sigma_2] \end{array}$$

Using Lemma 1.3.10 for $[\sigma_1] \xrightarrow{[\rho_1]^*[\rho_2]} [\sigma_1] \oplus_{[\sigma]} [\sigma_2] \xleftarrow{[\rho_2]^*[\rho_1]} [\sigma_2]$ can extend this to morphisms in $\mathbb{T}$

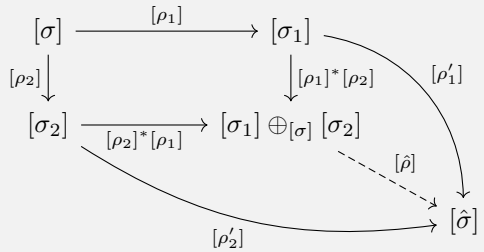
$$\sigma_1 \xrightarrow{\rho_1^*\rho_2} \sigma_1 \oplus_\sigma \sigma_2 \xleftarrow{\rho_2^*\rho_1} \sigma_2$$

We have that $\rho_1^*\rho_2 \circ \rho_1 = \rho_2^*\rho_1 \circ \rho_2$ from $[\rho_1]^*[\rho_2] \circ [\rho_1] = [\rho_2]^*[\rho_1] \circ [\rho_2]$.

Induced Morphism Consider the diagram where each square is independent.

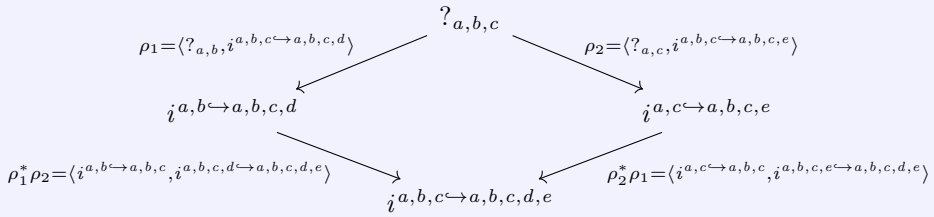


The $[\hat{\rho}]$ part of morphism $\hat{\rho}$ is induced by



which by Lemma 1.3.10 extends to a unique $\hat{\rho}$. $\mathcal{I}$-neutrality of the injections and composition of independent (product) squares follows from the independent structure on $\mathbb{W}$. □

Example 1.3.15. The following diagram is a L.I.C. square:



Notice that location variables d and e are not conflated because they are fresh w.r.t. $?_{a,b,c}$.

Remark 1.3.16. Applying Lemma 1.3.12 to the L.I.C. of morphisms $\sigma_1 \xleftarrow{\rho_1} \sigma \xrightarrow{\rho_2} \sigma_2$, we have that if ρ_1 is equipped with instantiation data $\eta : \underline{\mathbb{H}}(\lfloor\sigma_1\rfloor \ominus \lfloor\rho_1\rfloor, [\sigma_1])$, then $\rho_2^*\rho_1$ can be equipped with instantiation data

$$\eta_{\rho_2^*\rho_1} : \underline{\mathbb{H}}(\lfloor\sigma_1 \oplus_\sigma \sigma_2\rfloor \ominus \lfloor\rho_2^*\rho_1\rfloor, [\sigma_1 \oplus_\sigma \sigma_2])$$

Example 1.3.17. Consider the square from Example 1.3.15. If morphism ρ_1 is equipped with instantiation data $\eta : \underline{\mathbb{H}}(\{a,b\}, \{a,b,c,d\})$, we equip $\rho_2^*\rho_1$ with instantiation data $\underline{\mathbb{H}}(\{b\}, \{a,b,c,d,e\})$ by $\underline{\mathbb{H}}(i^{b\hookrightarrow a,b}, i^{a,b,c,d\hookrightarrow a,b,c,d,e})(\eta)$.

Lemma 1.3.18 (Pasting Law for Local Independent Coproducts)**.** In a diagram $\begin{array}{ccccc} \sigma & \xrightarrow{\rho_1} & \sigma_1 & \xrightarrow{\rho_3} & \sigma_3 \\ {\scriptstyle\rho_1'}\downarrow & & \downarrow{\scriptstyle\rho_2} & & \downarrow{\scriptstyle\rho_3'} \\ \sigma_1' & \xrightarrow[\rho_2']{} & \sigma_2 & \xrightarrow[\rho_3'']{} & \sigma_3' \end{array}$ where the left square is a L.I.C., then the whole square is a L.I.C. iff the right square is.

Proof. By Lemma 1.3.11, to show that a commuting square is a L.I.C., it is enough to show that the morphisms out of the span are jointly surjective at the $\lceil - \rceil$, and $\lfloor - \rfloor$ level, and further that we have a pullback at the $\lceil - \rceil$ level.
For the $\to$ direction, it is clear that if any $\lceil \rho_3'' \rceil$ and $\lceil \rho_3' \rceil$ are not jointly surjective then the whole square cannot be as well which is a contradiction. The same applies to the $\lfloor - \rfloor$ level, and so the right square must be jointly surjective. We must further show that we have a pullback at the $\lceil - \rceil$ level, and so consider $\lceil \rho_3'' \rceil(b) = \lceil \rho_3' \rceil(a)$. From the left square being a L.I.C. we have that $\lceil \rho_2' \rceil$ and $\lceil \rho_2 \rceil$ are jointly surjective. So if $\lceil \rho_2 \rceil(b') = b$ then we are done. Otherwise we have $\lceil \rho_2' \rceil(b') = b$, and by the outer square being L.I.C., it is a pullback and therefor, b' and a are related by $\lceil \sigma \rceil$, which implies that they are related by $\lceil \sigma_1 \rceil$ also.
For the $\leftarrow$ direction, consider a $a : C \in \lceil \sigma_3' \rceil$, and we show that it is mapped from $\lceil \sigma_1' \rceil$ or $\lceil \sigma_3 \rceil$. By the right square being jointly surjective, either a is mapped from $\lceil \sigma_3 \rceil$ or it comes from $\lceil \sigma_2 \rceil$. If it comes from $\lceil \sigma_2 \rceil$, then by the left square being L.I.C. either it is mapped from $\lceil \sigma_1' \rceil$ or it is mapped from $\lceil \sigma_1 \rceil$ and since it is a commuting square, it must also be mapped from $\lceil \sigma_3 \rceil$. The same reasoning applies to $\lfloor - \rfloor$. That the outer square is also a pullback at the $\lceil - \rceil$ level follows from pullback pasting. □

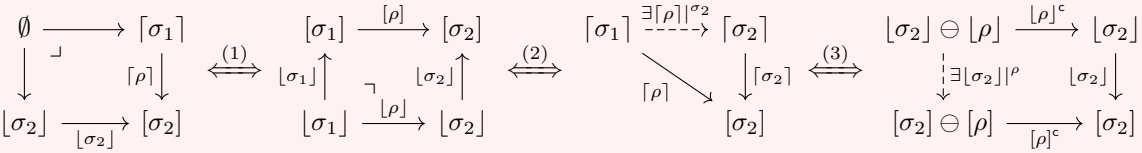
1.3.3 Variable Preserving Morphisms

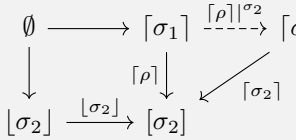
We identify a class of morphisms in $\mathbb{T}$: morphisms that do not instantiate any location variables from $\lceil \sigma_1 \rceil$. We call these morphisms variable preserving since every variable in $\lceil \sigma_1 \rceil$ is also a variable in $\lceil \sigma_2 \rceil$.

Definition 1.3.19 (Variable Preserving Morphisms)**.** A morphism $\rho : \sigma_1 \to \sigma_2$ is variable preserving if it satisfies $\lfloor \sigma_2 \rfloor \otimes_{\lceil \sigma_2 \rceil} \lceil \sigma_1 \rceil = \emptyset$.

Example 1.3.20. The morphism $?_a \xrightarrow{\langle ?_b, i^{a \hookrightarrow a,b,c} \rangle} i^{b \hookrightarrow a,b,c}$ is a variable preserving morphism since variable a from the domain is mapped to a variable in the codomain. The morphism $?_a \xrightarrow{\langle ?_a, 1_a \rangle} 1_a$ is not a variable preserving morphism since variable a is instantiated by the morphism.

Lemma 1.3.21 (Variable Preserving Property Equivalence)**.** Let $\rho : \sigma_1 \to \sigma_2$. We have the following equivalences

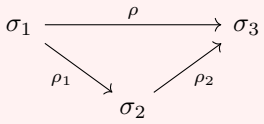


Proof.
For (1), consider the canonical isomorphism $\lceil \sigma_1 \rceil \cong \lfloor \sigma_1 \rfloor \oplus \lceil \sigma_1 \rceil$.
For $\overset{(2)}{\Longrightarrow}$ consider $\lceil \rho \rceil : \lceil \sigma_1 \rceil \to \lceil \sigma_2 \rceil \cong \lfloor \sigma_2 \rfloor \oplus \lceil \sigma_2 \rceil$. If $\lceil \rho \rceil$ maps to $\lfloor \sigma_2 \rfloor$ we get a contradiction from the pullback.
For $\overset{(2)}{\Longleftarrow}$, we have the diagram  which is a pullback since $\mathsf{Img}(\lceil \sigma_2 \rceil) \cap \mathsf{Img}(\lfloor \sigma_2 \rfloor) = \emptyset$ which makes the inner square a pullback.
For $\overset{(3)}{\Longrightarrow}$ the second pullback gives $\lfloor \sigma_2 \rfloor \lfloor \rho \rfloor^{\mathsf{c}}(a : C) \notin \mathsf{Img}([\rho])$.
For $\overset{(3)}{\Longleftarrow}$, we show $\lfloor \sigma_2 \rfloor \otimes_{\lceil \sigma_2 \rceil} \lceil \sigma_1 \rceil = \emptyset$, and so consider $\lceil \rho \rceil(a : C) = \lfloor \sigma_2 \rfloor(b : C)$. From ρ being a commuting square we have $b : C \in \lfloor \sigma_2 \rfloor \ominus \lfloor \rho \rfloor$. But then $\lfloor \sigma_2 \rfloor(b) \in \lceil \sigma_2 \rceil \ominus \lceil \rho \rceil$ which is a contradiction as it must also be equal to $\lceil \rho \rceil(a)$. □

Notation. For a variable preserving morphism $\sigma_1 \xrightarrow{\rho} \sigma_2$, let $\rho^{\mathsf{c}} : (\sigma_2 \ominus \rho) \to \sigma_2$ denote the morphism in $\mathbb{T}$ given by $\xRightarrow{(3)}$ in Lemma 1.3.21.

Lemma 1.3.22 (Variable Preservation Closed Under Composition)**.** Variable preserving morphisms are closed under composition.

Proof. Follows directly by pullback pasting. □

Lemma 1.3.23 (Variable Preserving Cancellation)**.** In  if ρ and ρ_2 are variable preserving, so is ρ_1.

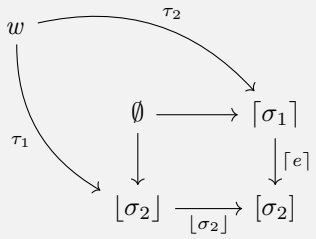
Proof. Follows directly by pullback pasting. □

Lemma 1.3.24 (Equalizers are Variable Preserving)**.** In an equalizer diagram

$$\sigma_1 \xrightarrow{e} \sigma_2 \underset{\rho_2}{\overset{\rho_1}{\rightrightarrows}} \sigma_3$$

e is variable preserving.

Proof.
We want to show $\lfloor \sigma_2 \rfloor \otimes_{\lceil \sigma_2 \rceil} \lceil \sigma_1 \rceil = \emptyset$ and so consider



Notice that the following square e' commutes

$$\begin{array}{ccc} \lfloor \sigma_1 \rfloor \oplus \lceil \sigma_1 \rceil \cong \lceil \sigma_1 \rceil & \xrightarrow{\lceil e \rceil} & \lceil \sigma_2 \rceil \\ {\scriptstyle 1 \oplus \tau_2}\uparrow & & \uparrow{\scriptstyle \lfloor \sigma_2 \rfloor} \\ \lfloor \sigma_1 \rfloor \oplus w & \xrightarrow[[\lfloor e \rfloor, \tau_1]]{} & \lfloor \sigma_2 \rfloor \end{array}$$

and the copairing $[\lfloor e \rfloor, \tau_1]$ is mono

$$\begin{aligned} [\lfloor e \rfloor, \tau_1] \circ \theta_1 = [\lfloor e \rfloor, \tau_1] \circ \theta_2 &\iff \\ \lfloor \sigma_2 \rfloor \circ [\lfloor e \rfloor, \tau_1] \circ \theta_1 = \lfloor \sigma_2 \rfloor \circ [\lfloor e \rfloor, \tau_1] \circ \theta_2 &\iff \\ [e] \circ 1 \oplus \tau_2 \circ \theta_1 = [e] \circ 1 \oplus \tau_2 \circ \theta_2 &\iff \\ 1 \oplus \tau_2 \circ \theta_1 = 1 \oplus \tau_2 \circ \theta_2 &\iff \\ \theta_1 = \theta_2 & \end{aligned}$$

Further notice that e' commutes with ρ_1, ρ_2 from $\lceil\rho_1\rceil \circ \lceil e\rceil = \lceil\rho_1\rceil \circ \lceil e\rceil$

$$\begin{aligned}
\lfloor\rho_1\rfloor \circ \tau_1 = \lfloor\rho_2\rfloor \circ \tau_1 &\iff \\
\lfloor\sigma_3\rfloor \circ \lfloor\rho_1\rfloor \circ \tau_1 = \lfloor\sigma_3\rfloor \circ \lfloor\rho_2\rfloor \circ \tau_1 &\iff \\
\lceil\rho_1\rceil \circ \lfloor\sigma_2\rfloor \circ \tau_1 = \lceil\rho_2\rceil \circ \lfloor\sigma_2\rfloor \circ \tau_1 &\iff \\
\lceil\rho_1\rceil \circ \lceil e\rceil \circ \tau_2 = \lceil\rho_2\rceil \circ \lceil e\rceil \circ \tau_2 = & \\
\lceil\rho_1\rceil \circ \lceil e\rceil \circ \lceil\sigma_1\rceil \circ \tau_2 = \lceil\rho_2\rceil \circ \lceil e\rceil \circ \lceil\sigma_1\rceil \circ \tau_2 &
\end{aligned}$$

and so the universal property of the equalizer gives us an injective morphism $\lfloor\sigma_1\rfloor \oplus w \to \lfloor\sigma_1\rfloor$, which means $w = \emptyset$. □

Lemma 1.3.25 (Regular Monos in $\mathbb{T}$)**.** A morphism $\rho : \sigma_1 \to \sigma_2$ in $\mathbb{T}$ is regular mono iff ρ is variable preserving.

Proof.
$\to$) Follows directly from Lemma 1.3.24.
$\leftarrow$) We show that this is an equalizer diagram:

$$\sigma_1 \xrightarrow{\rho} \sigma_2 \underset{\rho^*\rho_r}{\overset{\rho^*\rho_l}{\rightrightarrows}} \sigma_2 \oplus_{\sigma_1} \sigma_2$$

From Lemma 1.3.21 and Lemma 1.3.27 we have $\lfloor\sigma_2\rfloor|^\rho$ and we can simplify the diagram to

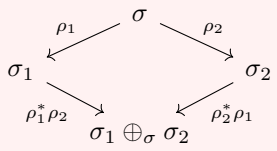
$$\begin{array}{ccccc}
 & & [\sigma_1] \oplus ([\sigma_2] \ominus [\rho]) & \underset{1\oplus i_2}{\overset{1\oplus i_1}{\rightrightarrows}} & [\sigma_1] \oplus ([\sigma_2] \ominus [\rho]) \oplus ([\sigma_2] \ominus [\rho]) \\
\sigma_2 & \cong & \uparrow{\scriptstyle \lfloor\sigma_1\rfloor \oplus \lfloor\sigma_2\rfloor|^\rho} & & \uparrow{\scriptstyle \lfloor\sigma_1\rfloor \oplus \lfloor\sigma_2\rfloor|^\rho \oplus \lfloor\sigma_2\rfloor|^\rho} \quad \cong \ \sigma_2 \oplus_{\sigma_1} \sigma_2 \\
 & & \lfloor\sigma_1\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\rho\rfloor) & \underset{1\oplus i_2}{\overset{1\oplus i_1}{\rightrightarrows}} & \lfloor\sigma_1\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\rho\rfloor) \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\rho\rfloor)
\end{array}$$

Consider $(1 \oplus i_1)\circ \cong \circ\phi = (1 \oplus i_2)\circ \cong \circ\phi$. Then ϕ must factor through ρ and uniqueness follows from ρ being mono. □

Lemma 1.3.26 (Isos are Variable Preserving)**.** All isomorphisms are variable preserving morphisms.

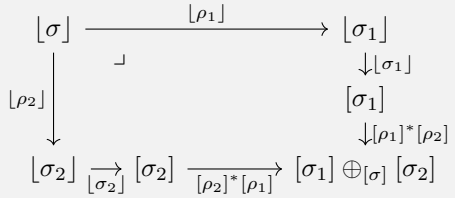
Proof. Follows from Lemma 1.3.25. □

Lemma 1.3.27 (Local Independent Coproducts and Variable Preserving Property)**.** In a L.I.C. where ρ_2 is a variable preserving morphism



we have that $\sigma_1 \oplus_\sigma \sigma_2 \cong \lfloor\sigma_1\rfloor \oplus_{\lfloor\sigma\rfloor} \lfloor\sigma_2\rfloor \xrightarrow{\lfloor\sigma_1\rfloor \oplus \lfloor\sigma_2\rfloor|^{\rho_2}} [\sigma_1] \oplus_{[\sigma]} [\sigma_2]$ and $\rho_1^*\rho_2$ is variable preserving as well.

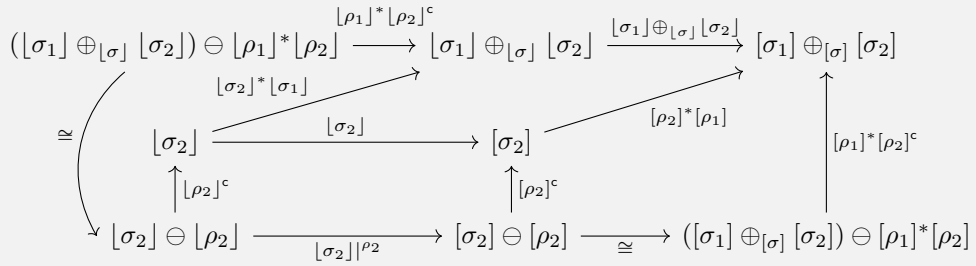
Proof.
Notice that this is a pullback square



By noticing the canonical isomorphism

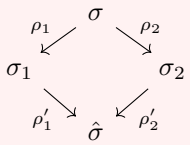
$$[\sigma_1] \oplus_{[\sigma]} [\sigma_2] \cong \lfloor\sigma\rfloor \oplus \lceil\sigma\rceil \oplus ([\sigma_1] \ominus [\rho_1]) \oplus ([\sigma_2] \ominus [\rho_2])$$

and that $[\rho_1]^*[\rho_2] \circ \lfloor\sigma_1\rfloor$ maps to $\lfloor\sigma\rfloor$, $\lceil\sigma\rceil$ and $[\sigma_1] \ominus [\rho_1]$, whereas $[\rho_2]^*[\rho_1] \circ \lfloor\sigma_2\rfloor$ maps to $\lfloor\sigma\rfloor$ and $([\sigma_2] \ominus [\rho_2])$ only (via $\lfloor\sigma_2\rfloor^{\rho_2}$).
That the induced morphism $\lfloor\sigma_1\rfloor \oplus_{\lfloor\sigma\rfloor} \lfloor\sigma_2\rfloor$ is $\lfloor\sigma_1\rfloor \oplus \lfloor\sigma_2\rfloor^{\rho_2}$ follows from uniqueness ($\lfloor\sigma_1\rfloor$ here has the isomorphic type $\lfloor\sigma\rfloor \oplus (\lfloor\sigma_1\rfloor \ominus \lfloor\rho_1\rfloor) \to [\sigma] \oplus ([\sigma_1] \ominus [\rho_1])$).
That $\rho_1^*\rho_2$ is variable preserving follows from



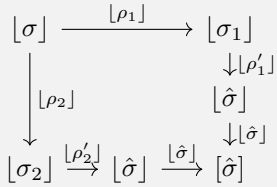
□

Lemma 1.3.28 (Pullbacks and Variable Preserving Property)**.** Let ρ_2 be variable preserving in

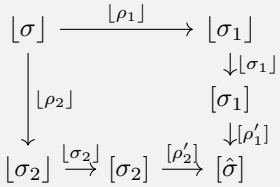


Then the above diagram is a pullback iff its $[-]$-image is a pullback.

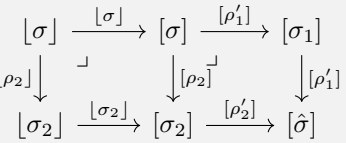
Proof. The $\to$ direction is immediate by Lemma 1.3.8. For $\leftarrow$, by Lemma A.1.1 it is enough to show that



is a pullback which is the same as



which is given by the pullback



and Lemma A.1.4. □

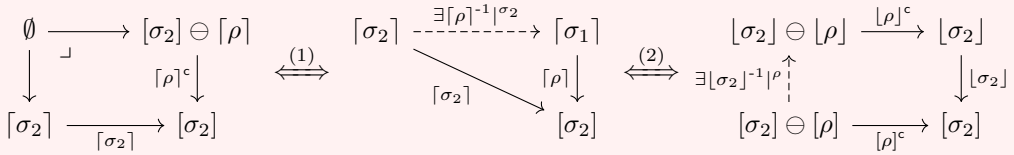
1.3.4 Variable Reflecting Morphisms

We identify another class of morphisms in $\mathbb{T}$: morphisms that do not introduce location variables. We call these morphisms variable reflecting since every variable in $\lceil\sigma_2\rceil$ comes from $\lceil\sigma_1\rceil$.

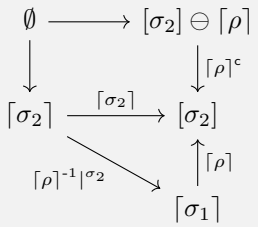
Definition 1.3.29 (Variable Reflecting Morphisms)**.** A morphism $\rho : \sigma_1 \to \sigma_2$ is variable reflecting if it satisfies $(\lfloor \sigma_2 \rfloor \ominus \lceil \rho \rceil) \otimes_{\lfloor \sigma_2 \rfloor} \lceil \sigma_2 \rceil = \emptyset$.

Example 1.3.30. The morphism $?_{a,b} \xrightarrow{\langle ?_a, 1_{a,b} \rangle} i^{a \hookrightarrow a,b}$ is a variable reflecting morphism since variable b from the codomain comes from the domain. The morphism $?_a \xrightarrow{\langle ?_a, i^{a \hookrightarrow a,b} \rangle} i^{a \hookrightarrow a,b}$ is not a variable reflecting morphism since variable b is fresh.

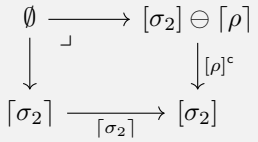
Lemma 1.3.31 (Variable Reflecting Property Equivalence)**.** Let $\rho : \sigma_1 \to \sigma_2$. We have the following equivalences



Proof. For $\overset{(1)}{\Longrightarrow}$, consider $\lceil \rho \rceil : \lceil \sigma_1 \rceil \to [\sigma_2] \cong \lfloor \sigma_2 \rfloor \oplus \lceil \sigma_2 \rceil$. If there exists $a : C \in \lceil \sigma_2 \rceil$ that $\lceil \rho \rceil$ does not map to, it must also be an element of $[\sigma_2] \ominus \lceil \rho \rceil$ by definition. But then the pullback gives us a contradiction. For $\overset{(1)}{\Longleftarrow}$, we have a diagram



which is a pullback since $\mathsf{Img}(\lceil \rho \rceil) \cap \mathsf{Img}(\lceil \rho \rceil^{\mathsf{c}}) = \emptyset$. For $\overset{(2)}{\Longrightarrow}$, first notice that this is a pullback



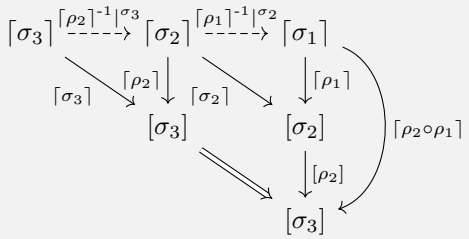
given by the inclusion $[\sigma_2] \ominus [\rho] \hookrightarrow [\sigma_2] \ominus \lceil \rho \rceil$ and pullback pasting. Now consider the morphism

$$[\sigma_2] \ominus [\rho] \xrightarrow{[\rho]^{\mathsf{c}}} [\sigma_2] \cong \lfloor \sigma_1 \rfloor \oplus (\lfloor \sigma_2 \rfloor \ominus \lfloor \rho \rfloor) \oplus \lceil \sigma_2 \rceil$$

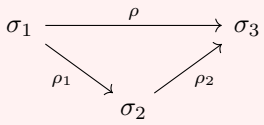
From ρ being a commuting square it cannot map to $\lfloor \sigma_1 \rfloor$, and from the pullback it can also not map to $\lceil \sigma_2 \rceil$. Lastly for $\overset{(2)}{\Longleftarrow}$, consider $\lceil \rho \rceil : \lceil \sigma_1 \rceil \to [\sigma_2] \cong \lfloor \sigma_2 \rfloor \oplus \lceil \sigma_2 \rceil$, and a $a : C \in \lceil \sigma_2 \rceil$ that is not mapped to from $\lceil \rho \rceil$. This gives us that either $a \in [\sigma_2] \ominus [\rho]$ which is a contradiction from $\lfloor \sigma_2 \rfloor^{-1}|^{\rho}$ or it comes from $\lfloor \sigma_1 \rfloor$ which is a contradiction from ρ being a commutative square. □

Lemma 1.3.32 (Composition of Variable Reflecting Morphisms)**.** Variable reflecting morphisms are closed under composition.

Proof. For variable reflecting morphisms $\sigma_1 \xrightarrow{\rho_1} \sigma_2 \xrightarrow{\rho_2} \sigma_3$ we have



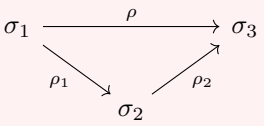
□

Lemma 1.3.33 (Variable Reflecting Cancellation)**.** In  if ρ is variable reflecting, so is ρ_2.

Proof. We show $\begin{array}{ccc} \lfloor\sigma_3\rfloor \ominus \lfloor\rho_2\rfloor & \xrightarrow{\lfloor\rho_2\rfloor^c} & \lfloor\sigma_3\rfloor \\ {\scriptstyle \exists \lfloor\sigma_3\rfloor^{-1}\mid\rho_2}\uparrow & & \downarrow{\scriptstyle \lfloor\sigma_3\rfloor} \\ [\sigma_3] \ominus [\rho_2] & \xrightarrow[[\rho_2]^c]{} & [\sigma_3] \end{array}$. The variable reflecting property of ρ gives us a $([\sigma_3]\ominus [\rho]) \dashrightarrow (\lfloor\sigma_3\rfloor \ominus \lfloor\rho\rfloor)$ and using the canonical isos we have

$$[\sigma_3] \ominus [\rho_2] \to [\sigma_3] \ominus [\rho] \to \lfloor\sigma_3\rfloor \ominus \lfloor\rho\rfloor \cong (\lfloor\sigma_3\rfloor \ominus \lfloor\rho_2\rfloor) \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\rho_1\rfloor)$$

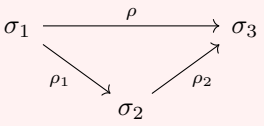
and its image must lie in $\lfloor\sigma_3\rfloor \ominus \lfloor\rho_2\rfloor$ by the commuting square of ρ_2. □

Lemma 1.3.34 (Variable Preserving with Reflecting Cancellation)**.** In  if ρ is variable preserving and ρ_1 is reflecting, then ρ_2 is variable preserving.

Proof. The variable preserving property of ρ gives us a $\lfloor\sigma_3\rfloor\ominus\lfloor\rho\rfloor \dashrightarrow [\sigma_3]\ominus[\rho]$, the variable reflecting property of ρ_1 gives us a $[\sigma_2] \ominus [\rho_1] \dashrightarrow \lfloor\sigma_2\rfloor \ominus \lfloor\rho_1\rfloor$, and together with the canonical isos we have

$$\lfloor\sigma_3\rfloor \ominus \lfloor\rho_2\rfloor \to \lfloor\sigma_3\rfloor \ominus \lfloor\rho\rfloor \to [\sigma_3] \ominus [\rho] \cong ([\sigma_3] \ominus [\rho_2]) \oplus ([\sigma_2] \ominus [\rho_1]) \to ([\sigma_3] \ominus [\rho_2]) \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\rho_1\rfloor)$$

and its image must lie in $[\sigma_3] \ominus [\rho_2]$. □

Lemma 1.3.35 (Variable Reflecting with Preserving Cancellation)**.** In  if ρ is variable reflecting and ρ_2 is variable preserving, then ρ_1 is variable reflecting.

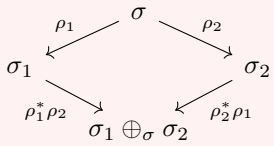
Proof. The variable reflecting property of ρ gives us a $([\sigma_3] \ominus [\rho]) \dashrightarrow (\lfloor\sigma_3\rfloor \ominus \lfloor\rho\rfloor)$, and the variable preserving property of ρ_2 gives us a $\lfloor\sigma_3\rfloor \ominus \lfloor\rho_2\rfloor \dashrightarrow [\sigma_3] \ominus [\rho_2]$, and together with the canonical isos

we have

$$[\sigma_2] \ominus [\rho_1] \to [\sigma_3] \ominus [\rho] \to \lfloor \sigma_3 \rfloor \ominus \lfloor \rho \rfloor \cong (\lfloor \sigma_3 \rfloor \ominus \lfloor \rho_2 \rfloor) \oplus (\lfloor \sigma_2 \rfloor \ominus \lfloor \rho_1 \rfloor) \to ([\sigma_3] \ominus [\rho_2]) \oplus (\lfloor \sigma_2 \rfloor \ominus \lfloor \rho_1 \rfloor)$$

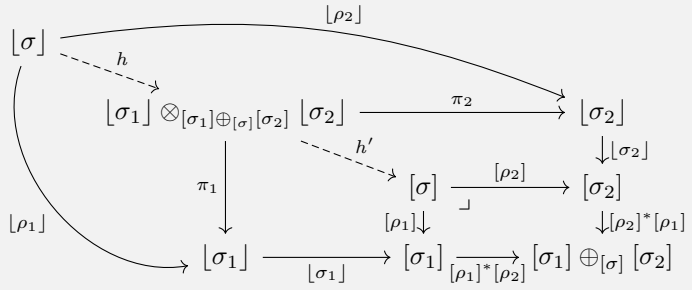
and its image must lie in $\lfloor \sigma_2 \rfloor \ominus \lfloor \rho_1 \rfloor$. □

Lemma 1.3.36 (Local Independent Coproducts and Variable Reflecting Morphisms)**.** In a L.I.C. square where ρ_2 is variable reflecting:

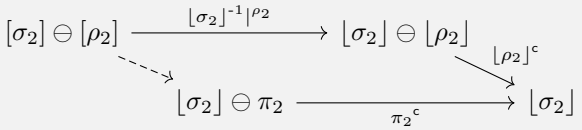


we have that $\rho_1^* \rho_2$ is variable reflecting as well.

Proof. Consider



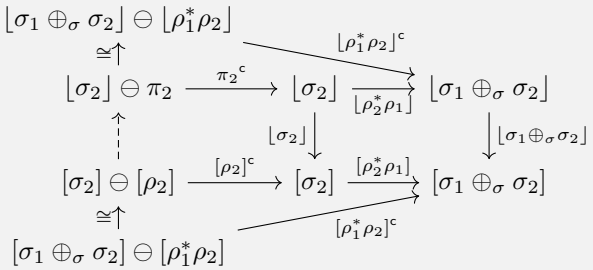
We first show that $\lfloor \sigma_2 \rfloor^{-1} | ^{\rho_2}$ factors through $\lfloor \sigma_2 \rfloor \ominus \pi_2$



and so consider $a : C \in [\sigma_2] \ominus [\rho_2]$. From the diagram we have that $a : C \in [\sigma_2] \ominus (\lfloor \sigma_2 \rfloor \circ \pi_2)$ and consider the canonical isomorphism

$$[\sigma_2] \ominus (\lfloor \sigma_2 \rfloor \circ \pi_2) \cong ([\sigma_2] \ominus \lfloor \sigma_2 \rfloor) \oplus (\lfloor \sigma_2 \rfloor \ominus \pi_2)$$

Morphism $\lfloor \sigma_2 \rfloor^{-1} | ^{\rho_2}$ tells us that a is in the image of $\lfloor \sigma_2 \rfloor$, and so it must come from $\lfloor \sigma_2 \rfloor \ominus \pi_2$. We then show that $\rho_1^* \rho_2$ is variable reflecting by observing that the following diagram commutes



□

1.3.5 Factorisation System

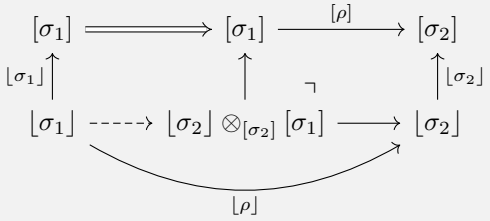
Lemma 1.3.37 (Epimorphisms in $\mathbb{T}$)**.** A morphism $\rho : \sigma_1 \to \sigma_2$ in $\mathbb{T}$ is epi iff $[\rho]$ is surjective.

Proof.
$\to$) Consider $\rho : \sigma_1 \to \sigma_2$ such that for every $l\, r$, $l \circ \rho = r \circ \rho \implies l = r$. Take the L.I.C. of $\sigma_2 \xleftarrow{\rho} \sigma_1 \xrightarrow{\rho} \sigma_2$, which gives us two morphisms $\rho^*\rho_l, \rho^*\rho_r : \sigma_2 \to \sigma_2 \oplus_{\sigma_1} \sigma_2$ that satisfy $\rho^*\rho_l \circ \rho = \rho^*\rho_r \circ \rho$. From the epi property of ρ, we have that $[\rho^*\rho_l] = [\rho^*\rho_r]$ which implies that $[\rho]$ is surjective by definition of $[\sigma_1] \oplus_{[\sigma]} [\sigma_2]$.
$\leftarrow$) Let l, r such that $l \circ \rho = r \circ \rho$. This means that $[l] \circ [\rho] = [r] \circ [\rho]$ and so $[l] = [r]$ which also implies $\lfloor l \rfloor = \lfloor r \rfloor$. □

Notation. We denote epis using $\twoheadrightarrow$ and regular monos with $\rightarrowtail_{\mathsf{r}}$.

Lemma 1.3.38 (Orthogonal Factorisation System in $\mathbb{T}$)**.** The classes of epis and regular monos (variable preserving) morphisms gives rise to an orthogonal factorisation system in $\mathbb{T}$.

Proof. The intuition here is that the epi part will represent the instantiations of the location variables coming from its domain and the regular mono part will be the introduction of all other locations. A morphism $\rho : \sigma_1 \to \sigma_2$ is factored as follows



□

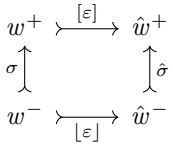
Example 1.3.39. We factor the morphism $?_a \xrightarrow{\langle ?_{a,b}, i^{a \hookrightarrow a,b,c} \rangle} i^{a,b \hookrightarrow a,b,c}$ as

$$?_a \xrightarrow{\langle ?_a, 1_a \rangle} 1_a \xrightarrow{\langle i^{a \hookrightarrow a,b}, i^{a \hookrightarrow a,b,c} \rangle} i^{a,b \hookrightarrow a,b,c}$$

1.4 The category of template instantiations $\mathbb{TI}$

We extend category $\mathbb{T}$ with instantiation data. This is where our template heap functor will be indexed over.

- Objects: Sort-preserving injective renamings between worlds $w^- \rightarrowtail w^+$.
- Morphisms: Injective, sort-preserving $\lfloor\varepsilon\rfloor$ and $[\varepsilon]$ such that



  and instantiation data $\eta_\varepsilon : \underline{\mathbb{H}}(\hat{w}^- \ominus \lfloor\varepsilon\rfloor, \hat{w}^+)$.

We have a forgetful functor $u : \mathbb{TI} \to \mathbb{T}$, and an inclusion functor $1_ : \mathbb{WI} \hookrightarrow \mathbb{TI}$ which maps to its identity morphism. We inherit the terminology of $\mathbb{T}$ and talk about concrete locations and location variables, we overload functors $\lfloor - \rfloor, [-] : \mathbb{TI} \to \mathbb{W}$ by precomposing them with u, and talk about variable preserving and reflecting morphisms by noticing that u reflects these properties.

Definition 1.4.1 (Composition of morphisms in $\mathbb{TI}$)**.** Composition of $\sigma_1 \xrightarrow{\varepsilon_1} \sigma_2 \xrightarrow{\varepsilon_2} \sigma_3$ is defined as composition of the underlying injections and composition of the instantiation data is given by

$$\eta_{\varepsilon_2 \circ \varepsilon_1} : \underline{\mathbb{H}}(\lfloor \sigma_3 \rfloor \ominus \lfloor \varepsilon_2 \circ \varepsilon_1 \rfloor, \lceil \sigma_3 \rceil)$$
$$a : C \mapsto \begin{cases} [\![\phi(C)]\!][\varepsilon_2](\pi_{a'}\eta_{\varepsilon_1}) & \text{if } \lfloor \varepsilon_2 \rfloor(a') = a \\ \pi_a \eta_{\varepsilon_2} & \text{if } a \in \lfloor \sigma_3 \rfloor \ominus \lfloor \varepsilon_2 \rfloor \end{cases}$$

Remark 1.4.2. Every $\sigma, \eta \in \underline{\mathbb{H}}(\lfloor \sigma \rfloor, \lceil \sigma \rceil)$ induces a morphism $?_\eta$

$$\begin{array}{ccc} \emptyset & \xrightarrow{?_{\lceil \sigma \rceil}} & \lceil \sigma \rceil \\ {\scriptstyle ?_\emptyset}\uparrow & & \uparrow{\scriptstyle \lfloor \sigma \rfloor} \\ \emptyset & \xrightarrow[?_{\lfloor \sigma \rfloor}]{} & \lfloor \sigma \rfloor \end{array}$$

Definition 1.4.3 (Template Heap Functor)**.** We define the template heap functor $\mathbb{TH} : \mathbb{TI} \to \mathsf{Set}$ as

$$\mathbb{TH}(\sigma_1 \xrightarrow{\varepsilon} \sigma_2) : \underline{\mathbb{H}}(\lfloor \sigma_1 \rfloor, \lceil \sigma_1 \rceil) \to \underline{\mathbb{H}}(\lfloor \sigma_2 \rfloor, \lceil \sigma_2 \rceil)$$
$$\mathbb{TH}(\varepsilon)(\eta') = \eta_{\varepsilon \circ ?_{\eta'}}$$

Note that $\eta_\varepsilon : \mathbb{TH}(\lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon \rfloor \xrightarrow{\lfloor \varepsilon \rfloor^c} \lfloor \sigma_2 \rfloor \xrightarrow{\lfloor \sigma_2 \rfloor} \lceil \sigma_2 \rceil)$ and we have that $\mathbb{TH} \cong \mathbb{TI}(?_\emptyset, -)$.

Definition 1.4.4 (Variable Extension Functor)**.** We lift the variable extension functor from $\mathbb{T}$ to $\mathbb{TI}$

$$(- \oplus w) : \mathbb{TI} \to \mathbb{TI}$$
$$(\sigma \oplus w) = i_1 \circ \lfloor \sigma \rfloor$$
$$(\varepsilon \oplus w) = \langle (u\varepsilon \oplus w), \underline{\mathbb{H}}\langle 1, i_1 \rangle(\eta_\varepsilon) \rangle$$

Lemma 1.4.5 (Inhabitation in $\mathbb{TI}$)**.** A set of sorts is inhabited (Definition 1.2.1) iff for every template σ, there exists a concrete template $1_{w'}$ and a morphism $\sigma \xrightarrow{\varepsilon} 1_{w'}$.

Proof. For the $\to$ direction, consider $\lceil \sigma \rceil$ and so for every $a : C \in \lceil \sigma \rceil$ we have a $a : C \to w$. By induction on world $\lceil \sigma \rceil$, this gives us a $\lceil \sigma \rceil \to \hat{w}$ and a heap $\eta : \mathbb{H}(\hat{w})$ by taking the disjoint sum of the worlds and heaps. We use this data to construct a morphism from σ to $1_{\lfloor \sigma \rfloor \oplus \hat{w}}$.
For the $\leftarrow$ direction, we have a morphism $?_{a:C} \xrightarrow{\varepsilon} 1_w$ for every $a : C$ which is exactly the data needed. □

Lemma 1.4.6 (Monomorphisms in $\mathbb{TI}$)**.** Every morphism in $\mathbb{TI}$ is mono.

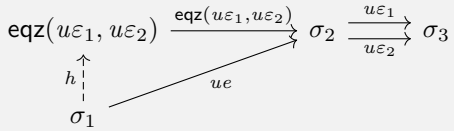
Proof. We show $\hat{\varepsilon} : \sigma_2 \to \hat{\sigma}$ is mono. Consider $\hat{\varepsilon} \circ \varepsilon_1 = \hat{\varepsilon} \circ \varepsilon_2$. We have that $u\varepsilon_1 = u\varepsilon_2$ from Lemma 1.3.4. It remains to show $\eta_{\varepsilon_1} = \eta_{\varepsilon_2}$, so take $a : C \in \lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon_1 \rfloor$.

$$\pi_a \eta_{\varepsilon_1} = \pi_a \eta_{\varepsilon_2} \iff$$
$$[\![\phi(C)]\!][\hat{\varepsilon}](\pi_a \eta_{\varepsilon_1}) = [\![\phi(C)]\!][\hat{\varepsilon}](\pi_a \eta_{\varepsilon_2}) \iff$$
$$\pi_{\lfloor \hat{\varepsilon} \rfloor(a)}(\eta_{\varepsilon \circ \varepsilon_1}) = \pi_{\lfloor \hat{\varepsilon} \rfloor(a)}(\eta_{\varepsilon \circ \varepsilon_2})$$

□

Lemma 1.4.7 (Forgetful Functor Preserves Equalizers)**.** $u : \mathbb{TI} \to \mathbb{T}$ preserves equalizers.

Proof. Let $e : \sigma_1 \to \sigma_2$ be an equalizer of ε_1, ε_2. We show that σ_1 is canonically isomorphic to $\mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)$ when extended with appropriate instantiation data.



We equip h and $\mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)$ by splitting η_e using the canonical isomorphism

$$\lfloor \sigma_2 \rfloor \ominus \lfloor u\varepsilon \rfloor \cong (\lfloor \sigma_2 \rfloor \ominus \mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)) \oplus (\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor) \ominus \lfloor h \rfloor)$$

and we show that the $(\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor) \ominus \lfloor h \rfloor)$ part of data η_e has locations in $\mathsf{eqz}([u\varepsilon_1], [u\varepsilon_2])$. So consider $a : C \in \mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor) \ominus \lfloor h \rfloor$, and $\pi_{\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)(a)}\eta_e$. Notice that

$$\begin{aligned}
\eta_{\varepsilon_1 \circ e} &= \eta_{\varepsilon_2 \circ e} \implies && \text{(def of } \mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)) \\
\pi_{\lfloor \varepsilon_1 \rfloor(\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)(a))}\eta_{\varepsilon_1 \circ e} &= \pi_{\lfloor \varepsilon_2 \rfloor(\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)(a))}\eta_{\varepsilon_2 \circ e} \implies && \text{(def of } \eta_{\varepsilon' \circ \varepsilon}) \\
[\![\phi(C)]\!][\varepsilon_1](\pi_{\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)(a)}\eta_e) &= [\![\phi(C)]\!][\varepsilon_2](\pi_{\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)(a)}\eta_e)
\end{aligned}$$

and so for every $b : C' \in [\sigma_2]$, we have

$$b \text{ is in } \pi_{\mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)(a)}\eta_e \implies [\varepsilon_1](b) = [\varepsilon_2](b)$$

We show that $\eta_{\varepsilon_1 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)} = \eta_{\varepsilon_2 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)}$ and so consider $a : C \in \lfloor \sigma_3 \rfloor \ominus \lfloor \varepsilon_1 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2) \rfloor$ and proceed by cases

- $a \in \lfloor \sigma_3 \rfloor \ominus \lfloor u\varepsilon_1 \rfloor$ and $a \in \lfloor \sigma_3 \rfloor \ominus \lfloor u\varepsilon_2 \rfloor$)

$$\begin{aligned}
\pi_a\eta_{\varepsilon_1 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)} &= \pi_a\eta_{\varepsilon_2 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)} \iff \\
\pi_a\eta_{\varepsilon_1} &= \pi_a\eta_{\varepsilon_2} \iff \\
\pi_a\eta_{\varepsilon_1 \circ e} &= \pi_a\eta_{\varepsilon_2 \circ e}
\end{aligned}$$

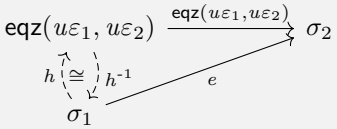
- $a \in \lfloor \sigma_3 \rfloor \ominus \lfloor u\varepsilon_1 \rfloor$, $a' \in \lfloor \sigma_2 \rfloor \ominus \mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)$, $a = \lfloor \varepsilon_2 \rfloor(a'))$

$$\begin{aligned}
\pi_a\eta_{\varepsilon_1 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)} &= \pi_a\eta_{\varepsilon_2 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)} \iff \\
\pi_a\eta_{\varepsilon_1} &= [\![\phi(C)]\!][\varepsilon_2](\pi_{a'}\eta_{\mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)}) \iff \\
\pi_a\eta_{\varepsilon_1} &= [\![\phi(C)]\!][\varepsilon_2](\pi_{a'}\eta_e) \iff \\
\pi_a\eta_{\varepsilon_1 \circ e} &= \pi_a\eta_{\varepsilon_2 \circ e}
\end{aligned}$$

- $a' \in \lfloor \sigma_2 \rfloor \ominus \mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)$, $a \in \lfloor \sigma_3 \rfloor \ominus \lfloor u\varepsilon_2 \rfloor$, $a = \lfloor \varepsilon_1 \rfloor(a'))$ Symmetric to case above.
- $a_1 \in \lfloor \sigma_2 \rfloor \ominus \mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)$, $a_2 \in \lfloor \sigma_2 \rfloor \ominus \mathsf{eqz}(\lfloor u\varepsilon_1 \rfloor, \lfloor u\varepsilon_2 \rfloor)$, $\lfloor \varepsilon_1 \rfloor(a_1) = a = \lfloor \varepsilon_2 \rfloor(a_2))$

$$\begin{aligned}
\pi_a\eta_{\varepsilon_1 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)} &= \pi_a\eta_{\varepsilon_2 \circ \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)} \iff \\
[\![\phi(C)]\!][\varepsilon_1](\pi_{a_1}\eta_{\mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)}) &= [\![\phi(C)]\!][\varepsilon_2](\pi_{a_2}\eta_{\mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)}) \iff \\
[\![\phi(C)]\!][\varepsilon_1](\pi_{a_1}\eta_e) &= [\![\phi(C)]\!][\varepsilon_2](\pi_{a_2}\eta_e) \iff \\
\pi_a\eta_{\varepsilon_1 \circ e} &= \pi_a\eta_{\varepsilon_2 \circ e}
\end{aligned}$$

As a result, by the UP of e we have a $h^{\text{-}1} : \mathsf{eqz}(u\varepsilon_1, u\varepsilon_2) \to \sigma_1$ in $\mathbb{TI}$. It remains to show that $h^{\text{-}1}$ is the inverse of h, that is



which follows from e and $\mathsf{eqz}(u\varepsilon_1, u\varepsilon_2)$ being mono. □

Lemma 1.4.8 (Forgetful Functor Reflects Pullbacks)**.** $u : \mathbb{TI} \to \mathbb{T}$ reflects pullbacks.

Proof. Consider a diagram $\begin{array}{ccccc} & & \sigma & & \\ & {\scriptstyle p_1}\swarrow & & \searrow{\scriptstyle p_2} & \\ \sigma_1 & & & & \sigma_2 \\ & {\scriptstyle \varepsilon_1}\searrow & & \swarrow{\scriptstyle \varepsilon_2} & \\ & & \hat{\sigma} & & \end{array}$ s.t. its u-image is a pullback. Given $\varepsilon_1 : \sigma' \to \sigma_1$, $\varepsilon_2 : \sigma' \to \sigma_2$ s.t. $\varepsilon_1 \circ \varepsilon_1' = \varepsilon_2 \circ \varepsilon_2'$, we get a $h : \sigma' \to \sigma$ in $\mathbb{T}$.
We equip h with data using the canonical iso

$$\lfloor \sigma_1 \rfloor \ominus \lfloor \varepsilon_1' \rfloor \cong (\lfloor \sigma_1 \rfloor \ominus \lfloor p_1 \rfloor) \oplus (\lfloor \sigma \rfloor \ominus \lfloor h \rfloor)$$

by giving $a : C \in \lfloor \sigma \rfloor \ominus \lfloor h \rfloor$, the cell $\pi_{\lfloor p_1 \rfloor(a)} \eta_{\varepsilon_1'}$. Notice that we can factor its locations through $[\sigma]$ since

$$\begin{aligned} [\![\phi(C)]\!][\varepsilon_1](\pi_{\lfloor p_1 \rfloor(a)} \eta_{\varepsilon_1'}) &= \pi_{\lfloor \varepsilon_1 \circ p_1 \rfloor(a)} \eta_{\varepsilon_1 \circ \varepsilon_1'} \\ &= \pi_{\lfloor \varepsilon_2 \circ p_2 \rfloor(a)} \eta_{\varepsilon_2 \circ \varepsilon_2'} \\ &= [\![\phi(C)]\!][\varepsilon_2](\pi_{\lfloor p_2 \rfloor(a)} \eta_{\varepsilon_2'}) \end{aligned}$$

and so for every $b : C' \in [\sigma_1]$, we have

$$b \text{ is in } \pi_{\lfloor p_1 \rfloor(a)} \eta_{\varepsilon_1'} \implies [\varepsilon_1](b) = [\varepsilon_2](b)$$

and so we can factor its locations through $[p_1]^{\text{-}1}$ since its u-image is a pullback.
To show that $\varepsilon_1' = p_1 \circ h$, consider $a : C \in \lfloor \sigma_1 \rfloor \ominus \lfloor p_1 \rfloor$. Since the $\lfloor - \rfloor$ image is a pullback, it must be that $\lfloor \varepsilon_1 \rfloor(a) \in \lfloor \hat{\sigma} \rfloor \ominus \lfloor \varepsilon_2 \rfloor$. This implies that

$$[\![\phi(C)]\!][\varepsilon_1](\pi_a \eta_{p_1}) = \pi_a \eta_{\varepsilon_1'} = [\![\phi(C)]\!][\varepsilon_1](\pi_a \eta_{\varepsilon_1'})$$

which implies $\pi_a \eta_{p_1} = \pi_a \eta_{\varepsilon_1'}$. The case for $\varepsilon_2' = p_1 \circ h$ is symmetric and uniqueness of h follows from π_1 being mono. □

Lemma 1.4.9 (Location Functor Preserves Pullbacks)**.** $[-] : \mathbb{TI} \to \mathbb{W}$ preserves pullbacks.

Proof. Consider $\begin{array}{ccc} \sigma & \xrightarrow{p_2} & \sigma_2 \\ {\scriptstyle p_1}\downarrow & \lrcorner & \downarrow{\scriptstyle \varepsilon_2} \\ \sigma_1 & \xrightarrow[\varepsilon_1]{} & \hat{\sigma} \end{array}$. If there are $a_1 : C \in [\sigma_1]$ and $a_2 : C \in [\sigma_2]$ s.t. $[\varepsilon_1](a_1) = [\varepsilon_2](a_2)$, we can form a new $\sigma' = \lfloor \sigma \rfloor \to [\sigma] \oplus a : C$ that maps a to a_1 and a_2 and notice that the data of p_1 and p_2 will be the same. As a result, the new square with root σ' will commute and thus it must be that $a : C$ is already in σ. □

1.4.1 Reachability

A concept that becomes relevant when we introduce instantiation data with location variables is the notion of reachable and unreachable locations. We talk about (un)reachable concrete locations and (un)reachable location variables, where reachability refers to whether we can reach the location by traversing the heap starting from a given concrete location.

Definition 1.4.10 (Reachable Locations)**.** The reachable locations from $w \xrightarrow{\tau} [\sigma]$ w.r.t. a template heap $\eta : \mathbb{TH}(\sigma)$ is the least $w^{\mathsf{r}} \xrightarrow{\tau^{\mathsf{ur}}} [\sigma]$ such that

(i) World w factors through w^{r}

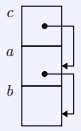
$$\begin{array}{ccc} w & \xrightarrow{\tau} & [\sigma] \\ & \searrow^{\exists \tau^{\mathsf{r}}} & \uparrow{\scriptstyle \tau^{\mathsf{ur}}} \\ & & w^{\mathsf{r}}k \end{array}$$

(ii) For every $b : C \in w^{\mathsf{r}} \otimes_{[\sigma]} \lfloor \sigma \rfloor$ i.e., every reachable concrete location, the locations in cell $d = \pi_{\lfloor \sigma \rfloor_* \tau^{\mathsf{ur}}(b)} \eta$ are contained in w^{r}

$$\begin{array}{ccc} 1 & \xrightarrow{d} & [\![\phi(C)]\!][\sigma] \\ & \searrow^{\exists d^{\mathsf{r}}} & \uparrow{\scriptstyle [\![\phi(C)]\!]\tau^{\mathsf{ur}}} \\ & & [\![\phi(C)]\!]w^{\mathsf{r}} \end{array}$$

Least here means that for any other compatible $\tau' : w' \to [\sigma]$, there exists a unique $h : w^{\mathsf{r}} \to w'$ that satisfies $\tau' \circ h = \tau^{\mathsf{ur}}$.
We refer to w^{r} as the reachable locations and $[\sigma] \ominus \tau^{\mathsf{ur}}$ as the unreachable ones.

Example 1.4.11. Consider $\tau = i^{a \hookrightarrow a,b,c}$ and template heap over $1_{a,b,c}$ being:



The least τ^{ur} is $i^{a,b \hookrightarrow a,b,c}$ since c is not reachable from a. Notice that $i^{a \hookrightarrow a,b,c}$ is not the least as it does not satisfy condition (ii) for a.

Reachability in terms of Heap Walks We give an alternative, algorithmic description of reachability (Definition 1.4.10) in terms of *heap walks* which describe finite, sequential traversals of the heap.

Definition 1.4.12 (Heap Walks)**.** For a heap $\eta \in \mathbb{TH}(\sigma)$, a sequence

$$a_0 : C_0, \ldots, a_n : C_n, c : C \in \lfloor \sigma \rfloor \times \lfloor \sigma \rfloor^* \times [\sigma]$$

is called a *heap walk* if

$$\forall i < n.\, \lfloor \sigma \rfloor (a_{i+1}) \mathsf{\ is\ in\ } \pi_{a_i} \eta \wedge c \mathsf{\ is\ in\ } \pi_{a_n} \eta$$

and it contains *no duplicate locations*

$$\forall i, j \leq n.\, i \neq j \implies a_i \neq a_j \wedge \forall i \leq n.\, \lfloor \sigma \rfloor (a_i) \neq c$$

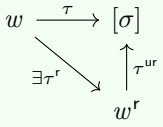
We informally use the proposition (a is in d) to state that location a is accessed inside cell d. For brevity, we avoid defining this directly but the intuition should be clear.

Example 1.4.13. Consider the heap from Example 1.4.11. Its heap walks are (i) c, a, b, (ii) c, a, (iii) a, b

Definition 1.4.14 (Reachable Location via a Heap Walk)**.** A location $c : C \in [\sigma]$ is said to be reachable from $a : C' \in [\sigma]$ in heap $\eta : \mathbb{TH}(\sigma)$ if $a = c$ or there exists a heap walk $a' : C', \overline{b}, c : C$ via η with $\lfloor\sigma\rfloor(a') = a$.

Definition 1.4.15 (Reachable Locations via Heap Walks)**.** The reachable locations from $w \xrightarrow{\tau} [\sigma]$ w.r.t. a template heap $\eta : \mathbb{TH}(\sigma)$ is a $w^{\mathsf{r}} \xrightarrow{\tau^{\mathsf{ur}}} [\sigma]$ that satisfies

(i) World w factors through w^{r}



(ii) For every $c : C \in [\sigma]$, $c \in \mathsf{Img}(\tau^{\mathsf{ur}}) \iff c$ is reachable from a $\tau(a)$

Lemma 1.4.16 (Reachable Locations Property Equivalence)**.** The two definitions of reachability, Definition 1.4.10 and Definition 1.4.15, are equivalent.

Proof. Property (i) of both definitions is identical, so we only show that property (ii) in the two definitions is equivalent.
Definition 1.4.10 $\implies$ Definition 1.4.15)
Direction $\Rightarrow$) Let $c : C \in w^{\mathsf{r}}$. If $\tau^{\mathsf{ur}}(c) \in \mathsf{Img}(\tau)$ we are done as then c is reachable from itself. If $\tau^{\mathsf{ur}}(c) \notin \mathsf{Img}(\tau)$, we enumerate all heap walks $a_0 : C_0, \ldots, a_n : C_n, \tau^{\mathsf{ur}}(c) : C$ of at most $|w^{\mathsf{r}}|$ length, that further satisfy $\forall i.\, \sigma(a_i) \in \mathsf{Img}(\tau^{\mathsf{ur}})$. There must exist at least one such heap walk as then $w^{\mathsf{r}} \setminus \{c\}$ would be the least world. If any of these heap walks has $\sigma(a_0) \in \mathsf{Img}(\tau)$ we are done. Otherwise collect all a_i in all the enumerated heap walks and remove them from w^{r}, call this $\hat{w}^{\mathsf{r}}$. We show that $\hat{w}^{\mathsf{r}}$ would be the least world which would be a contradiction. Take a $b \in \hat{w}^{\mathsf{r}} \otimes_{[\sigma]} \lfloor\sigma\rfloor$, and consider the locations in d_b. If d_b contains a location b' which is not in $\hat{w}^{\mathsf{r}}$ it would mean that b' is part of a heap walk generated above. Since the heap walks do not contain duplicates it must be less than $|w^{\mathsf{r}}|$ and thus we can extend that heap walk with b. So that would be a contradiction that $b \in \hat{w}^{\mathsf{r}}$. And so $\hat{w}^{\mathsf{r}}$ is the least world which is itself a contradiction and so there must exist at least one heap walk with $\lfloor\sigma\rfloor(a_0) \in \mathsf{Img}(\tau)$.
Direction $\Leftarrow$) We have that $c : C$ is reachable from some $\tau(a)$. If $a = c$ we are done. Otherwise we have a heap walk $a_0 : C_0, \ldots, a_n : C_n, c : C$ with $\lfloor\sigma\rfloor(a_0) \in \mathsf{Img}(\tau)$. Proceed by induction on n. If $n = 0$, then $\lfloor\sigma\rfloor(a_0) \in \mathsf{Img}(\tau)$, and so it is also in $\mathsf{Img}(\tau^{\mathsf{ur}})$. If $a_n \in w^{\mathsf{r}}$ and $a_{n+1} \in d_{a_n}$, then a_{n+1} must also be in w^{r} by the fact that d_{a_n} factors through w^{r}. So every a_i is in w^{r}, and thus c must also be in w^{r}.
Definition 1.4.15 $\implies$ Definition 1.4.10) Consider a $a : C \in w^{\mathsf{r}} \otimes_{[\sigma]} \lfloor\sigma\rfloor$, and $d_a \in [\![\phi(C)]\!][\sigma]$. Since a is in w^{r}, a is reachable from a $\tau(a')$ by assumption, and any location in d_a will also be reachable by extending the heap walk with that new location. As a result, by the bi-implication, the new location will also be in w^{r}. □

Lemma 1.4.17 (Decidability of Reachability)**.** There exists an algorithm that computes $w^{\mathsf{r}} \xrightarrow{\tau^{\mathsf{ur}}} [\sigma]$ for every $w \xrightarrow{\tau} [\sigma]$.

Proof. Intuitively, the algorithm will traverse the heap from $w \otimes_{[\sigma]} \lfloor\sigma\rfloor$ and return all the locations found including $\mathsf{Img}(\tau)$. □

1.4.2 Connectivity

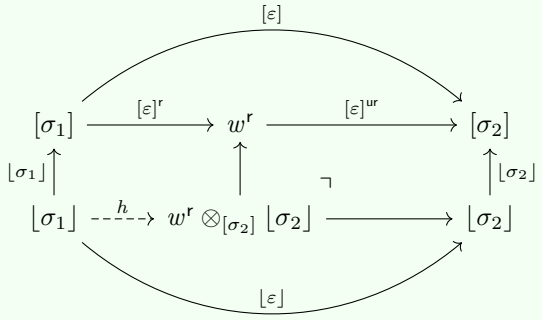
We extend the notion of reachability to morphisms in $\mathbb{TI}$ to characterise morphisms that satisfy (i) all cells in the instantiation data are reachable from locations in its domain, or (ii) none of the cells in the instantiation data are reachable from locations in its domain. We appropriately call morphisms that satisfy (i) as connected, and (ii) as disconnected.

We first give a factorisation of morphisms in $\mathbb{TI}$, by factoring out the reachable and unreachable components of a morphism.

Definition 1.4.18 (Factorisation in $\mathbb{TI}$)**.** For a morphism $\varepsilon : \sigma_1 \to \sigma_2$, we consider the reachable locations from $[\varepsilon]$ w.r.t. $\eta_\varepsilon : \mathbb{TH}(\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor \xrightarrow{\lfloor\varepsilon\rfloor^{\mathsf{c}}} \lfloor\sigma_2\rfloor \xrightarrow{\lfloor\sigma_2\rfloor} [\sigma_2])$. By Lemma 1.4.17 we get

$$[\sigma_1] \xrightarrow{[\varepsilon]^{\mathsf{r}}} w^{\mathsf{r}} \xrightarrow{[\varepsilon]^{\mathsf{ur}}} [\sigma_2]$$

We extend $[\varepsilon]^{\mathsf{r}}$ and $[\varepsilon]^{\mathsf{ur}}$ into morphisms in $\mathbb{TI}$ by taking the *largest possible choice* for the concrete locations, that is, we consider the pullback



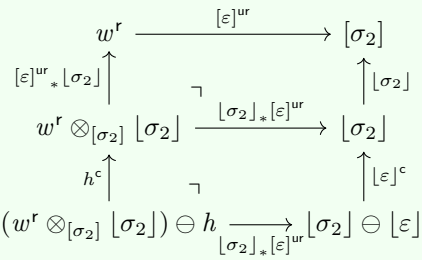
where the instantiation data are given by the canonical isomorphism

$$\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor \cong (\lfloor\sigma_2\rfloor \ominus \lfloor\sigma_2\rfloor_*[\varepsilon]^{\mathsf{ur}}) \oplus (w^{\mathsf{r}} \otimes_{[\sigma_2]} \lfloor\sigma_2\rfloor \ominus h)$$

We show that locations $(w^{\mathsf{r}} \otimes_{[\sigma_2]} \lfloor\sigma_2\rfloor) \ominus h$ have cells that access only w^{r} from

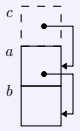
$$(w^{\mathsf{r}} \otimes_{[\sigma_2]} \lfloor\sigma_2\rfloor) \ominus h \cong w^{\mathsf{r}} \otimes_{[\sigma_2]} (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor)$$

which follows from pullback pasting



and so Definition 1.4.10 gives us the factoring of the cells through w^{r}.

Notation. For a morphism $\varepsilon : \sigma_1 \to \sigma_2$, we write $\mathsf{c}(\varepsilon) : \sigma_1 \to \sigma_{\mathsf{c}(\varepsilon)}$ and $\mathsf{uc}(\varepsilon) : \sigma_{\mathsf{c}(\varepsilon)} \to \sigma_2$ for its factorisation given by Definition 1.4.18. We also write $\sigma_{\mathsf{uc}(\varepsilon)} := \sigma_2 \ominus \mathsf{uc}(\varepsilon)$.

Example 1.4.19. Consider the morphism $?_a \xrightarrow{\langle ?_{\{a,b\}}, i^{a \hookrightarrow a,b,c} \rangle} i^{a,b \hookrightarrow a,b,c}$ with data 

Its reachable/unreachable factoring given by Definition 1.4.18 is

$$?_a \xrightarrow{\langle ?_{\{a,b\}}, i^{a \hookrightarrow a,b} \rangle} 1_{a,b} \xrightarrow{\langle i^{a,b \hookrightarrow a,b,c}, i^{a,b \hookrightarrow a,b,c} \rangle} i^{a,b \hookrightarrow a,b,c}$$

with data 

for the reachable morphism.

Definition 1.4.20 (Connected morphisms)**.** A morphism $\varepsilon : \sigma_1 \to \sigma_2$ is said to be connected if $\mathsf{uc}(\varepsilon)$ is an isomorphism.

Definition 1.4.21 (Disconnected morphisms)**.** A morphism $\varepsilon : \sigma_1 \to \sigma_2$ is said to be disconnected if $\mathsf{c}(\varepsilon)$ is an isomorphism.

Lemma 1.4.22 (Factoring Connectedness)**.** For a morphism $\varepsilon : \sigma_1 \to \sigma_2$, its $\mathsf{c}(\varepsilon)$ part is connected.

Proof. We have the diagram
$$\begin{array}{ccccc} \sigma_1 & & \xrightarrow{\mathsf{c}(\varepsilon)} & & \sigma_{\mathsf{c}(\varepsilon)} \xrightarrow{\mathsf{uc}(\varepsilon)} \sigma_2 \\ & \searrow^{\mathsf{c}(\mathsf{c}(\varepsilon))} & & \nearrow_{\mathsf{uc}(\mathsf{c}(\varepsilon))} & \\ & & \sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))} & & \end{array}$$
and we want to show that $\mathsf{uc}(\mathsf{c}(\varepsilon))$ is iso.
It is enough to show that $[\mathsf{uc}(\mathsf{c}(\varepsilon))]$ is iso, as $\lfloor \mathsf{uc}(\mathsf{c}(\varepsilon)) \rfloor$ is given by the pullback

$$\begin{array}{ccc} [\sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))}] & \xrightarrow{[\mathsf{uc}(\mathsf{c}(\varepsilon))]} & [\sigma_{\mathsf{c}(\varepsilon)}] \\ \uparrow & \lrcorner & \uparrow \lfloor \sigma_{\mathsf{c}(\varepsilon)} \rfloor \\ \lfloor \sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))} \rfloor & \xrightarrow[\lfloor \mathsf{uc}(\mathsf{c}(\varepsilon)) \rfloor]{} & \lfloor \sigma_{\mathsf{c}(\varepsilon)} \rfloor \end{array}$$

and so if $[\mathsf{uc}(\mathsf{c}(\varepsilon))]$ is iso, so is $\lfloor \mathsf{uc}(\mathsf{c}(\varepsilon)) \rfloor$.
We proceed by showing that

$$[\sigma_1] \xrightarrow{[\mathsf{c}(\mathsf{c}(\varepsilon))]} [\sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))}] (\xrightarrow{[\mathsf{uc}(\mathsf{c}(\varepsilon))]} [\sigma_{\mathsf{c}(\varepsilon)}] \xrightarrow{[\mathsf{uc}(\varepsilon)]} [\sigma_2])$$

satisfies Definition 1.4.10 for $[\sigma_1] \xrightarrow{[\varepsilon]} [\sigma_2]$ and η_ε. As a result, by the least property of Definition 1.4.10 for $[\mathsf{c}(\varepsilon)]$, we get that $[\mathsf{uc}(\mathsf{c}(\varepsilon))]$ must be iso.
Property (i) is immediate, and so consider property (ii). Consider $a : C \in (\lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon \rfloor) \otimes_{[\sigma_2]} [\sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))}]$, where the pullback is given by

$$\begin{array}{ccccc} [\sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))}] & \xrightarrow{[\mathsf{uc}(\mathsf{c}(\varepsilon))]} & [\sigma_{\mathsf{c}(\varepsilon)}] & \xrightarrow{[\mathsf{uc}(\varepsilon)]} & [\sigma_2] \\ \uparrow & \lrcorner & \uparrow & \lrcorner & \uparrow \lfloor \sigma_2 \rfloor \\ \lfloor \sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))} \rfloor & \xrightarrow[\lfloor \mathsf{uc}(\mathsf{c}(\varepsilon)) \rfloor]{} & \lfloor \sigma_{\mathsf{c}(\varepsilon)} \rfloor & \xrightarrow[\lfloor \mathsf{uc}(\varepsilon) \rfloor]{} & \lfloor \sigma_2 \rfloor \\ \lfloor \mathsf{c}(\mathsf{c}(\varepsilon)) \rfloor^{\mathsf{c}} \uparrow & \lrcorner & \lfloor \mathsf{c}(\varepsilon) \rfloor^{\mathsf{c}} \uparrow & \lrcorner & \lfloor \varepsilon \rfloor^{\mathsf{c}} \uparrow \\ \lfloor \sigma_{\mathsf{c}(\mathsf{c}(\varepsilon))} \rfloor \ominus \lfloor \mathsf{c}(\mathsf{c}(\varepsilon)) \rfloor & \xrightarrow{\lfloor \mathsf{uc}(\mathsf{c}(\varepsilon)) \rfloor} & \lfloor \sigma_{\mathsf{c}(\varepsilon)} \rfloor \ominus \lfloor \mathsf{c}(\varepsilon) \rfloor & \xrightarrow{\lfloor \mathsf{uc}(\varepsilon) \rfloor} & \lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon \rfloor \end{array}$$

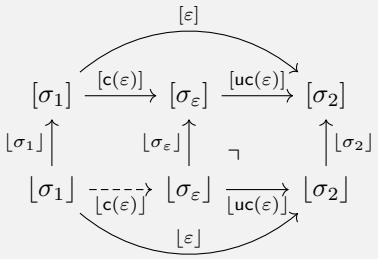
which we use to prove that cell a has locations in $[\sigma_{\mathsf{c}(\mathsf{c}())\varepsilon}]$

$$\begin{aligned}\pi_{\lfloor \mathsf{uc}(\varepsilon)\rfloor \lfloor \mathsf{uc}(\mathsf{c}(\varepsilon))\rfloor (a)}\eta_\varepsilon &= [\![\phi(C)]\!][\mathsf{uc}(\varepsilon)](\pi_{\lfloor \mathsf{uc}(\mathsf{c}(\varepsilon))\rfloor (a)}\eta_{\mathsf{c}(\varepsilon)}) \\ &= [\![\phi(C)]\!][\mathsf{uc}(\varepsilon)]([\![\phi(C)]\!][\mathsf{uc}(\mathsf{c}(\varepsilon))](\pi_a \eta_{\mathsf{c}(\mathsf{c}(\varepsilon))}))\end{aligned}$$

□

Lemma 1.4.23 (Factoring and Variable Preservation)**.** For a morphism $\varepsilon : \sigma_1 \to \sigma_2$, its $\mathsf{uc}(\varepsilon)$ part is variable preserving.

Proof. Morphism $\mathsf{uc}(\varepsilon)$ is given by the following diagram



which by Lemma 1.3.21 since $\mathsf{uc}(\varepsilon)$ is a pullback it is also variable preserving. □

Lemma 1.4.24 (Epimorphisms in $\mathbb{TI}$)**.** A morphism $\varepsilon : \sigma_1 \to \sigma_2$ in $\mathbb{TI}$ is epi iff ε is connected.

Proof.
$\to$) From Definition 1.4.18 we can factor ε into $\sigma_1 \xrightarrow{\mathsf{c}(\varepsilon)} \sigma_\varepsilon \xrightarrow{\mathsf{uc}(\varepsilon)} \sigma_2$ and to show that ε is connected it is enough to show that $[\mathsf{uc}(\varepsilon)]$ is surjective.
By Lemma 1.4.23 we form morphisms $1 \oplus i_1$ and $1 \oplus i_2$ in $\mathbb{TI}$

$$\begin{array}{ccc} [\sigma_\varepsilon] \oplus ([\sigma_2] \ominus [\mathsf{uc}(\varepsilon)]) & \overset{1\oplus i_1}{\underset{1\oplus i_2}{\rightrightarrows}} & [\sigma_\varepsilon] \oplus ([\sigma_2] \ominus [\mathsf{uc}(\varepsilon)])^2 \\ \uparrow \lfloor\sigma_\varepsilon\rfloor \oplus \lfloor\sigma_2\rfloor|^{\mathsf{uc}(\varepsilon)} & & \uparrow \lfloor\sigma_\varepsilon\rfloor \oplus \lfloor\sigma_2\rfloor|^{\mathsf{uc}(\varepsilon)^2} \\ \lfloor\sigma_\varepsilon\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\mathsf{uc}(\varepsilon)\rfloor) & \overset{1\oplus i_1}{\underset{1\oplus i_2}{\rightrightarrows}} & \lfloor\sigma_\varepsilon\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\mathsf{uc}(\varepsilon)\rfloor)^2 \end{array}$$

$\sigma_2 \cong_{\mathsf{r}}$

with data $\eta_{1\oplus i_i} = \underline{\mathbb{H}}(\cong, 1 \oplus i_{i \bmod 2+1} \circ [\cong_{\mathsf{r}}])(\eta_{\mathsf{uc}(\varepsilon)})$. The data $\eta_{(1\oplus i_i)\circ \mathsf{uc}(\varepsilon)}$ results in two copies of $\eta_{\mathsf{uc}(\varepsilon)}$ and so $\eta_{(1\oplus i_1)\circ \mathsf{uc}(\varepsilon)} = \eta_{(1\oplus i_2)\circ \mathsf{uc}(\varepsilon)}$. As a result

$$\begin{aligned}(1 \oplus i_1) \circ \mathsf{uc}(\varepsilon) = (1 \oplus i_2) \circ \mathsf{uc}(\varepsilon) &\implies \\ (1 \oplus i_1) \circ \varepsilon = (1 \oplus i_2) \circ \varepsilon &\implies \\ 1 \oplus i_1 = 1 \oplus i_2 &\implies \\ [\mathsf{uc}(\varepsilon)] \text{ is surjective} &\implies \\ \mathsf{uc}(\varepsilon) \text{ is iso}&\end{aligned}$$

$\leftarrow$) Consider $\varepsilon_1 \circ \varepsilon = \varepsilon_2 \circ \varepsilon$. We show that $[\varepsilon_1] = [\varepsilon_2]$ which in turn implies $\varepsilon_1 = \varepsilon_2$.
Consider $a' : C \in [\sigma_2]$, which by connectedness gives us a $a : C \in [\sigma_\varepsilon]$ s.t. $[\mathsf{uc}(\varepsilon)](a) = a'$. If a is in the image of $[\mathsf{c}(\varepsilon)]$ we are done. Otherwise by Lemma 1.4.16 we have a heap walk $b_0 : C_0, \ldots, b_n : C_n, a : C$ along $\eta_{\sigma_\varepsilon}$ where $\lfloor\sigma_\varepsilon\rfloor(b_0) \in \mathsf{Img}([\mathsf{c}(\varepsilon)])$.

We proceed by induction on the length of the sequence to show that

$$\forall i \leq n. [\varepsilon_1]\lfloor\sigma_\varepsilon\rfloor(b_i) = [\varepsilon_2]\lfloor\sigma_\varepsilon\rfloor(b_i)$$

$n = 0$) Then b_0 is in the image of $[\mathsf{c}(\varepsilon)]$ and so $[\varepsilon_1]\lfloor\sigma_\varepsilon\rfloor(b_0) = [\varepsilon_2]\lfloor\sigma_\varepsilon\rfloor(b_0)$ by $\varepsilon \circ \varepsilon_1 = \varepsilon \circ \varepsilon_2$.
$n = n' + 1$) We have

$$\begin{aligned}
[\varepsilon_1]\lfloor\sigma_\varepsilon\rfloor(b_{n'}) = [\varepsilon_2]\lfloor\sigma_\varepsilon\rfloor(b_{n'}) = & && (\mathsf{IH})\\
\lfloor\sigma_2\rfloor\lfloor\varepsilon_1\rfloor(b_{n'}) = \lfloor\sigma_2\rfloor\lfloor\varepsilon_2\rfloor(b_{n'}) \implies & && (\text{by } \lfloor\sigma_2\rfloor \text{ is injective})\\
\lfloor\varepsilon_1\rfloor(b_{n'}) = \lfloor\varepsilon_2\rfloor(b_{n'}) \implies & && (\text{by } \eta_{\varepsilon_1\circ\varepsilon} = \eta_{\varepsilon_2\circ\varepsilon})\\
[\![\phi(C_{n'})]\!][\varepsilon_1](d_{n'}) = [\![\phi(C_{n'})]\!][\varepsilon_2](d_{n'}) \implies & && (\text{and since } b_n \,\mathsf{is\,in}\, d_{n'})\\
[\varepsilon_1]\lfloor\sigma_\varepsilon\rfloor(b_n) = [\varepsilon_2]\lfloor\sigma_\varepsilon\rfloor(b_n) & &&
\end{aligned}$$

Since a is in d_n and $[\varepsilon_1]\lfloor\sigma_\varepsilon\rfloor(b_n) = [\varepsilon_2]\lfloor\sigma_\varepsilon\rfloor(b_n)$, we have that $[\varepsilon_1](a) = [\varepsilon_2](a)$. □

Lemma 1.4.25 (Composition of Connected Morphisms)**.** Connected morphisms are closed under composition.

Proof. Follows immediately from Lemma 1.4.24 and epis being closed under composition. □

Lemma 1.4.26 (Regular Monos in $\mathbb{TI}$)**.** A morphism $\varepsilon : \sigma_1 \to \sigma_2$ in $\mathbb{TI}$ is regular mono iff ε is variable preserving.

Proof.
$\to$) If ε is part of an equalizer diagram, by Lemma 1.4.7, if we forget the data we still have an equalizer diagram in $\mathbb{T}$. By Lemma 1.3.25, $u\varepsilon$ will be variable preserving and since variable preservation is reflected by u, ε is variable preserving.
$\leftarrow$) We show that this is an equalizer diagram (Lemma 1.3.21 gives us $\lfloor\sigma_2\rfloor|^\varepsilon$, and $\cong^{\mathsf{c}}$ is the canonical isomorphism induced)

$$\begin{array}{ccc}
[\sigma_1] \oplus ([\sigma_2] \ominus [\varepsilon]) & \underset{1\oplus i_2}{\overset{1\oplus i_1}{\rightrightarrows}} & [\sigma_1] \oplus ([\sigma_2] \ominus [\varepsilon]) \oplus ([\sigma_2] \ominus [\varepsilon]) \\
\uparrow \lfloor\sigma_1\rfloor \oplus \lfloor\sigma_2\rfloor|^\varepsilon & & \uparrow \lfloor\sigma_1\rfloor \oplus \lfloor\sigma_2\rfloor|^\varepsilon \oplus \lfloor\sigma_2\rfloor|^\varepsilon \\
\lfloor\sigma_1\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor) & \underset{1\oplus i_2}{\overset{1\oplus i_1}{\rightrightarrows}} & \lfloor\sigma_1\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor) \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor)
\end{array}$$

$\sigma_2 \quad \cong^{\mathsf{c}}$

We equip $1 \oplus i_1$ with instantiation data $\underline{\mathbb{H}}(\cong^1, 1 \oplus i_2 \circ [\cong^{\mathsf{c}}])(\eta_\varepsilon)$ by noticing the isomorphism

$$(\lfloor\sigma_1\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor) \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor)) \ominus (1 \oplus i_1) \cong^1 \lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor$$

We do the same for $1 \oplus i_2$, i.e., we use $\underline{\mathbb{H}}(\cong^2, 1 \oplus i_1 \circ [\cong^{\mathsf{c}}])(\eta_\varepsilon)$.
To show that $1 \oplus i_1 \circ \cong^{\mathsf{c}} \circ\varepsilon = 1 \oplus i_2 \circ \cong^{\mathsf{c}} \circ\varepsilon$, we need to show that the data match. Notice that

$$(\lfloor\sigma_1\rfloor \oplus (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor)^2) \ominus (1 \oplus i_1 \circ \lfloor\varepsilon\rfloor) \cong (\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor)^2$$

and $\eta_{1\oplus i_1 \circ \cong^{\mathsf{c}}\varepsilon} = \eta_{1\oplus i_2 \circ \cong^{\mathsf{c}}\varepsilon}$ as they have two identical copies of $\underline{\mathbb{H}}(1, \cong^{\mathsf{c}})(\eta_\varepsilon)$.
Now consider $(1 \oplus i_1) \circ \cong^{\mathsf{c}} \circ\varepsilon' = (1 \oplus i_2) \circ \cong^{\mathsf{c}} \circ\varepsilon'$. Then $u\varepsilon'$ must factor through $u\varepsilon$ and $\eta_{\varepsilon'}$ can be split between the factoring through the canonical isomorphism. Uniqueness of this factoring follows from ε being mono. □

Lemma 1.4.27 (Regular Monos are closed under composition in $\mathbb{TI}$)**.** Regular monos in $\mathbb{TI}$ are closed under composition.

Proof. Follows from Lemma 1.4.26 and Lemma 1.4.28, u preserving and reflecting variable preserving morphisms and Lemma 1.3.22. □

Lemma 1.4.28 (Variable Preservation and Disconnectedness)**.** A morphism $\varepsilon : \sigma_1 \to \sigma_2$ is variable preserving iff it is disconnected.

Proof. The $\leftarrow$ direction follows from Lemma 1.4.23 and since $\mathsf{c}(\varepsilon)$ is iso it is also variable preserving, and variable preservation is closed under composition. For the $\to$ direction, by Lemma 1.4.23 and Lemma 1.3.23 we have that $\mathsf{c}(\varepsilon)$ is variable preserving. Since $\mathsf{c}(\varepsilon)$ is epi and regular mono it is also iso. □

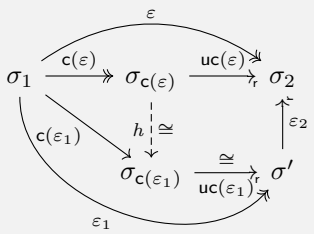
Lemma 1.4.29 (Factoring Disconnectedness)**.** For a morphism $\varepsilon : \sigma_1 \to \sigma_2$, its $\mathsf{uc}(\varepsilon)$ part is disconnected.

Proof. By Lemma 1.4.23, $\mathsf{uc}(\varepsilon)$ is variable preserving which by Lemma 1.4.28 means it disconnected. □

1.4.3 Factorisation System

Lemma 1.4.30 (Factorisation is Unique)**.** The connectivity (epi-regular mono) factorisation in $\mathbb{TI}$ is unique up to unique isomorphism.

Proof. We show that the $\sigma_1 \xrightarrow{\mathsf{c}(\varepsilon)} \sigma_{\mathsf{c}(\varepsilon)} \xrightarrow{\mathsf{uc}(\varepsilon)} \sigma_2$ factorisation is unique up to unique iso, so consider



and we show that there exists a unique iso $h : \sigma_{\mathsf{c}(\varepsilon)} \to \sigma_{\mathsf{c}(\varepsilon_1)}$ that commutes in the diagram above. We first show that

$$[\sigma_1] \xrightarrow{[\mathsf{c}(\varepsilon_1)]} [\sigma_{\mathsf{c}(\varepsilon_1)}](\xrightarrow{\mathsf{uc}(\varepsilon_1)} \sigma' \xrightarrow{[\varepsilon_2]} \sigma_2)$$

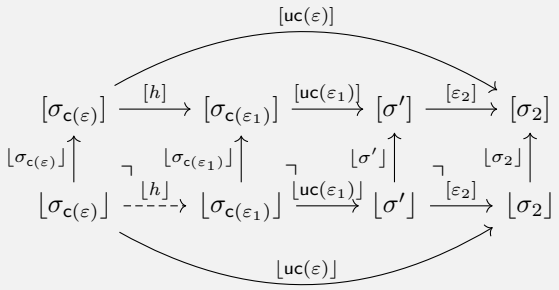
satisfies Definition 1.4.10 for $[\sigma_1] \xrightarrow{[\varepsilon]} [\sigma_2]$ and η_ε.
Property (i) is obvious, and for property (ii) consider $(\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor) \otimes_{[\sigma_2]} [\sigma_{\mathsf{c}(\varepsilon_1)}]$, given by

$$\begin{array}{ccccc}
[\sigma_{\mathsf{c}(\varepsilon_1)}] & \xrightarrow{[\mathsf{uc}(\varepsilon_1)]} & [\sigma'] & \xrightarrow{[\varepsilon_2]} & [\sigma_2] \\
\uparrow{\scriptstyle \lfloor\sigma_{\mathsf{c}(\varepsilon_1)}\rfloor} & \urcorner & \uparrow{\scriptstyle \lfloor\sigma'\rfloor} & \urcorner & \uparrow{\scriptstyle \lfloor\sigma_2\rfloor} \\
\lfloor\sigma_{\mathsf{c}(\varepsilon_1)}\rfloor & \xrightarrow{\lfloor\mathsf{uc}(\varepsilon_1)\rfloor} & \lfloor\sigma'\rfloor & \xrightarrow{\lfloor\varepsilon_2\rfloor} & \lfloor\sigma_2\rfloor \\
\uparrow{\scriptstyle \lfloor\mathsf{c}(\varepsilon_1)\rfloor^{\mathsf{c}}} & \urcorner & \uparrow{\scriptstyle \lfloor\varepsilon_1\rfloor^{\mathsf{c}}} & \urcorner & \uparrow{\scriptstyle \lfloor\varepsilon\rfloor^{\mathsf{c}}} \\
\lfloor\sigma_{\mathsf{c}(\varepsilon_1)}\rfloor \ominus \lfloor\mathsf{c}(\varepsilon_1)\rfloor & \xrightarrow{\lfloor\mathsf{uc}(\varepsilon_1)\rfloor} & \lfloor\sigma'\rfloor \ominus \lfloor\varepsilon_1\rfloor & \xrightarrow{\lfloor\varepsilon_2\rfloor} & \lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon\rfloor
\end{array}$$

and we show that locations in every $\pi_{\lfloor\varepsilon_2\rfloor\lfloor\mathsf{uc}(\varepsilon_1)\rfloor(a)}\eta_\varepsilon$ factor through $[\sigma_{\mathsf{c}(\varepsilon_1)}]$

$$\begin{aligned}
\pi_{\lfloor\varepsilon_2\rfloor\lfloor\mathsf{uc}(\varepsilon_1)\rfloor(a)}\eta_\varepsilon &= \pi_{\lfloor\varepsilon_2\rfloor\lfloor\mathsf{uc}(\varepsilon_1)\rfloor(a)}\eta_{\varepsilon_2\circ\varepsilon_1} \\
&= [\![\phi(C)]\!][\varepsilon_2](\pi_{\lfloor\mathsf{uc}(\varepsilon_1)\rfloor(a)}\eta_{\varepsilon_1}) \\
&= [\![\phi(C)]\!][\varepsilon_2 \circ \mathsf{uc}(\varepsilon_1)](\pi_a \eta_{\mathsf{c}(\varepsilon_1)})
\end{aligned}$$

By the UP of reachable locations we have $[h] : [\sigma_{\mathsf{c}(\varepsilon)}] \to [\sigma_{\mathsf{c}(\varepsilon_1)}]$, which we extend to $\mathbb{TI}$ by



Notice that since the outer square and the right (combined) square are pullbacks, so is the left square, and $\mathsf{c}(\varepsilon_1) = h \circ \mathsf{c}(\varepsilon)$ and $\mathsf{uc}(\varepsilon) = \varepsilon_2 \circ \mathsf{uc}(\varepsilon_1) \circ h$.
We further show that

$$[\sigma_1] \xrightarrow{[\mathsf{c}(\varepsilon)]} [\sigma_{\mathsf{c}(\varepsilon)}](\xrightarrow{[h]} [\sigma_{\mathsf{c}(\varepsilon_1)}] \xrightarrow{[\mathsf{uc}(\varepsilon_2)]} [\sigma'])$$

satisfies Definition 1.4.10 for $[\sigma_1] \xrightarrow{[\varepsilon_1]} [\sigma']$ and η_{ε_1}.
Property (i) is obvious, and for property (ii) consider $(\lfloor\sigma'\rfloor \ominus \lfloor\varepsilon_1\rfloor) \otimes_{[\sigma']} [\sigma_{\mathsf{c}(\varepsilon)}]$, given by

$$\begin{array}{ccccc}
[\sigma_{\mathsf{c}(\varepsilon)}] & \xrightarrow{[h]} & [\sigma_{\mathsf{c}(\varepsilon_1)}] & \xrightarrow{[\mathsf{uc}(\varepsilon_2)]} & [\sigma'] \\
\uparrow{\scriptstyle \lfloor\sigma_{\mathsf{c}(\varepsilon)}\rfloor} & \urcorner & \uparrow{\scriptstyle \lfloor\sigma_{\mathsf{c}(\varepsilon_1)}\rfloor} & \urcorner & \uparrow{\scriptstyle \lfloor\sigma'\rfloor} \\
\lfloor\sigma_{\mathsf{c}(\varepsilon)}\rfloor & \xrightarrow{\lfloor h\rfloor} & [\sigma_{\mathsf{c}(\varepsilon_1)}] & \xrightarrow{\lfloor\mathsf{uc}(\varepsilon_2)\rfloor} & \lfloor\sigma'\rfloor \\
\uparrow{\scriptstyle \lfloor\mathsf{c}(\varepsilon)\rfloor^{\mathsf{c}}} & \urcorner & \uparrow{\scriptstyle \lfloor\mathsf{c}(\varepsilon_1)\rfloor^{\mathsf{c}}} & \urcorner & \uparrow{\scriptstyle \lfloor\varepsilon_1\rfloor^{\mathsf{c}}} \\
\lfloor\sigma_{\mathsf{c}(\varepsilon)}\rfloor \ominus \lfloor\mathsf{c}(\varepsilon)\rfloor & \xrightarrow{\lfloor h\rfloor} & \lfloor\sigma_{\mathsf{c}(\varepsilon_1)}\rfloor \ominus \lfloor\mathsf{c}(\varepsilon_1)\rfloor & \xrightarrow{\lfloor\mathsf{uc}(\varepsilon_2)\rfloor} & \lfloor\sigma'\rfloor \ominus \lfloor\varepsilon_1\rfloor
\end{array}$$

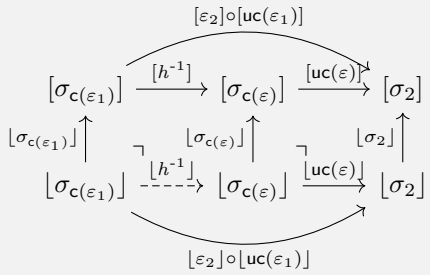
and we show that locations in every $\pi_{\lfloor\mathsf{uc}(\varepsilon_1)\rfloor\lfloor h\rfloor(a)}\eta_{\varepsilon_1}$ factor through $[\sigma_{\mathsf{c}(\varepsilon)}]$

$$\begin{aligned}
\pi_{\lfloor\mathsf{uc}(\varepsilon_1)\rfloor\lfloor h\rfloor(a)}\eta_{\varepsilon_1} &= \pi_{\lfloor\mathsf{uc}(\varepsilon_1)\rfloor\lfloor h\rfloor(a)}\eta_{\mathsf{uc}(\varepsilon_1)\circ\mathsf{c}(\varepsilon_1)} \\
&= [\![\phi(C)]\!][\mathsf{uc}(\varepsilon_1)](\pi_{\lfloor h\rfloor(a)}\eta_{\mathsf{c}(\varepsilon_1)}) && \text{(def of } \eta_{\mathsf{uc}(\varepsilon_1)\circ\mathsf{c}(\varepsilon_1)}) \\
&= [\![\phi(C)]\!][\mathsf{uc}(\varepsilon_1) \circ [h]](\pi_a \eta_{\mathsf{c}(\varepsilon)}) && \text{(UP of } [\sigma_{\mathsf{c}(\varepsilon)}])
\end{aligned}$$

As a result, we get a $[h^{-1}] : [\sigma_{\mathsf{c}(\varepsilon_1)}] \to [\sigma_{\mathsf{c}(\varepsilon)}]$ which is the inverse of $[h]$ by

$$\begin{aligned}
[h] \circ [h^{-1}] = 1 &\iff \\
[\mathsf{uc}(\varepsilon)_1] \circ [h] \circ [h^{-1}] = [\mathsf{uc}(\varepsilon_1)] &\iff \\
[\varepsilon_2] \circ [\mathsf{uc}(\varepsilon)_1] \circ [h] \circ [h^{-1}] = [\varepsilon_2] \circ [\mathsf{uc}(\varepsilon_1)] &\iff \\
[\mathsf{uc}(\varepsilon)] \circ [h^{-1}] = [\varepsilon_2] \circ [\mathsf{uc}(\varepsilon_1)]
\end{aligned}$$

and $[h] \circ [h^{-1}] \circ [h] = [h] \iff [h^{-1}] \circ [h] = 1$.
We extend $[h^{-1}]$ into a morphism in $\mathbb{T}$ by



where the left square is a pullback because the outer and right square are pullbacks, and regular monos compose (Lemma 1.4.27). As a result, $\lfloor h^{-1}\rfloor$ is also iso.
Uniqueness of iso h follows from uniqueness of the underlying components. □

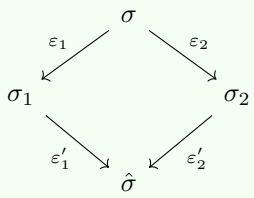
Lemma 1.4.31 (Orthogonal Factorisation System in $\mathbb{TI}$)**.** The classes of epis (connected) and regular monos (disconnected) morphisms gives rise to an orthogonal factorisation system in $\mathbb{TI}$.

Proof. Follows from Lemma 1.4.30, epis being closed under composition and including all isomorphisms, and regular monos in $\mathbb{TI}$ being closed under composition (Lemma 1.4.27) and include all isomorphisms. □

1.4.4 Unification

A natural question to ask when dealing with two template heaps that have a common set of public locations is whether they unify, that is, if there exist instantiations that we can apply to both heaps respectively such that they result in the same heap. Through the isomorphism $\mathbb{TH} \cong \mathbb{TI}(?_\emptyset, -)$, we simply phrase this question as whether a span $\sigma_1 \xleftarrow{\varepsilon_1} \sigma \xrightarrow{\varepsilon_2} \sigma_2$ has a cospan $\sigma_1 \xrightarrow{\varepsilon_1'} \hat{\sigma} \xleftarrow{\varepsilon_2'} \sigma_2$ such that the obvious square commutes.

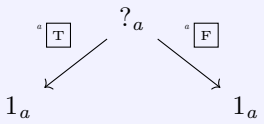
Definition 1.4.32 (Unifier of Morphisms in $\mathbb{TI}$)**.** We say that two morphisms $\sigma_1 \xleftarrow{\varepsilon_1} \sigma \xrightarrow{\varepsilon_2} \sigma_2$ unify if there exists a diagram



We call $\varepsilon_1', \varepsilon_2'$ a unifier of $\varepsilon_1, \varepsilon_2$.

Notice that some spans do not have any compatible cospan, that is, some pair of heaps do not unify as the following example shows.

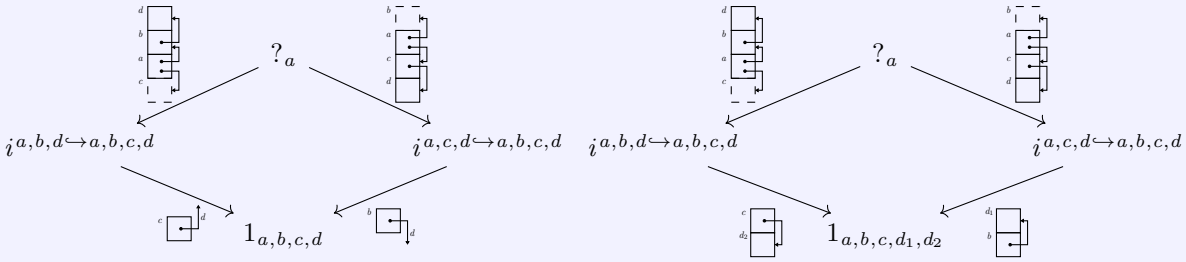
Example 1.4.33. Consider the following diagram where both morphisms instantiate the variable a but with different values (true (T) and false (F)).

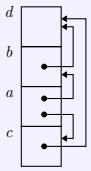
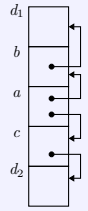


There cannot be any unifiers that would make the diagram commute as the resulting heaps will always be different.

General Unifiers We seek to identify the most general unifers of two morphisms $\sigma_1 \xleftarrow{\varepsilon_1} \sigma \xrightarrow{\varepsilon_2} \sigma_2$, that is, the least cospan $\sigma_1 \xrightarrow{\varepsilon_1'} \hat{\sigma} \xleftarrow{\varepsilon_2'} \sigma_2$ such that every other cospan factors uniquely thorough it. Unfortunately such M.G.U. does not always exist as the following counterexample shows.

Example 1.4.34. Consider the following two unifers



The unifier on the left results in the heap  whereas the one on the right results in 

which are mutually incomparable.

Even though most general unifiers do not always exist, we do have that every unifier can be factored though "special" unifiers that are themselves mutually incomparable. We call these special unifiers *general unifiers*.

Definition 1.4.35 (General Unifiers)**.** We call a unifier $\sigma_1 \xrightarrow{\varepsilon_1'} \hat{\sigma} \xleftarrow{\varepsilon_2'} \sigma_2$ general if $\lfloor\sigma_1\rfloor \xrightarrow{\lfloor\varepsilon_1'\rfloor} \lfloor\hat{\sigma}\rfloor \xleftarrow{\lfloor\varepsilon_2'\rfloor} \lfloor\sigma_2\rfloor$ are jointly surjective and $[\sigma_1] \xrightarrow{[\varepsilon_1']} [\hat{\sigma}] \xleftarrow{[\varepsilon_2']} [\sigma_2]$ are jointly surjective.

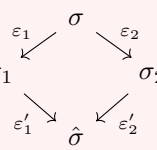
Remark 1.4.36. By Lemma 1.3.11, this is equivalent to saying that a unifier is general if it is generated by a jointly surjective cospan $[\sigma_1] \xrightarrow{[\varepsilon_1']} [\hat{\sigma}] \xleftarrow{[\varepsilon_2']} [\sigma_2]$.

Example 1.4.37. Both unifiers from Example 1.4.34 are general unifiers. The unifier on the left is generated by

$$a, b, c, d \xrightarrow{1} a, b, c, d \xleftarrow{1} a, b, c, d$$

whereas the one on the right by

$$a, b, c \oplus d \xrightarrow{1\oplus i_1} a, b, c, d_1, d_2 \xleftarrow{1\oplus i_2} a, b, c \oplus d$$

Lemma 1.4.38 (Data of General Unifiers)**.** If  is a general unifier then $\eta_{\varepsilon_1'}$ and $\eta_{\varepsilon_2'}$ are completely determined by η_{ε_1} and η_{ε_2}.

Proof. W.l.g. we show that $\eta_{\varepsilon_2'}$ is completely determined by η_{ε_1} and the other direction is symmetric. Since $\lfloor\varepsilon_1'\rfloor$ and $\lfloor\varepsilon_2'\rfloor$ are jointly surjective, every $a : C \in \lfloor\hat{\sigma}\rfloor \ominus \lfloor\varepsilon_2'\rfloor$ satisfies that there exists a unique $a' : C \in \lfloor\sigma_1\rfloor \ominus \lfloor\varepsilon_1\rfloor$ s.t. $\lfloor\varepsilon_1'\rfloor(a') = a$.

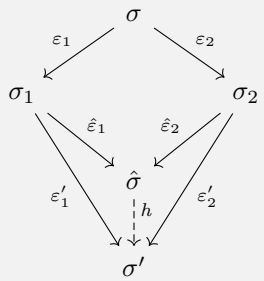
This implies by definition of $\eta_{\varepsilon_2' \circ \varepsilon_2}$ and $\eta_{\varepsilon_2' \circ \varepsilon_2} = \eta_{\varepsilon_1' \circ \varepsilon_1}$ that

$$\pi_a \eta_{\varepsilon_2'} = \pi_a \eta_{\varepsilon_2' \circ \varepsilon_2} = \pi_{\lfloor \varepsilon_1' \rfloor (a')} \eta_{\varepsilon_1' \circ \varepsilon_1} = [\![\phi(C)]\!][\varepsilon_1'](\pi_{a'} \eta_{\varepsilon_1})$$

□

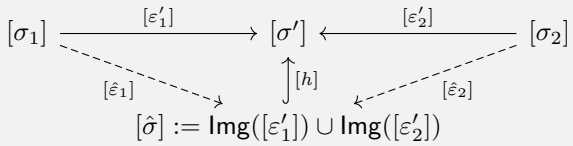
Lemma 1.4.39 (General Unifier Property)**.** Every unifier of $\sigma_1 \xleftarrow{\varepsilon_1} \sigma \xrightarrow{\varepsilon_2} \sigma_2$ factors uniquely (up to iso) through a general unifier.

Proof. For a unifier $\sigma_1 \xrightarrow{\varepsilon_1'} \sigma' \xleftarrow{\varepsilon_2'} \sigma_2$ we show the following diagram:



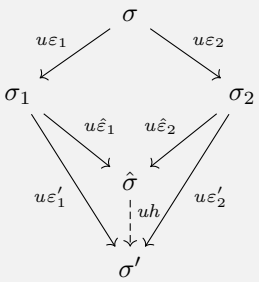
We define the general unifier $\sigma_1 \xrightarrow{\hat{\varepsilon}_1} \hat{\sigma} \xleftarrow{\hat{\varepsilon}_2} \sigma_2$ as follows.

1. We define $[\hat{\varepsilon}_1]$ and $[\hat{\varepsilon}_2]$ as



   It is clear that they are jointly surjective.

2. We generate morphisms $u\hat{\varepsilon}_1$ and $u\hat{\varepsilon}_2$ from $[\hat{\varepsilon}_1]$ and $[\hat{\varepsilon}_2]$ using Lemma 1.3.10, and so by the universal property of generated morphisms we have the following diagram in $\mathbb{T}$



   That $u\hat{\varepsilon}_1 \circ u\varepsilon_1 = u\hat{\varepsilon}_2 \circ u\varepsilon_2$ follows from the outer square and inner triangles commuting.

3. For the instantiation data we show that $\eta_{\varepsilon_i'}$ can be factored into η_h and $\eta_{\hat{\varepsilon}_i}$. W.l.g. let $i = 1$. Through the canonical isomorphism

$$\lfloor \sigma' \rfloor \ominus \lfloor u\varepsilon_1' \rfloor \cong (\lfloor \sigma' \rfloor \ominus \lfloor uh \rfloor) \oplus (\lfloor \hat{\sigma} \rfloor \ominus \lfloor u\hat{\varepsilon}_1 \rfloor)$$

   we have heaps

$$\eta_{\hat{\varepsilon}_1} : \underline{\mathbb{H}}(\lfloor \hat{\sigma} \rfloor \ominus \lfloor \hat{\varepsilon}_1 \rfloor, [\sigma']) \quad \text{and} \quad \eta_h : \underline{\mathbb{H}}(\lfloor \sigma' \rfloor \ominus \lfloor h \rfloor, [\sigma'])$$

   We further show for every $a : C \in \lfloor \hat{\sigma} \rfloor \ominus \hat{\varepsilon}_1$ that $d = \pi_a \eta_{\hat{\varepsilon}_1}$ satisfies

$$\begin{array}{ccc} 1 & \xrightarrow{d} & [\![\phi(C)]\!][\sigma'] \\ & \searrow^{\exists \hat{d}} & \uparrow{\scriptstyle [h]} \\ & & [\![\phi(C)]\!][\hat{\sigma}] \end{array}$$

By Lemma 1.3.11 we have that $\lfloor \hat{\varepsilon}_1 \rfloor$ and $\lfloor \hat{\varepsilon}_2 \rfloor$ are jointly surjective, and since $a : C \in \lfloor \hat{\sigma} \rfloor \ominus \lfloor \hat{\varepsilon}_1 \rfloor$, and $u\hat{\varepsilon}_1 \circ u\varepsilon_1 = u\hat{\varepsilon}_2 \circ u\varepsilon_2$ we must have a $a' : C \in \lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon_2 \rfloor$ s.t. $\lfloor \hat{\varepsilon}_2 \rfloor(a') = a$. And so

$$\begin{aligned} \pi_a \eta_{\hat{\varepsilon}_1} &= \pi_{\lfloor h \rfloor(a)} \eta_{\varepsilon_1'} \\ &= \pi_{\lfloor h \rfloor(a)} \eta_{\varepsilon_1' \circ \varepsilon_1} \\ &= \pi_{\lfloor h \rfloor(a)} \eta_{\varepsilon_2' \circ \varepsilon_2} \\ &= \pi_{\lfloor h \rfloor(\lfloor \hat{\varepsilon}_2 \rfloor(a'))} \eta_{\varepsilon_2' \circ \varepsilon_2} \\ &= \pi_{\lfloor \varepsilon_2' \rfloor(a')} \eta_{\varepsilon_2' \circ \varepsilon_2} \\ &= [\![\phi(C)]\!][\varepsilon_2'](\pi_{a'} \eta_{\varepsilon_2}) \\ &= [\![\phi(C)]\!][h]([\![\phi(C)]\!][\hat{\varepsilon}_2](\pi_{a'} \eta_{\varepsilon_2})) \end{aligned}$$

Lastly, $\eta_{\hat{\varepsilon}_1 \circ \varepsilon_1} = \eta_{\hat{\varepsilon}_2 \circ \varepsilon_2}$ follows from h being mono (Lemma 1.4.6).

To show uniqueness, consider another general unifier $\sigma_1 \xrightarrow{\check{\varepsilon}_1} \check{\sigma} \xleftarrow{\check{\varepsilon}_2} \sigma_2$.
Notice that the induced morphism from $\check{\sigma} \to \sigma'$ must factor through $\hat{\sigma}$ in an isomorphic way

$$\begin{array}{ccccc} \sigma_1 & \xrightarrow{\hat{\varepsilon}_1} & \hat{\sigma} & \xleftarrow{\hat{\varepsilon}_2} & \sigma_2 \\ & \searrow^{\check{\varepsilon}_1} & \cong \uparrow & \swarrow_{\check{\varepsilon}_2} & \\ & & \check{\sigma} & & \end{array}$$

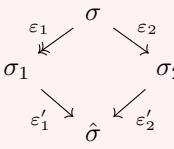
It remains to show that their data match which follows from Lemma 1.4.38. □

Lemma 1.4.40 (General Unifiers of Identities)**.** The general unifiers of a span $\sigma \xleftarrow{1} \sigma \xrightarrow{\varepsilon} \sigma'$ is only cospan $\sigma \xrightarrow{\varepsilon} \sigma' \xleftarrow{1} \sigma'$ (up to iso).

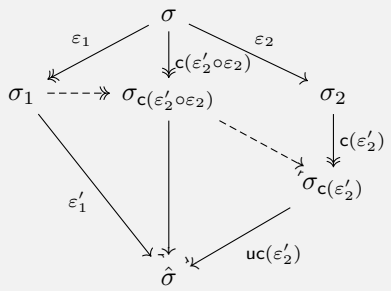
Proof. To show that cospan $\sigma \xrightarrow{\varepsilon} \sigma' \xleftarrow{1} \sigma'$ is a general unifier, we need it to be jointly surjective at the level $\lfloor - \rfloor$ and $[-]$, which is immediate from the identity morphism. That every other unifier factors through this (and so it is unique up to iso), is immediate. □

Lemma 1.4.41 (Pasting Law for General Unifiers)**.** In a diagram $\begin{array}{ccccc} \sigma & \xrightarrow{\varepsilon_1} & \sigma_1 & \xrightarrow{\varepsilon_3} & \sigma_3 \\ {\scriptstyle \varepsilon_1'}\downarrow & & \downarrow{\scriptstyle \varepsilon_2} & & \downarrow{\scriptstyle \varepsilon_3'} \\ \sigma_1' & \xrightarrow[\varepsilon_2']{} & \sigma_2 & \xrightarrow[\varepsilon_3'']{} & \sigma_3' \end{array}$ where the left square is a general unifier, then the whole square is a general unifier iff the right square is.

Proof. For the $\to$ direction, it is clear that if any $[\varepsilon_3'']$ and $[\varepsilon_3']$ are not jointly surjective then the whole square cannot be as well which is a contradiction. The same applies to the $\lfloor - \rfloor$ level, and so the right square must be a general unifier.
For the $\leftarrow$ direction, consider a $a : C \in [\sigma_3']$. Then by the right square being a general unifier, either a is mapped from $[\sigma_3]$ or it comes from $[\sigma_2]$. If it comes from $[\sigma_2]$, then by the left square being a general unifier either it is mapped from σ_1' or it is mapped from σ_1 and since it is a commuting square, it must also be mapped from σ_3. The same reasoning applies to $\lfloor - \rfloor$. □

Lemma 1.4.42 (General Unifiers Preserve Connectivity)**.** If  is a general unifier and ε_1 is connected, then so is ε_2'.

Proof. Consider a $a : C \in [\hat{\sigma}] \ominus [\mathsf{uc}(\varepsilon_2')]$. This means that $a : C \in [\hat{\sigma}] \ominus [\varepsilon_2']$, and so there exists $a' : C \in [\sigma_1]$ s.t. $[\varepsilon_1'](a') = a$ by definition of general unifiers (Definition 1.4.35). From the factorisation we have a diagram which leads to a contradiction that it commutes unless $[\hat{\sigma}] \ominus [\mathsf{uc}(\varepsilon_2')] = \emptyset$.



Since $\lceil \mathsf{uc}(\varepsilon_2') \rceil$ is iso and $\mathsf{uc}(\varepsilon_2')$ is a regular mono, we have that $\lfloor \mathsf{uc}(\varepsilon_2') \rfloor$ is iso and thus connected. □

Lemma 1.4.43 (Decidability of Unification)**.** There exists an algorithm that decides for a span $\sigma_1 \xleftarrow{\varepsilon_1} \sigma \xrightarrow{\varepsilon_2} \sigma_2$ if they unify or not. If they unify, it returns a non-empty finite set of general unifiers $\mathsf{GU}(\varepsilon_1, \varepsilon_2) \subseteq_{\mathsf{fin}} \coprod_{\hat{\sigma} \in \mathsf{Obj}(\mathbb{T})} \mathbb{TI}(\sigma_1, \hat{\sigma}) \times \mathbb{TI}(\sigma_2, \hat{\sigma})$ that contains all its general unifiers up to isomorphism.

Proof. By Lemma 1.4.39 we have that every unifier factors through a general one and there are only a finite number of isomorphic classes of general unifiers by definition of their jointly surjective property. As a result, the algorithm will enumerate every isomorphic class of jointly surjective cospans $[\sigma_1] \xrightarrow{\tau_1} \hat{w} \xleftarrow{\tau_2} [\sigma_2]$ by considering a representative for each class (note that $\mathsf{Obj}(\mathbb{W})$ is countable when Sort is, which makes it possible to constructively pick one).
It will then equip every representative cospan with data given by Lemma 1.3.12 for every side which by Lemma 1.4.38 tells us that it should be the only data that we have to consider. All that is left to check for it to be a general unifier is that it forms a commuting square in $\mathbb{TI}$. If the algorithm finds no such commuting generated cospans then it must be that the span does not unify. □

Unification and Local Independence Structure

Definition 1.4.44 (Local Independent Unifiers)**.** A unifier is called a local independent unifier (L.I.U.) if its u image is a local independent square.

Definition 1.4.45 (Local Independent General Unifiers)**.** A unifier is called a local independent general unifier (L.I.G.U.) if its u image is a local independent coproduct square.

Lemma 1.4.46 (Local Independent General Unifier is General)**.** A local independent general unifier is a general unifier.

Proof. To show that a L.I.G.U. $\sigma_1 \xrightarrow{\varepsilon_1'} \hat{\sigma} \xrightarrow{\varepsilon_2'} \sigma_2$ is a general unifier it is enough to show that it is generated by $[\sigma_1] \xrightarrow{[\varepsilon_1']} [\hat{\sigma}] \xleftarrow{[\varepsilon_2']} [\sigma_2]$ and moreover this cospan is jointly surjective. This is the case as L.I.C. in $\mathbb{T}$ is generated from the L.I.C. in $\mathbb{W}$ which is itself a jointly surjective cospan. □

Lemma 1.4.47 (Uniqueness of Local Independent General Unifiers)**.** L.I.G.U. are unique up to iso.

Proof. Any two L.I.G.U. are iso in $\mathbb{T}$, by L.I.C. being unique up to iso. That their data must also be match follows from Lemma 1.4.38. □

Lemma 1.4.48 (Local Independent General Unifiers Property)**.** The general unifier of a local independent unifier is a local independent general unifier.

Proof. The general unifier of a L.I.U. $\sigma_1 \xrightarrow{\varepsilon_1'} \sigma' \xleftarrow{\varepsilon_2'} \sigma_2$ is given by the generated cospan over

$$\begin{array}{ccccc} [\sigma_1] & \xrightarrow{[\varepsilon_1']} & [\sigma'] & \xleftarrow{[\varepsilon_2']} & [\sigma_2] \\ & \searrow & \uparrow & \swarrow & \\ & & \mathsf{Img}([\varepsilon_1']) \cup \mathsf{Img}([\varepsilon_2']) & & \end{array}$$

And since it is a local independent unifier we have

$$\begin{array}{ccccc} [\sigma_1] & \xrightarrow{[\varepsilon_1]^*[\varepsilon_2]} & [\sigma_1] \oplus_{[\sigma]} [\sigma_2] & \xleftarrow{[\varepsilon_2]^*[\varepsilon_1]} & [\sigma_2] \\ & \searrow & \cong \uparrow & \swarrow & \\ & & \mathsf{Img}([\varepsilon_1']) \cup \mathsf{Img}([\varepsilon_2']) & & \end{array}$$

which is exactly the cospan that generates L.I.G.U. □

Lemma 1.4.49 (Pasting Law for Local Independent General Unifiers)**.** In a diagram $\begin{array}{ccccc} \sigma & \xrightarrow{\varepsilon_1} & \sigma_1 & \xrightarrow{\varepsilon_3} & \sigma_3 \\ {\scriptstyle\varepsilon_1'}\downarrow & & \downarrow{\scriptstyle\varepsilon_2} & & \downarrow{\scriptstyle\varepsilon_3'} \\ \sigma_1' & \xrightarrow[\varepsilon_2']{} & \sigma_2 & \xrightarrow[\varepsilon_3'']{} & \sigma_3' \end{array}$ where the left square is a L.I.G.U., then the whole square is a L.I.G.U. iff the right square is.

Proof. Immediate from Lemma 1.3.18. □

Lemma 1.4.50. Every span of the form $\sigma_1 \xleftarrow{\varepsilon_1} \sigma \xrightarrow{\varepsilon_2}_r \sigma_2$ has a local independent general unifier.

Proof. We have that the L.I.C. in $\mathbb{T}$ commutes $\begin{array}{ccccc} & & \sigma & & \\ & {\scriptstyle u\varepsilon_1}\swarrow & & \searrow{\scriptstyle u\varepsilon_2} & \\ \sigma_1 & & & & \sigma_2 \\ & {\scriptstyle u\varepsilon_1^* u\varepsilon_2}\searrow & & \swarrow{\scriptstyle u\varepsilon_2^* u\varepsilon_1} & \\ & & \sigma_1 \oplus_\sigma \sigma_2 & & \end{array}$. We need to show that it also commutes in $\mathbb{TI}$ with data given by Remark 1.3.16.
Notice that by Lemma 1.3.27, we have that $\lfloor \sigma_1 \oplus_\sigma \sigma_2 \rfloor$ is given by

$$\begin{array}{ccc} \lfloor \sigma \rfloor & \xrightarrow{\lfloor u\varepsilon_2 \rfloor} & \lfloor \sigma_2 \rfloor \\ {\scriptstyle \lfloor u\varepsilon_1 \rfloor}\downarrow & & \downarrow{\scriptstyle \lfloor u\varepsilon_2^* u\varepsilon_1 \rfloor} \\ \lfloor \sigma_1 \rfloor & \xrightarrow[\lfloor u\varepsilon_1^* u\varepsilon_2 \rfloor]{} & \lfloor \sigma_1 \rfloor \oplus_{\lfloor \sigma \rfloor} \lfloor \sigma_2 \rfloor \end{array}$$

and $u\varepsilon_1^* u\varepsilon_2$ is variable preserving.
We equip $u\varepsilon_1^* u\varepsilon_2$ with data $\underline{\mathbb{H}}(\cong, [u\varepsilon_2^* u\varepsilon_1])(\eta_{\varepsilon_2})$ with the canonical iso

$$(\lfloor \sigma_1 \rfloor \oplus_{\lfloor \sigma \rfloor} \lfloor \sigma_2 \rfloor) \ominus \lfloor u\varepsilon_1^* u\varepsilon_2 \rfloor \cong \lfloor \sigma_2 \rfloor \ominus \lfloor u\varepsilon_2 \rfloor$$

We equip $u\varepsilon_2^* u\varepsilon_1$ with data in the same way. This gives that $\eta_{u\varepsilon_1^* \varepsilon_2 \circ \varepsilon_1} = \eta_{u\varepsilon_2^* \varepsilon_1 \circ \varepsilon_2}$. □

1.4.5 Preservation of Connectivity

Given $\sigma_1 \xrightarrow{\varepsilon} \sigma_2 \xrightarrow{\varepsilon'} \sigma_3$, we ask whether the connectivity of ε is preserved, that is, whether every location introduced by ε will remain reachable in the composite $\varepsilon' \circ \varepsilon$.

Definition 1.4.51 (Connectedness Preserving Morphisms)**.** A morphism $\varepsilon_2 : \sigma_2 \to \sigma_3$ preserves connectivity of a morphism $\varepsilon_1 : \sigma_1 \twoheadrightarrow \sigma_2$ if it satisfies $[\sigma_2] \otimes_{[\sigma_3]} ([\sigma_3] \ominus [\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)]) = \emptyset$.

Lemma 1.4.52 (Connectedness Preserving Morphisms Property Equivalence)**.** Let $\sigma_1 \xrightarrow{\varepsilon_1} \sigma_2 \xrightarrow{\varepsilon_2} \sigma_3$. We have the following equivalence.

$$\begin{array}{ccc} \emptyset & \longrightarrow & [\sigma_3] \ominus [\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)] \\ \downarrow & \lrcorner & \downarrow{\scriptstyle [\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)]^{\mathsf{c}}} \\ [\sigma_2] & \xrightarrow[[\varepsilon_2]]{} & [\sigma_3] \end{array} \quad \Leftrightarrow \quad \begin{array}{ccc} [\sigma_2] & & \\ {\scriptstyle \exists \mathsf{r}_{\varepsilon_1}^{\varepsilon_2 \circ \varepsilon_1}}\downarrow & \searrow^{[\varepsilon_2]} & \\ [\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}] & \xrightarrow[[\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)]]{} & [\sigma_3] \end{array}$$

Proof.
⇒) Consider the canonical iso $[\sigma_3] \cong [\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}] \oplus ([\sigma_3] \ominus [\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)])$. Now if any $a : C \in [\varepsilon_2]$ maps to $[\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}]$, we would get a contradiction $a : C \to \emptyset$ and so it must factor through $[\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}]$.
⇐) We $\mathsf{Img}([\varepsilon_2]) \cap \mathsf{Img}([\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)]^{\mathsf{c}})$ that must be empty as the image of $[\varepsilon_2]$ factors through $[\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}]$. □

Lemma 1.4.52 tells us that connectivity preservation is exactly uniqueness of factorisation (Lemma 1.4.30), and so every morphism preserves connectivity as the following lemma shows.

Lemma 1.4.53 (All Morphisms are Connectedness Preserving)**.** Every $\varepsilon_2 : \sigma_2 \to \sigma_3$ preserves the connectivity of every $\varepsilon_1 : \sigma_1 \twoheadrightarrow \sigma_2$.

Proof. By Lemma 1.4.30, we have

$$\begin{array}{ccccccc} & & & & \overset{\varepsilon_2 \circ \varepsilon_1}{\frown} & & \\ \sigma_1 & \xrightarrow{\varepsilon_1} & \sigma_2 & \xrightarrow{\mathsf{c}(\varepsilon_2)} & \sigma_{\mathsf{c}(\varepsilon_2)} & \xrightarrow{\mathsf{uc}(\varepsilon_2)} & \sigma_3 \\ & {\scriptstyle \mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}\searrow & & & \downarrow{\scriptstyle \cong} & \nearrow{\scriptstyle \mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)} & \\ & & & & \sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)} & & \end{array}$$

which proves preservation of connectivity by Lemma 1.4.52. □

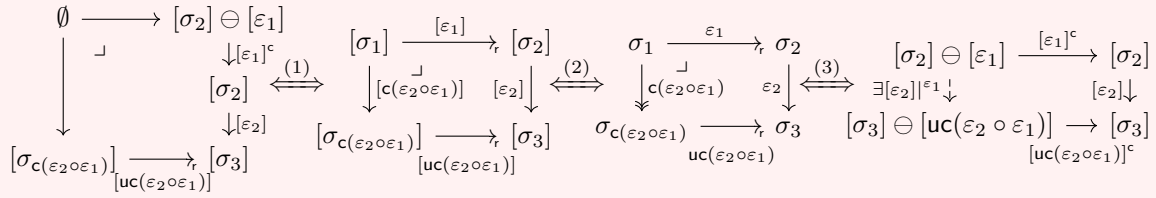
1.4.6 Preservation of Disconnectivity

Similarly, we ask if a morphism $\varepsilon' : \sigma_2 \twoheadrightarrow \sigma_3$ preserves a disconnected morphism $\varepsilon : \sigma_1 \to_{\mathsf{r}} \sigma_2$, that is, if every location in $[\sigma_2] \ominus [\varepsilon]$ remains unreachable in $\varepsilon' \circ \varepsilon$.

Such morphisms can only instantiate variables coming from σ_1 with data that do not point to locations $[\sigma_2] \ominus [\varepsilon]$ in order to preserve the unreachability of $[\sigma_2] \ominus [\varepsilon]$ in the composite $\varepsilon_2 \circ \varepsilon_1$.

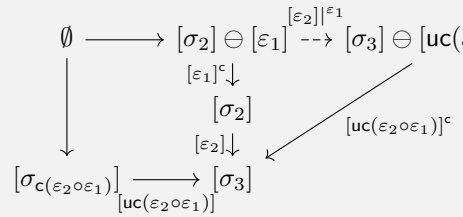
Definition 1.4.54 (Disconnectedness Preserving Morphisms)**.** A morphism $\varepsilon_2 : \sigma_2 \to \sigma_3$ preserves the disconnectivity of a morphism $\varepsilon_1 : \sigma_1 \to_{\mathsf{r}} \sigma_2$ if it satisfies $([\sigma_2] \ominus [\varepsilon_1]) \otimes_{[\sigma_3]} [\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}] = \emptyset$.

Lemma 1.4.55 (Disconnectedness Preserving Morphisms Property Equivalence)**.** Let $\sigma_1 \xrightarrow{\varepsilon_1}_{\mathsf{r}} \sigma_2 \xrightarrow{\varepsilon_2} \sigma_3$. We have the following equivalence.



Proof. For (1), we consider the canonical isomorphism $[\sigma_2] \cong [\sigma_1] \oplus ([\sigma_2] \ominus [\varepsilon_1])$. For $\overset{(1)}{\Longrightarrow}$ we cannot have anything factoring through $([\sigma_2] \ominus [\varepsilon_1])$ as we get a contradiction and for $\overset{(1)}{\Longleftarrow}$, if $[\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}] \xleftarrow{\pi_1} w \xrightarrow{\pi_2} ([\sigma_2] \ominus [\varepsilon_1])$ is the pullback on the left, we can form a square in the right one to get a unique injection $[\sigma_{\mathsf{c}(\varepsilon_1)}] \oplus w \dashrightarrow [\sigma_{\mathsf{c}(\varepsilon_1)}]$ and so w must be empty. For $\overset{(2)}{\Longrightarrow}$, by Lemma 1.3.28, the diagram in $\mathbb{T}$ is a pullback and by Lemma 1.4.8 it is also a pullback in $\mathbb{TI}$. For $\overset{(3)}{\Longleftarrow}$, by Lemma 1.4.9 the $[-]$-image is a pullback.

For $\overset{(3)}{\Longleftarrow}$, we have that



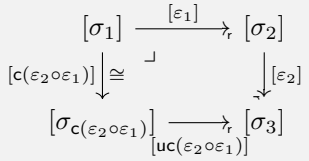
is a pullback as $\mathsf{Img}([\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)]) \cap \mathsf{Img}([\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)]^{\mathsf{c}}) = \emptyset$ and so the inner square must be a pullback.

For $\overset{(3)}{\Longrightarrow}$, the second pullback gives that $[\varepsilon_2]([\varepsilon_1]^{\mathsf{c}}(a)) \notin \mathsf{Img}([\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)])$.

□

Lemma 1.4.56 (Regular Monos are Disconnectedness Preserving)**.** For $\sigma_1 \xrightarrow{\varepsilon_1}_{\mathsf{r}} \sigma_2 \xrightarrow{\varepsilon_2}_{\mathsf{r}} \sigma_3$, ε_2 preserves disconnectivity of ε_1.

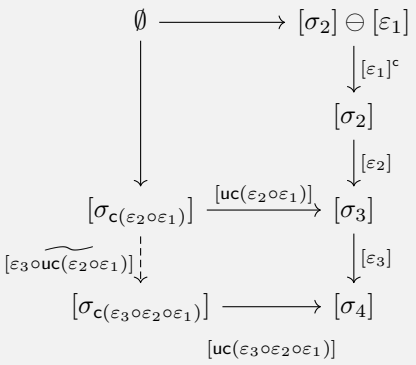
Proof. Notice that $\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)$ is epi and a regular mono and thus iso. As a result,



is a pullback by Lemma A.1.2. □

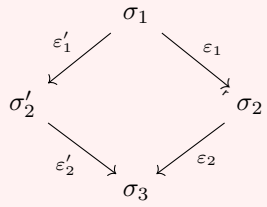
Lemma 1.4.57 (Disconnectedness Preservation is Downwards Closed)**.** For $\sigma_1 \xrightarrow{\varepsilon_1}_r \sigma_2 \xrightarrow{\varepsilon_2} \sigma_3 \xrightarrow{\varepsilon_3} \sigma_4$, if $\varepsilon_3 \circ \varepsilon_2$ preserves disconnectivity of ε_1 then ε_2 preserves disconnectivity of ε_1.

Proof. We have a diagram



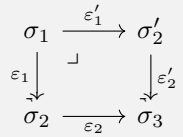
Where the dashed morphism is given by Lemma 1.4.31. That $\varepsilon_3 \circ \varepsilon_2$ preserves disconnectivity of ε_1 gives us that the outer square is a pullback, and by Lemma A.1.3, the inner square is also pullback which translates to ε_2 preserves disconnectivity of ε_1. □

Lemma 1.4.58 (L.I.G.U. and Disconnectedness Preservation)**.** Every L.I.G.U. diagram of the form

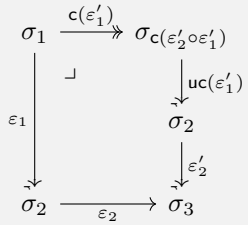


satisfies ε_2 preserves the disconnectivity of ε_1.

Proof. By Lemma 1.3.28 we have that the u-image is a pullback in $\mathbb{T}$, which by Lemma 1.4.8 it is a pullback in $\mathbb{TI}$.
So we have a pullback



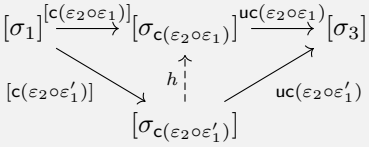
and notice ε_2' is r.mono by Lemma 1.3.27 and u reflecting variable preserving morphisms.
By Lemma A.1.4, we have that



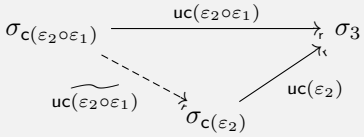
which proves that ε_2 preserves disconnectivity of ε_1 by Lemma 1.4.55. □

Lemma 1.4.59 (Disconnectedness Preserving Morphisms are Data Agnostic)**.** For $\sigma_1 \xrightarrow{\varepsilon_1}_r \sigma_2 \xrightarrow{\varepsilon_2} \sigma_3$ if ε_2 preserves disconnectivity of ε_1, then for all $\eta' : \underline{\mathbb{H}}(\lfloor\sigma_2\rfloor \ominus \lfloor\varepsilon_1\rfloor, \lfloor\sigma_2\rfloor)$, ε_2 preserves disconnectivity of $\varepsilon_1' = \langle u\varepsilon_1, \eta'\rangle$.

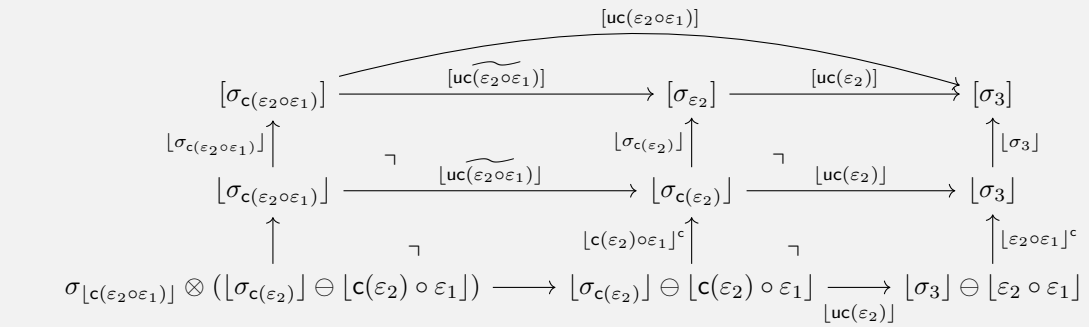
Proof. We first show that $[\sigma_1] \xrightarrow{[\mathsf{c}(\varepsilon_2\circ\varepsilon_1)]} [\sigma_{\mathsf{c}(\varepsilon_2\circ\varepsilon_1)}] \xrightarrow{[\mathsf{uc}(\varepsilon_2\circ\varepsilon_1)]} [\sigma_3]$ satisfies Definition 1.4.10 for $\eta_{\varepsilon_2\circ\varepsilon_1'}$. This would give us a h that satisfies



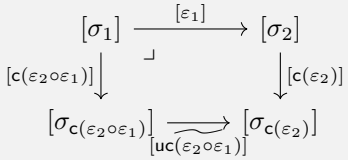
Property (i) follows immediately. For Property (ii), first by the lifting property we have



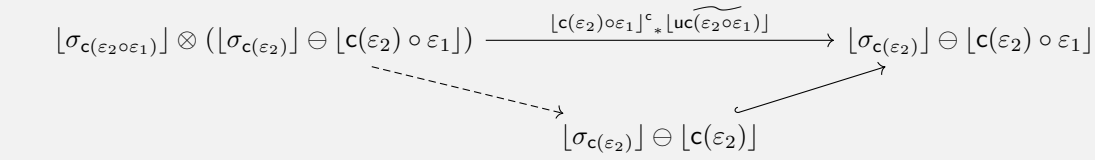
, and the pullback $[\sigma_{\mathsf{c}(\varepsilon_2\circ\varepsilon_1)}] \otimes_{[\sigma_3]} (\lfloor\sigma_3\rfloor \ominus \lfloor\varepsilon_2\circ\varepsilon_1\rfloor)$ is given by the pullback $\lfloor\sigma_{\mathsf{c}(\varepsilon_2\circ\varepsilon_1)}\rfloor \otimes_{\lfloor\sigma_{\mathsf{c}(\varepsilon_2)}\rfloor} (\lfloor\sigma_{\mathsf{c}(\varepsilon_2)}\rfloor \ominus \lfloor\mathsf{c}(\varepsilon_2)\circ\varepsilon_1\rfloor)$ from



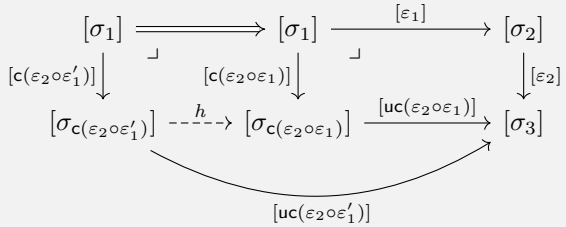
Now by Lemma 1.4.57, we have that $\mathsf{c}(\varepsilon_2)$ preserves disconnectivity of ε_1 and so



which implies that $\lfloor\mathsf{c}(\varepsilon_2)\circ\varepsilon_1\rfloor^{\mathsf{c}}{}_*\lfloor\widetilde{\mathsf{uc}(\varepsilon_2\circ\varepsilon_1)}\rfloor$ must factor as



As a result, all locations $[\sigma_{\mathsf{c}(\varepsilon_2\circ\varepsilon_1)}] \otimes_{[\sigma_3]} (\lfloor\sigma_3\rfloor \ominus \lfloor\varepsilon_2\circ\varepsilon_1\rfloor)$ in $\eta_{\varepsilon_2\circ\varepsilon_1'}$ point to cells in $\eta_{\mathsf{c}(\varepsilon_2)}$. Since $u\varepsilon_1 = u\varepsilon_1'$, the locations in these cells factor through $[\sigma_{\mathsf{c}(\varepsilon_2\circ\varepsilon_1)}]$ by Definition 1.4.10 for $[\mathsf{c}(\varepsilon_2\circ\varepsilon_1)]$ at $\eta_{\varepsilon_2\circ\varepsilon_1}$.
We now use h to show



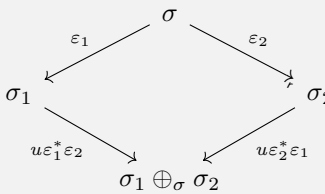
The right square is a pullback by Lemma 1.4.55, and the left square is a pullback by Lemma A.1.2. As a result by pullback pasting the outer square is a pullback giving us that ε_2 preserves the disconnectivity of ε_1'. □

Lemma 1.4.59 leads us to a definition of disconnectivity preservation w.r.t. morphisms in $\mathbb{T}$.

Definition 1.4.60 (Disconnectedness Preserving Morphisms w.r.t. $\mathbb{T}$)**.** We say that $\varepsilon_2 : \sigma_2 \to \sigma_3$ preserves the disconnectivity of $\sigma_1 \xrightarrow{\rho_1}_{\mathsf{r}} \sigma_2$ living in $\mathbb{T}$, if for all $\eta_{\rho_1} : \underline{\mathbb{H}}(\lfloor \sigma_2 \rfloor \ominus \lfloor \rho_1 \rfloor, [\sigma_2])$, ε_2 preserves the disconnectivity of $\langle \rho_1, \eta_{\rho_1} \rangle$.

Lemma 1.4.61 (Disconnectedness Preservation and L.I.C.)**.** For a span $\sigma_1 \xleftarrow{u\varepsilon_1} \sigma \xrightarrow{\rho_2}_{\mathsf{r}} \sigma_2$ in $\mathbb{T}$, the induced morphism in $\mathbb{TI}$, $\rho_2^* \varepsilon_1$, given by Lemma 1.3.14 and Remark 1.3.16, satisfies: $\rho_2^* \varepsilon_1$ preserves the disconnectivity of ρ_2.

Proof. For some $\eta_{\rho_2} : \underline{\mathbb{H}}(\lfloor \sigma_2 \rfloor \ominus \lfloor \rho_2 \rfloor, [\sigma_2])$, and $\varepsilon_2 = \langle \rho_2, \eta_{\rho_2} \rangle$, we have by Lemma 1.4.50 that



with data given by Remark 1.3.16 commutes in $\mathbb{TI}$. Since this is a L.I.G.U. we have by Lemma 1.4.58 that $u\varepsilon_2^* \varepsilon_1$ preserves the disconnectivity of ε_2. □

1.5 Families of Template Instantiations

We model finite collections of templates and their instantiations using the $\mathsf{Fam}(\mathbb{TI})$ construction, the free finite product completion of $\mathbb{TI}$.

- Objects: A finite set I, and an I-indexed set of templates $(\sigma_i \in \mathsf{Obj}(\mathbb{TI}))_{i \in I}$.
- Morphisms: A $(\sigma_i)_{i \in I} \xrightarrow{\phi} (\tau_j)_{j \in J}$ consists of a mapping $|\phi| : J \to I$ and a J-indexed set of morphisms $(\phi_j : \sigma_{|\phi|(j)} \to \tau_j)_{j \in J}$

We have a forgetful functor $|-| : \mathsf{Fam}(\mathbb{TI}) \to \mathsf{Fin}^{\mathsf{op}}$ that maps into the indexing sets.

Notice also that $\mathsf{Fam}(\mathbb{TI})$ is isomorphic to the opposite full subcategory of finite sums of covariant representable presheaves,

$$\mathbb{TI} \xrightarrow{y^{\mathsf{op}}} [\mathbb{TI}, \mathsf{Set}]^{\mathsf{op}}_{\sqcup} \cong \mathsf{Fam}(\mathbb{TI})$$

Which extends to an embedding of $\mathbb{TI}$ into it $\mathsf{Fam}(\mathbb{TI})$ by selecting a singleton family.

Notation. We use metavariable Σ to range over objects in $\mathsf{Fam}(\mathbb{TI})$ and use set membership notation $\sigma_i \in \Sigma$ to range over the templates in Σ. We also use list notation $[\sigma, \sigma, \tau]$ to write objects in $\mathsf{Fam}(\mathbb{TI})$.

Lemma 1.5.1 (Coequalizers in $\mathsf{Fam}(\mathbb{TI})$)**.** The category $\mathsf{Fam}(\mathbb{TI})$ has coequalizers.

Proof. Since $\mathsf{Fam}(\mathbb{TI}) \cong [\mathbb{TI}, \mathsf{Set}]^{\mathsf{op}}_{\sqcup}$ and inclusions reflect limits, it is enough to check that the equalizer of $\Sigma_2 \underset{\psi}{\overset{\phi}{\rightrightarrows}} \Sigma_1$ in $[\mathbb{TI}, \mathsf{Set}]$ can be represented as a finite sum of representables.
Equalizers are given pointwise by

$$\mathsf{eqz}(\phi, \psi)(\sigma) \cong \{inj_{j \in |\Sigma_2|}(\varepsilon : \sigma_j \to \sigma) : |\phi|(j) = |\psi|(j) \wedge \varepsilon \circ \phi_j = \varepsilon \circ \psi_j\}$$

Notice that since every ε is mono, we have $\phi_j = \psi_j$ for every such ε above.
So define the coequalizer $\mathsf{coeqz}(\phi, \psi)$ in $\mathsf{Fam}(\mathbb{TI})$ as exactly the σ_j that satisfy $\phi_j = \psi_j$

$$\mathsf{coeqz}(\phi, \psi) := \{\sigma_j \in \Sigma_2 : |\phi|(j) = |\psi|(j) \wedge \phi_j = \psi_j\}$$

□

Lemma 1.5.2 (Pushouts in $\mathsf{Fam}(\mathbb{TI})$)**.** The category $\mathsf{Fam}(\mathbb{TI})$ has pushouts given by the general unifiers of §1.4.4.

Proof. We define the pushout $\begin{array}{ccc} \Sigma & \xrightarrow{\psi} & \Sigma_2 \\ {\scriptstyle\phi}\downarrow & \ulcorner & \downarrow{\scriptstyle\psi^*\phi} \\ \Sigma_1 & \xrightarrow[\phi^*\psi]{} & \Sigma_1 \oplus_\Sigma \Sigma_2 \end{array}$ by noticing that pullbacks in $[\mathbb{TI}, \mathsf{Set}]$ are given pointwise and

$$\Sigma_1\sigma' \otimes_{\Sigma\sigma'} \Sigma_2\sigma' \cong \{\langle inj_{\sigma_i}(\sigma_i \xrightarrow{\varepsilon_i} \sigma'), inj_{\sigma_j}(\sigma_j \xrightarrow{\varepsilon_j} \sigma')\rangle : |\phi|(i) = |\psi|(j) \wedge \begin{array}{ccccc} & & \sigma_{|\phi|(i)} & & \\ & {\scriptstyle\phi_i}\swarrow & & \searrow{\scriptstyle\psi_j} & \\ \sigma_i & & & & \sigma_j \\ & {\scriptstyle\varepsilon_i}\searrow & & \swarrow{\scriptstyle\varepsilon_j} & \\ & & \sigma' & & \end{array} \}$$

By Lemma 1.4.39, every such $\varepsilon_i, \varepsilon_j$ pair factors through a general unifier, and so $\Sigma_1 \oplus_\Sigma \Sigma_2 \in \mathsf{Fam}(\mathbb{TI})$ must be isomorphic to the set of general unifiers of matching pairs i, j.
Specifically, we select all pairs that have matching indices $\{\langle i, j\rangle \in |\Sigma_1| \times |\Sigma_2| : |\phi|(i) = |\psi|(j)\}$ and for every such pair we have by Lemma 1.4.43 a finite set $\mathsf{GU}(\phi_i, \psi_j)$ which we turn into a finite family of templates $\Sigma_{\mathsf{GU}(\phi_i,\psi_j)}$ by projecting out the first component.

$$\Sigma_1 \oplus_\Sigma \Sigma_2 := \prod_{\langle i,j\rangle \in |\Sigma_1| \times |\Sigma_2| \,:\, |\phi|(i) = |\psi|(j)} \Sigma_{\mathsf{GU}(\phi_i,\psi_j)}$$

For every vertex of a general unifier, $\sigma_k \in \Sigma_{\mathsf{GU}(\phi_i,\psi_j)}$, we have morphisms $\varepsilon_i^k : \sigma_i \to \sigma_k$ and $\varepsilon_j^k : \sigma_j \to \sigma^k$, and so define the injections in $\Sigma_1 \oplus_\Sigma \Sigma_2$ as

$$\psi^*\phi := (\varepsilon_i^k)_{\langle i,j,k\rangle \in |\Sigma_1 \oplus_\Sigma \Sigma_2|} : \Sigma_1 \to \Sigma_1 \otimes_\Sigma \Sigma_2$$
$$\phi^*\psi := (\varepsilon_j^k)_{\langle i,j,k\rangle \in |\Sigma_1 \oplus_\Sigma \Sigma_2|} : \Sigma_2 \to \Sigma_1 \otimes_\Sigma \Sigma_2$$

The square $\phi^*\psi \circ \phi = \psi^*\phi \circ \psi$ commutes from $\varepsilon_i^k \circ \phi_i = \varepsilon_j^k \circ \psi_j$ for every $\langle i, j, k\rangle \in |\Sigma_1 \oplus_\Sigma \Sigma_2|$. □

Lemma 1.5.3. The embedding $y^{\mathsf{op}} : \mathbb{TI} \to \mathsf{Fam}(\mathbb{TI})$ preserves equalizers and pullbacks from $\mathbb{TI}$.

Proof. See Lemmas A.2.1 and A.2.2. □

Definition 1.5.4 (Connected Morphisms)**.** A morphism $\phi : \Sigma \to \Sigma'$ is said to be *connected* if every ϕ_j is connected.

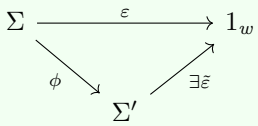
Definition 1.5.5 (Non-Overlapping Morphisms)**.** A morphism $\phi : \Sigma \to \Sigma'$ is said to be *non-overlapping* if every distinct pair of morphisms with the same root, $\sigma_i \xleftarrow{\phi_i} \sigma_k \xrightarrow{\phi_j} \sigma_j$, do not unify. That is, for every $i, j \in |\Sigma'|$ s.t. $i \neq j$ we have

$$|\phi|(i) = |\phi|(j) \implies \forall \sigma_i \xrightarrow{\varepsilon_i} \hat{\sigma}, \sigma_j \xrightarrow{\varepsilon_j} \hat{\sigma}.\, \varepsilon_i \circ \phi_i \neq \varepsilon_j \circ \phi_j$$

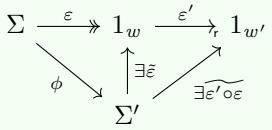
Lemma 1.5.6. A morphism $\phi : \Sigma \to \Sigma'$ is epi iff it is connected and non-overlapping.

Proof. An epi $\phi : \Sigma \to \Sigma' \in \mathsf{Fam}(\mathbb{TI})$ is a mono in $[\mathbb{TI}, \mathsf{Set}]_{\sqcup}$. Component-wise injectivity is exactly being connected and non-overlapping, and so it follows by Lemma A.2.3. □

Definition 1.5.7 (Total Morphisms)**.** We say that a morphism $\phi : \Sigma \to \Sigma'$ is *total* if for every morphism $\varepsilon : \Sigma \to 1_w$ there exists a $\tilde{\varepsilon} : \Sigma' \to 1_w$ such that



and further, this mapping is compatible w.r.t. connectivity



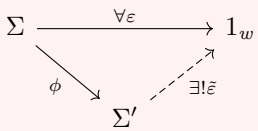
The induced morphism $\widetilde{\varepsilon' \circ \varepsilon}$ for a connected morphism ε and unconnected ε' factors through the induced morphism $\tilde{\varepsilon}$.

Lemma 1.5.8. Total morphisms are closed under composition.

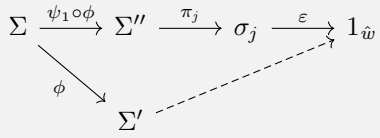
Proof. Easy exercise. □

Definition 1.5.9 (Complete Covers)**.** A *complete* cover is a connected, non-overlapping and total cover.

Lemma 1.5.10. When the set of sorts are inhabited (Definition 1.2.1), a morphism $\phi : \Sigma \to \Sigma'$ is complete iff totality is satisfied uniquely



Proof. For $\to$, we only have to show that totality is satisfied uniquely which is immediate from ϕ being epi.
For $\leftarrow$ direction, uniqueness of totality implies that the mapping is compatible w.r.t. connectivity and so it is a total morphism. To show that ϕ is also epi, consider $\psi_1, \psi_2 : \Sigma' \to \Sigma''$ s.t. $\psi_1 \circ \phi = \psi_2 \circ \phi$. Notice that for every $j \in |\Sigma''|$ we have



where ε is given by Lemma 1.4.5. By uniqueness we get $\varepsilon \circ \pi_j \circ \psi_1 = \varepsilon \circ \pi_j \circ \psi_2$ which implies by ε being mono that $\pi_j \circ \psi_1 = \pi_j \circ \psi_2$ for every $j \in |\Sigma'|$. Thus $\psi_1 = \psi_2$. □

Lemma 1.5.11 (Connectivity is stable under pushouts)**.** In a pushout diagram,

$$\begin{array}{ccc} \Sigma & \xrightarrow{\phi_2} & \Sigma_2 \\ {\scriptstyle \phi_1}\downarrow & \ulcorner & \downarrow{\scriptstyle \phi_2^* \phi_1} \\ \Sigma_1 & \xrightarrow[\phi_1^* \phi_2]{} & \Sigma_1 \oplus_\Sigma \Sigma_2 \end{array}$$

if ϕ_1 is connected, so is $\phi_2^* \phi_1$.

Proof. Directly by Lemma 1.4.42. □

Lemma 1.5.12 (Connectivity and Non-Overlapping stable under pushousts)**.** In a pushout diagram,

$$\begin{array}{ccc} \Sigma & \xrightarrow{\phi_2} & \Sigma_2 \\ {\scriptstyle \phi_1}\downarrow & \ulcorner & \downarrow{\scriptstyle \phi_2^* \phi_1} \\ \Sigma_1 & \xrightarrow[\phi_1^* \phi_2]{} & \Sigma_1 \oplus_\Sigma \Sigma_2 \end{array}$$

if ϕ_1 is connected and non-overlapping, so is $\phi_2^* \phi_1$.

Proof. Connected, non-overlapping morphisms are epis, and epis are stable under pushouts. □

Lemma 1.5.13 (Totality is stable under pushouts)**.** In a pushout diagram,

$$\begin{array}{ccc} \Sigma & \xrightarrow{\phi_2} & \Sigma_2 \\ {\scriptstyle \phi_1}\downarrow & \ulcorner & \downarrow{\scriptstyle \phi_2^* \phi_1} \\ \Sigma_1 & \xrightarrow[\phi_1^* \phi_2]{} & \Sigma_1 \oplus_\Sigma \Sigma_2 \end{array}$$

if ϕ_1 is total, so is $\phi_2^* \phi_1$.

Proof. To show totality for $\phi_2^* \phi_1$, it is enough to give an induced morphism $\tilde{\varepsilon} : \Sigma_1 \oplus_\Sigma \Sigma_2 \to 1_{\hat{w}}$ for every connected $\varepsilon : \Sigma_2 \to 1_{\hat{w}}$. By construction, the mapping will be compatible with connectivity. Totality of ϕ_1 at $\Sigma \xrightarrow{\phi_2} \Sigma_2 \xrightarrow{\varepsilon} 1_{\hat{w}}$ gives us a $\widetilde{\varepsilon \circ \phi_2}$ s.t. $\varepsilon \circ \phi_2 = \phi_1 \circ \widetilde{\varepsilon \circ \phi_2}$ and so by the UP of the pushout we get $\tilde{\varepsilon}$. □

Corollary 1.5.14 (Completeness is stable under pushouts)**.** In a pushout diagram,

$$\begin{array}{ccc} \Sigma & \xrightarrow{\phi_2} & \Sigma_2 \\ {\scriptstyle \phi_1}\downarrow & \ulcorner & \downarrow{\scriptstyle \phi_2^*\phi_1} \\ \Sigma_1 & \xrightarrow[\phi_1^*\phi_2]{} & \Sigma_1 \oplus_\Sigma \Sigma_2 \end{array}$$

if ϕ_1 is complete, so is $\phi_2^*\phi_1$.

2 The Finitary Monad

2.1 FGLS Monad

We give an alternative description of the FGLS monad $\mathcal{T} : [\mathbb{W}, \mathsf{Set}] \to [\mathbb{W}, \mathsf{Set}]$ using natural transformations. We utilise this presentation to define the finitary submonad from.

$$\mathcal{T}Xw \cong [\mathbb{WI}, \mathsf{Set}](\mathbb{W}(w, u(-)) \times \mathbb{H}, P(Xu \times \mathbb{H}))$$

where $u : \mathbb{WI} \to \mathbb{W}$ is the forgetful functor that forgets the heap data. We further define the functorial action, unit and join of the FGLS monad in terms of this naturality style.

Functorial action

Definition 2.1.1 (Functorial action of $\mathcal{T}$ on renamings)**.** Let $\sigma : w \rightarrowtail w'$ be a renaming in $\mathbb{W}$ in

$$\begin{aligned} &\mathcal{T}X\sigma : \mathcal{T}Xw \to \mathcal{T}Xw' \\ &\quad \phi \mapsto \mathbb{W}(w', u(-)) \times \mathbb{H} \xrightarrow{(-\circ\sigma)\times 1} \mathbb{W}(w, u(-)) \times \mathbb{H} \xrightarrow{\phi} P(Xu \times \mathbb{H}) \end{aligned}$$

Unit of Monad

Definition 2.1.2 (Unit of $\mathcal{T}$)**.**

$$\begin{aligned} &\eta_X : X \Rightarrow \mathcal{T}X \\ &\quad x \mapsto_w \mathbb{W}(w, u(-)) \times \mathbb{H} \xrightarrow{X(-)(x)\times 1} Xu \times \mathbb{H} \xrightarrow{\eta^P_{Xu\times\mathbb{H}}} P(Xu \times \mathbb{H}) \end{aligned}$$

where $X(-)(x)$ is simply the functorial action of X on x.

Join of Monad

Definition 2.1.3 (Multiplication of $\mathcal{T}$)**.**

$$\begin{aligned} &\mu_X : \mathcal{T}^2X \Rightarrow \mathcal{T}X \\ &\quad \phi \mapsto_w \mathbb{W}(w, u(-)) \times \mathbb{H} \xrightarrow{\phi} P(\mathcal{T}Xu \times \mathbb{H}) \xrightarrow{P(\mathsf{eval})} P^2(Xu \times \mathbb{H}) \xrightarrow{\mu^P_{Xu\times\mathbb{H}}} P(Xu \times \mathbb{H}) \end{aligned}$$

where $\mathsf{eval}\langle\phi, h\rangle \overset{w}{=} \phi_w\langle 1_w, h\rangle$.

2.2 Matching

2.2.1 Stencils

A *stencil* is a template heap with a selection of public locations. They represent collections of heaps with a common segment determined by the public locations.

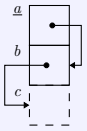
Definition 2.2.1 (Stencils)**.** Stencils are given by

$$\mathcal{S} : \mathbb{W}^{\mathsf{op}} \to [\mathbb{TI}, \mathsf{Set}]$$
$$\mathcal{S}_w = \mathbb{TI}(?_w, -) \cong \mathbb{W}(w, [-]) \times \mathbb{TH}$$

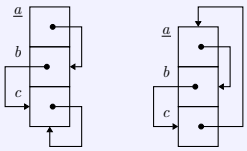
with the obvious functorial actions. We call the contravariant operation $\mathcal{S}_\sigma$ as *hiding*.
Concrete stencils are stencils over heaps, that is

$$\mathcal{S}_w 1_- = \mathbb{TI}(?_w, 1_-) \cong \mathbb{W}(w, u(-)) \times \mathbb{H} : \mathbb{WI} \to \mathsf{Set}$$

Example 2.2.2. Consider the following stencil given diagrammatically and assume all locations are of a single sort C (underlined locations are the public locations, dashed boxes represent location variables)



It represents heaps that have this segment as starting point from a, such as



2.2.2 Covers

Notation. Let metavariable Σ to range over finite families of templates, that is, objects in $\Sigma \in \mathsf{Obj}(\mathsf{Fam}(\mathbb{TI}))$.

Definition 2.2.3 (Cover)**.** Let Σ be a finite family of templates, and $w \in \mathsf{Obj}(\mathbb{W})$. A *cover* cov over $\langle \Sigma, w \rangle$ is an indexed family of stencils $\prod_{\sigma_i \in \Sigma} \mathcal{S}_w \sigma_i$, that is, it is a morphism $\mathsf{cov} : ?_w \to \Sigma \in \mathsf{Fam}(\mathbb{TI})$.

We inherit the terminology from §1.5, and talk about connected, non-overlapping, total and complete covers. We revisit these concepts here in their specialised form.

Definition 2.2.4 (Connected Cover)**.** A cover cov over $\langle w, \Sigma \rangle$ is said to be connected if every stencil $\pi_i \circ \mathsf{cov}$ for $\sigma_i \in \Sigma$ is connected.

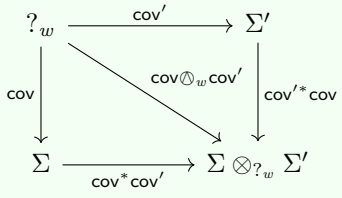
Definition 2.2.5 (Non-Overlapping Cover)**.** A cover cov over $\langle \Sigma, w \rangle$ is said to be *non-overlapping* if for every distinct $\sigma_i, \sigma_j \in \Sigma$, we have

$$\forall \sigma_i \xrightarrow{\varepsilon_i} \hat{\sigma}, \sigma_j \xrightarrow{\varepsilon_j} \hat{\sigma}.\, \varepsilon_i \circ \pi_i \circ \mathsf{cov} \neq \varepsilon_j \circ \pi_j \circ \mathsf{cov}$$

Definition 2.2.6 (Total Covers)**.** A cover S over $\langle \Sigma, w \rangle$ is *total* if for every connected, concrete stencil $\varepsilon : ?_w \to 1_{w'}$, there exists a $\sigma_i \in \Sigma$, $\tilde{\varepsilon} : \sigma_i \to 1_{w'}$ such that $\tilde{\varepsilon} \circ \pi_i \circ \mathsf{cov} = \varepsilon$.

Definition 2.2.7 (Complete Covers)**.** A *complete* cover is a connected, non-overlapping and total cover.

Definition 2.2.8 (Cover Intersection)**.** Given a cover cov over $\langle w, \Sigma\rangle$ and a cover cov' over $\langle w, \Sigma'\rangle$ we define their intersection $\mathsf{cov} \mathbin{\textcircled{\scriptsize \&}}_w \mathsf{cov}'$ as the biggest cover that is included in both of them.
It is given by pushouts in $\mathsf{Fam}(\mathbb{TI})$



Hiding Locations in a Cover We extend the contravariant hiding action on stencils to covers.

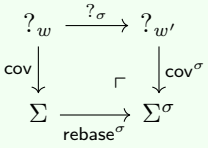
Definition 2.2.9 (Hiding in Covers)**.** For a renaming $\sigma : w \rightarrowtail w'$, and a cover cov over $\langle w', \Sigma\rangle$, we have a cover over $\langle w, \Sigma\rangle$ by applying the hiding action on every stencil

$$\mathsf{hide}_\sigma(\mathsf{cov}) := ?_w \xrightarrow{\sigma} ?_{w'} \xrightarrow{\mathsf{cov}} \Sigma$$

Hiding does not preserve connectivity, non-overlapping, or totality in general.

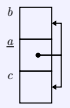
Rebasing a Cover We define a covariant action on renamings.

Definition 2.2.10. For a $\sigma : w \rightarrowtail w'$ and cover cov over $\langle w, \Sigma\rangle$, we can rebase cov to a cover cov^σ over $\langle w', \Sigma^\sigma\rangle$ for some induced family of templates Σ^σ.
We define the construction using pushouts in $\mathsf{Fam}(\mathbb{TI})$



Cover rebasing preserves connectivity, non-overlapping property (when cover is connected), totality, and thus completeness (see Lemmas 1.5.11 to 1.5.13 and Corollary 1.5.14).

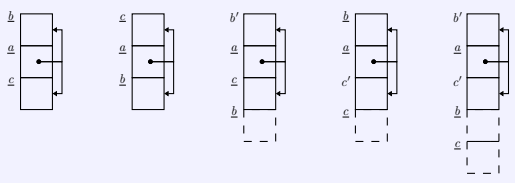
Example 2.2.11. Consider a singleton cover over $\langle a : C, [1_{a,b,c}]\rangle$ given by



and we rebase it over renaming $a : C \hookrightarrow a : C, b : C, c : C$, where $\phi(C) = 1 + \mathsf{ref}_C \times \mathsf{ref}_C$. The resulting cover is over

$$\langle \{a : C, b : C, c : C\}, [1_{a,b,c}, 1_{a,b,c}, i^{a,b',c\hookrightarrow a,b',c,b}, i^{a,b,c'\hookrightarrow a,b,c',c}, i^{a,b',c'\hookrightarrow a,b',c',b,c}]\rangle$$

with stencils



2.3 Evaluating

2.3.1 Allocation Morphisms

We call morphisms in $\mathbb{T}$ and $\mathbb{TI}$ that are both variable preserving and reflecting as allocation morphisms. We intent to capture the idea that allocations only introduce concrete locations and do not interfere with location variables.

Definition 2.3.1 (Allocation Morphisms)**.** A morphism $\rho : \sigma_1 \to \sigma_2$ is an allocation morphism if it satisfies $\lfloor \sigma_2 \rfloor \otimes_{[\sigma_2]} \lceil \sigma_1 \rceil = \emptyset$ and $\lceil \sigma_2 \rceil \otimes_{[\sigma_2]} \lceil \sigma_1 \rceil = \emptyset$.

Example 2.3.2. The morphism $?_a \xrightarrow{\langle ?_b, i^{a \hookrightarrow a,b} \rangle} i^{b \hookrightarrow a,b}$ is an allocation morphism since it does not instantiate nor introduce any variables.

Notation. An allocation morphism is represented using the arrow $\xrightarrow{}_{\nu}$.

Lemma 2.3.3 (Composition of Allocation Morphisms)**.** Allocation morphisms are closed under composition.

Proof. Follows from Lemma 1.3.22 and Lemma 1.3.32. □

Lemma 2.3.4 (Codomain of Allocation Morphisms)**.** If $\rho : \sigma_1 \to \sigma_2$ is an allocation morphism, then

$$\sigma_2 \cong \begin{array}{c} \lfloor \sigma_2 \rfloor \oplus \lceil \sigma_1 \rceil \\ \uparrow i_1 \\ \lfloor \sigma_2 \rfloor \end{array}$$

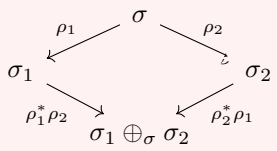
Proof. We have the canonical isomorphism

$$\sigma_2 \cong \begin{array}{c} \lfloor \sigma_2 \rfloor \oplus \lceil \sigma_2 \rceil \\ \uparrow i_1 \\ \lfloor \sigma_2 \rfloor \end{array}$$

and notice that $\lceil \sigma_1 \rceil \cong \lceil \sigma_2 \rceil$, from $\lceil \rho \rceil \circ \lceil \rho \rceil^{-1}|^{\sigma_2} \circ \lceil \rho \rceil|^{\sigma_2} = \lceil \rho \rceil \circ 1$ and $\lceil \sigma_2 \rceil \circ \lceil \rho \rceil|^{\sigma_2} \circ \lceil \rho \rceil^{-1}|^{\sigma_2} = \lceil \sigma_2 \rceil \circ 1$. □

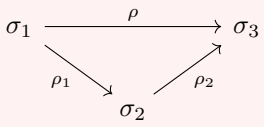
Notation. If ρ is an allocation morphism, let $\cong^{\rho}_{\nu}$ be the isomorphism given by Lemma 2.3.4.

Lemma 2.3.5 (Local Independent Coproducts and Allocation Morphisms)**.** In a L.I.C. square where ρ_2 is an allocation morphism



we have that $\rho_1^*\rho_2$ is an allocation morphism as well.

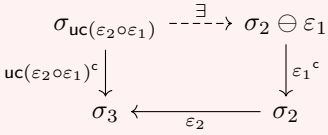
Proof. Follows from Lemma 1.3.27 and Lemma 1.3.36. □

Lemma 2.3.6 (Allocation Morphism Cancellation)**.** In 

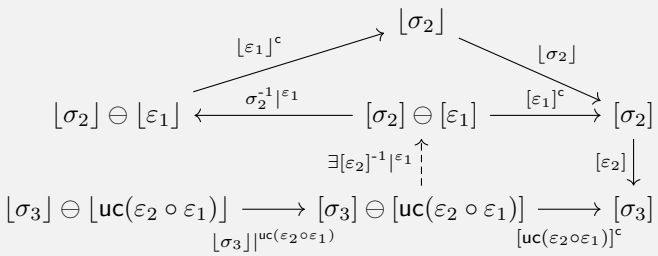
if ρ is an allocation morphism then ρ_1 is an allocation morphism iff ρ_2 is.

Proof. For ($\rightarrow$), it follows from Lemmas 1.3.33 and 1.3.34, and for ($\leftarrow$) it follows from Lemmas 1.3.23 and 1.3.35. □

Lemma 2.3.7 (Allocation Morphisms and Epis)**.** Every $\sigma_1 \xrightarrow{\varepsilon_1} \sigma_2 \xrightarrow{\varepsilon_2} \sigma_3$ has the following diagram in $\mathbb{T}$



Proof. Notice that if it holds at $[-]$ level it also holds at $\lfloor - \rfloor$ from

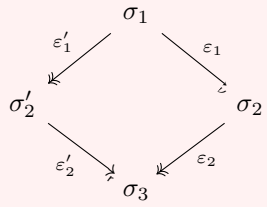


We construct $[\sigma_3] \ominus [\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)] \dashrightarrow [\sigma_2] \ominus \lfloor \varepsilon_1 \rfloor$, using the canonical iso

$$([\sigma_{\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)}] \ominus [\mathsf{c}(\varepsilon_2 \circ \varepsilon_1)]) \oplus ([\sigma_3] \ominus [\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)]) \cong ([\sigma_2] \ominus [\varepsilon_1]) \oplus ([\sigma_3] \ominus [\varepsilon_2])$$

Observe that if there exists $a, a' : C$ s.t. $[\mathsf{uc}(\varepsilon_2 \circ \varepsilon_1)^{\mathsf{c}}](a) = [\varepsilon_2]^{\mathsf{c}}(a')$ then there exists $b : C$, s.t. $[\mathsf{uc}(\varepsilon_2)][\mathsf{c}(\varepsilon_2)]^{\mathsf{c}}(b) = a'$ since ε_2 is connected. As a result, we have a $b : C$ that is not in the image of $[\sigma_2]$ and is reachable in ε_2. By Definition 1.4.15, we have a heap walk that leads to b with root location variable in $\lceil \sigma_2 \rceil$ and since ε_1 is an allocation morphism by Lemma 1.3.31, this variable comes from $\lceil \sigma_1 \rceil$. As a result, b must be reachable in $\varepsilon_2 \circ \varepsilon_1$ which is a contradiction as it maps to the unreachable segment by $[\mathsf{uc}(\varepsilon_2)][\mathsf{c}(\varepsilon_2)]^{\mathsf{c}}$. □

Lemma 2.3.8 (Disconnectedness Preservation and L.I.G.U.)**.** A diagram of the form



s.t. ε_2 preserves the disconnectivity of ε_1 implies that it is a L.I.G.U.

Proof. We first show that ε_2' is variable reflecting. From Lemmas 2.3.7 and 1.3.31 we have

$$\begin{array}{ccc} [\sigma_3] \ominus [\varepsilon_2'] & \dashrightarrow & [\sigma_2] \ominus [\varepsilon_1] \\ {\scriptstyle \lfloor \sigma_3 \rfloor | ^{\varepsilon_2'}} \uparrow & & \downarrow {\scriptstyle \lfloor \sigma_2 \rfloor^{-1} | ^{\varepsilon_1}} \\ \lfloor \sigma_3 \rfloor \ominus \lfloor \varepsilon_2' \rfloor & \dashrightarrow & \lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon_1 \rfloor \end{array}$$

which gives us a morphism $[\sigma_3] \ominus [\varepsilon_2'] \to \lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon_1 \rfloor$ and notice that post-composing this with $\lfloor \varepsilon_2 \rfloor \circ \lfloor \varepsilon_1 \rfloor^{\mathsf{c}}$ must reside in $\lfloor \sigma_3 \rfloor \ominus \lfloor \varepsilon_2' \rfloor$ for otherwise the square of ε_2' would not commute.
By Lemma 1.3.27, we need to show that both $\lfloor - \rfloor$ and $[-]$ images are L.I.C. Notice that by Lemma 1.4.55, its $[-]$-image is a pullback, and by Lemma 1.3.28 we have that the u-image is a pullback in $\mathbb{T}$, and so the $\lfloor - \rfloor$-image is a pullback as well.

So it enough to show that the $\lfloor - \rfloor$ and $[-]$ images of cospan $\sigma_2' \xrightarrow{\varepsilon_2'} \sigma_3 \xleftarrow{\varepsilon_2} \sigma_2$ are jointly surjective. We show that $\lfloor - \rfloor$-image of the cospan is jointly surjective which implies that the $[-]$-image is jointly surjective by the variable reflecting property of ε_2' (Lemma 1.3.31).
Joint surjectivity at $\lfloor - \rfloor$ is immediate from Lemma 2.3.7, every $a : C \in \lfloor \sigma_3 \rfloor \ominus \lfloor \mathsf{uc}(\varepsilon_2 \circ \varepsilon_1) \rfloor$ factors through $\lfloor \sigma_2 \rfloor \ominus \lfloor \varepsilon_1 \rfloor$ and so it is in the image of $\lfloor \varepsilon_2 \rfloor$. □

Definition 2.3.9 (Category of Template Allocations)**.** We consider $\mathsf{alloc}(\mathbb{T}) \subseteq \mathbb{T}$, the wide subcategory of $\mathbb{T}$ where morphisms are allocation morphisms. Identities are allocation morphisms and by Lemma 2.3.3, they are also closed under composition. Similarly, we consider $\mathsf{alloc}(\mathbb{TI}) \subseteq \mathbb{TI}$ and notice that we have a forgetful functor $\mathsf{alloc}(\mathbb{TI}) \xrightarrow{\mathsf{alloc}(u)} \mathsf{alloc}(\mathbb{T})$.

2.3.2 Resultlets

Resultlets represent collections of results in the spirit of the FGLS monad.

Definition 2.3.10. For a $A : \mathbb{TI} \to \mathsf{Set}$ and template σ, we define the resultlet functor R_σ^A as

$$\mathsf{R}_\sigma^A := \sigma/\mathsf{alloc}(u) \xrightarrow{U} \mathsf{alloc}(\mathbb{TI}) \hookrightarrow \mathbb{TI} \xrightarrow{A} \mathsf{Set}$$

A resultlet r over $\langle \sigma, A \rangle$ is an element of $\mathsf{Res}A\sigma$

$$r \in \mathsf{Res}A\sigma := \coprod_{\sigma \xrightarrow{\rho'} \sigma' \in \sigma/\mathsf{alloc}(u)} \mathsf{R}_\sigma^A \rho' = \coprod_{\sigma \xrightarrow{\rho'} \sigma' \in \sigma/\mathsf{alloc}(u)} A(\sigma')$$

Remark 2.3.11. For FGLS we are interested in situations where $A = X[-] \times \mathbb{TH}$ for a $X : \mathbb{W} \to \mathsf{Set}$. A resultlet over $\langle \sigma, X[-] \times \mathbb{TH} \rangle$ then consists of

- An allocation morphism $\sigma \xrightarrow{\rho} \sigma'$, which by Lemma 2.3.4 is determined only by $\lfloor \sigma \rfloor \xrightarrow{\lfloor \rho \rfloor} \lfloor \sigma' \rfloor$.
- An X-valued return value $x : X[\sigma']$ which by Lemma 2.3.4 is isomorphic to a value in $X(\lfloor \sigma' \rfloor \oplus \lceil \sigma \rceil)$.

- A template heap $\eta : \underline{\mathbb{H}}(\sigma')$, which is isomorphic to a heaplet $\underline{\mathbb{H}}(\lfloor\sigma'\rfloor, \lfloor\sigma'\rfloor \oplus \lceil\sigma\rceil)$.

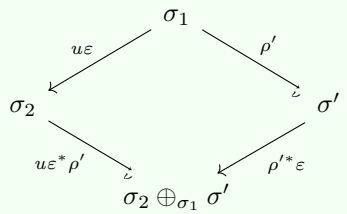
We define actions of Res on morphisms in $[\mathbb{TI}, \mathsf{Set}]$ and $\mathbb{TI}$ but we stress that some of these actions are not functorial. In §2.3.3 we quotient Res to form a functor out of it.

Definition 2.3.12 (Action on morphisms in $[\mathbb{TI}, \mathsf{Set}]$)**.** For a natural transformation $\alpha : A \to B$, and every $\sigma \in \mathsf{Obj}(\mathbb{T})$

$$\mathsf{Res}(\alpha)_\sigma : \mathsf{Res}A\sigma \to \mathsf{Res}B\sigma$$
$$inj_{\sigma \xrightarrow{\rho'}_{\iota} \sigma'}(a) \mapsto inj_{\rho'}(\alpha_{\sigma'}(a))$$

It changes a resultlet over $\langle\sigma, A\rangle$ to one over $\langle\sigma, B\rangle$.

Definition 2.3.13 (Covariant action on morphisms in $\mathbb{TI}$)**.** For a $\varepsilon : \sigma_1 \to \sigma_2 \in \mathbb{TI}$ and a resultlet $inj_{\sigma_1 \xrightarrow{\rho'}_{\iota} \sigma'} a$ we consider the following L.I.C. in $\mathbb{T}$



Notice that by Lemma 2.3.5, $u\varepsilon^*\rho'$ is an allocation morphism. We further equip $\rho'^*\varepsilon$ with data using Remark 1.3.16 and so define the covariant action $\mathsf{Res}A\varepsilon$ as

$$\mathsf{Res}A\varepsilon : \mathsf{Res}A\sigma_1 \to \mathsf{Res}A\sigma_2$$
$$inj_{\rho'} a \mapsto inj_{u\varepsilon^*\rho'} A(\rho'^*\varepsilon)(a)$$

Definition 2.3.14 (Contravariant action on allocation morphisms in $\mathbb{T}$)**.** For every allocation morphism $\sigma_2 \xrightarrow{\rho'}_{\iota} \sigma_1$ in $\mathbb{T}$ define

$$\mathsf{hide}^A_{\rho'} : \mathsf{Res}A\sigma_1 \to \mathsf{Res}A\sigma_2$$
$$inj_\rho a \mapsto inj_{\rho \circ \rho'} a$$

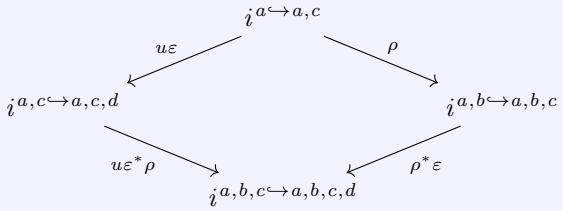
where $\rho \circ \rho'$ is an allocation morphism itself by Lemma 2.3.3.
We call this action *hiding* as it forgets some of the public locations.

Example 2.3.15. Consider the allocation morphism

$$i^{a \hookrightarrow a,c} \xrightarrow{\rho}_{\iota} i^{a,b \hookrightarrow a,b,c}$$

that introduces a new concrete location b, and a resultlet $inj_{\rho'} a$ where $a : A(i^{a,b \hookrightarrow a,b,c})$.

We can apply the covariant action $\varepsilon = i^{a \hookrightarrow a,c} \xrightarrow{\langle i^{a \hookrightarrow a,c}, i^{a,c \hookrightarrow a,c,d}\rangle} 1_{a,c}$ that instantiates c with some data $\eta_\varepsilon : \underline{\mathbb{H}}(c, \{a, c, d\})$



to get the resultlet

$$inj_{u\varepsilon^*\rho} A(\rho^*\varepsilon)(a)$$

with data $\eta_{\rho^*\varepsilon}$ being $\underline{\mathbb{H}}(1, [u\varepsilon^*\rho])(\eta_\varepsilon)$.

Consider the allocation morphism $\rho' =?_c \xrightarrow{\langle ?_a, i^{c \hookrightarrow a,c} \rangle} i^{a \hookrightarrow a,c}$. We can apply the contravariant action ρ' to get the resultlet $inj_{\langle ?_{\{a,b\}}, i^{c \hookrightarrow a,b,c} \rangle} a$.

Monad Structure on Resultlets We equip resultlets with unit and multiplication structure.

Definition 2.3.16 (Unit of a Resultlet)**.** We define unit morphisms

$$\eta^A_\sigma : A\sigma \to \mathsf{Res}A\sigma$$
$$a \mapsto inj_{1_\sigma} a$$

Definition 2.3.17 (Multiplication of a Resultlet)**.** We define multiplication morphisms

$$\mu^A_\sigma : \mathsf{Res}(\mathsf{Res}A)\sigma \to \mathsf{Res}A$$
$$inj_\rho r \mapsto \mathsf{hide}^A_\rho r$$

2.3.3 Hiding Monad on Template Heaps

Definition 2.3.18 (Hiding Monad on Template Heaps)**.** We generalise the hiding monad $\mathbb{P}$ from [Kam+17] to template heaps by appropriately quotienting the type of resultlets

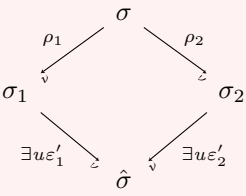
$$\mathbb{TP} : [\mathbb{TI}, \mathsf{Set}] \to [\mathbb{TI}, \mathsf{Set}]$$
$$(\mathbb{TP}A)(\sigma) = colimit\,(\mathsf{R}^A_\sigma) \ni q_{\sigma \xrightarrow{\rho'} \sigma'} a$$

which satisfies the natural (monad) isomorphism for every $A : \mathbb{TI} \to \mathsf{Set}$

$$(\mathbb{TP}A) \circ 1_- \overset{\mathbb{P}}{\cong} \mathbb{P}(A \circ 1_-) \;:\; \mathbb{WI} \to \mathsf{Set}$$

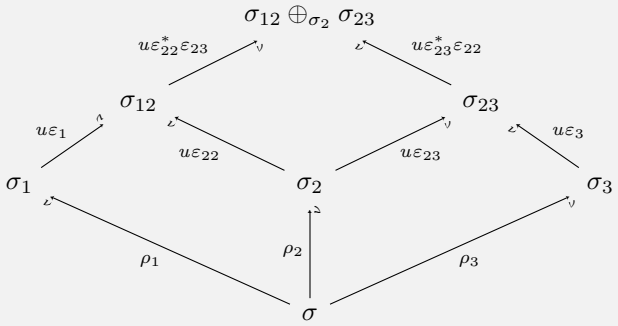
We inject resultlets into the monad using the colimit injections $inj^{\mathsf{R}^A_\sigma}_{\rho'} : \mathsf{R}^A_\sigma \rho' \to \mathbb{TP}A\sigma$

$$q^A_\sigma : \mathsf{Res}A\sigma \to \mathbb{TP}A\sigma$$
$$inj_{\rho'} a \mapsto inj^{\mathsf{R}^A_\sigma}_{\rho'}(a)$$

Lemma 2.3.19 (Quotienting in the Hiding Monad on Template Heaps)**.** For $inj_{\sigma \xrightarrow{\rho_1} \sigma_1} a_1$, and $inj_{\sigma \xrightarrow{\rho_2} \sigma_2} a_2$, we have that $q_{\rho_1} a_1 = q_{\rho_2} a_2 \in \mathbb{TP} A\sigma$ iff there exist 

s.t. $A(\varepsilon_1')(a_1) = A(\varepsilon_2')(a_2)$.

Proof. This is a direct generalisation of the proof in Lemma 1.2.3. We first show that the RHS relation is an equivalence. Reflexivity and symmetry is immediate. For transitivity, we have

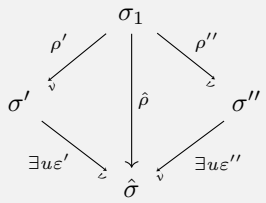


Notice that the top square commutes by Lemma 1.4.50 which implies that $inj_{\rho_1}(a_1) \sim inj_{\rho_3}(a_3)$. To prove that the generated relation by the colimit is exactly $\sim$, one shows direction $\rightarrow$ by inducting on the list of related elements, and for $\leftarrow$, notice that if $inj_{\rho_1}(a_1) \sim inj_{\rho_2}(a_2)$, then

$$inj_{\rho_1} a_1 \leq inj_{u\varepsilon_1 \circ \rho_1} A(\varepsilon_1)(a_1) \geq inj_{\rho_2} a_2$$

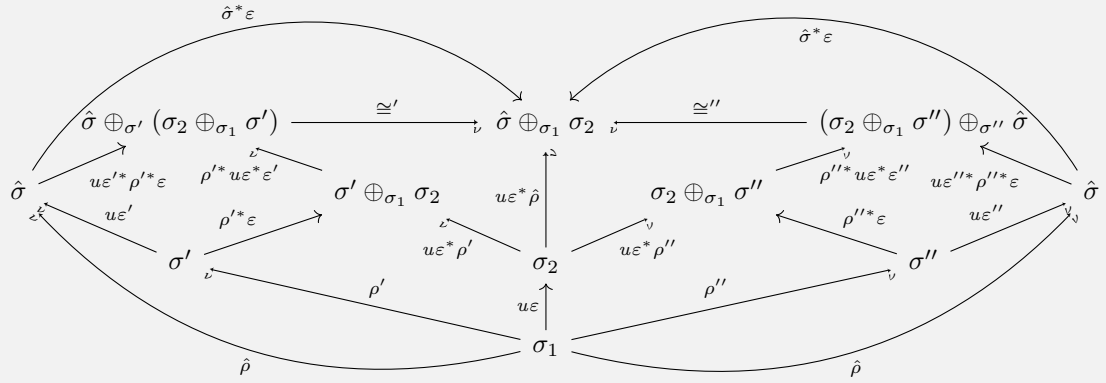
□

Lemma 2.3.20. $\mathbb{TP}$ together with the covariant actions on morphisms, unit and multiplication from Res form a monad.

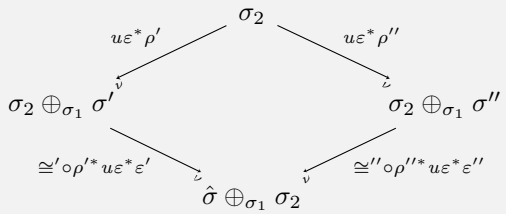
Proof.
We first show it is functorial on morphisms $\varepsilon : \sigma_1 \rightarrow \sigma_2 \in \mathbb{TI}$. Consider two related $inj_{\sigma_1 \xrightarrow{\rho} \sigma'} a'$, $inj_{\sigma_1 \xrightarrow{\rho'} \sigma''} a''$. By Lemma 2.3.19 we have 

with $A(\varepsilon')(a') = A(\varepsilon'')(a'')$.

We form the following diagram

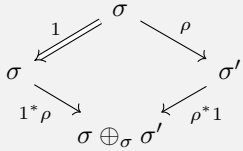


where the isomorphisms are given by the pasting property of L.I.C. in $\mathbb{T}$. The induced data given by Remark 1.3.16 of the morphisms also commute as they are uniquely determined by $\eta_\varepsilon, \eta_{\varepsilon'}, \eta_{\varepsilon''}$.
We relate $inj_{u\varepsilon^*\rho'} A(\rho'^*\varepsilon)(a')$ with $inj_{u\varepsilon^*\rho''} A(\rho''^*\varepsilon)(a'')$ by

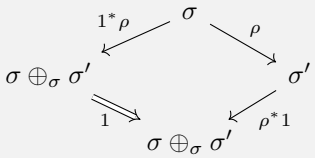


which give us

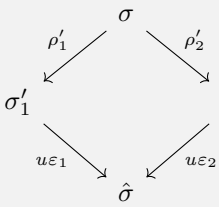
$$\begin{aligned} A(\cong' \circ \rho'^* u\varepsilon^*\varepsilon' \circ \rho'^*\varepsilon)(a') &= A(\cong' \circ u\varepsilon'^*\rho'^* u\varepsilon \circ u\varepsilon')(a') \\ &= A(\hat{\sigma}^*\varepsilon \circ u\varepsilon'')(a'') \\ &= A(\cong'' \circ u\varepsilon''^*\rho''^* u\varepsilon \circ u\varepsilon'')(a'') \\ &= A(\cong'' \circ \rho''^* u\varepsilon^*\varepsilon'' \circ \rho''^*\varepsilon)(a'') \end{aligned}$$

To show functoriality, for preservation of identities consider 

and so take



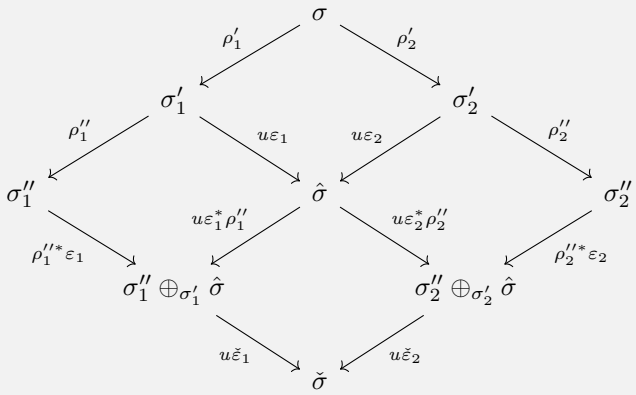
and for composition one considers pasting law for L.I.C.

That multiplication is quotient respecting, consider 

for some $q_1 = q_{\sigma'_1 \xrightarrow{\rho''_1} \sigma''_1} a''_1$, and $q_2 = q_{\sigma'_2 \xrightarrow{\rho''_2} \sigma''_2} a''_2$. with

$$\mathbb{TP}(A)(\varepsilon_1)(q_1) = \mathbb{TP}(A)(\varepsilon_2)(q_2)$$

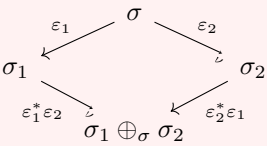
Unpacking this gives us a diagram



with $A(u\check{\varepsilon}_1 \circ \rho_1'^*\varepsilon_1)(a_1'') = A(u\check{\varepsilon}_2 \circ \rho_2''^*\varepsilon_2)(a_2'')$.
Which is exactly what we need to show that

$$\mathsf{hide}^A_{\rho_1'}\, q_1 = \mathsf{hide}^A_{\rho_2'}\, q_2$$

That the action on morphisms in $[\mathbb{TI}, \mathsf{Set}]$ is quotient respecting and natural is a simple exercise and the same goes for naturality of unit and multiplication and that they satisfy the monad laws. □

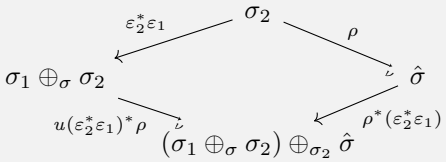
Lemma 2.3.21. Given a L.I.G.U. diagram  and a $r \in \mathbb{TP}A(\sigma_2)$, we have

$$\mathsf{hide}^A_{\varepsilon_1^*\varepsilon_2}(\mathbb{TP}(\varepsilon_2^*\varepsilon_1)(r)) = \mathbb{TP}(\varepsilon_1)(\mathsf{hide}^A_{\varepsilon_2}(r))$$

Proof. Let $r = q_{\sigma_2 \xrightarrow{\rho} \hat{\sigma}}\, a$, for some $a : A(\hat{\sigma})$. The LHS of the equation gives

$$q_{\sigma_1 \xrightarrow{u(\varepsilon_1^*\varepsilon_2)} \sigma_1 \oplus_\sigma \sigma_2 \xrightarrow{u(\varepsilon_2^*\varepsilon_1)^*\rho)} (\sigma_1 \oplus_\sigma \sigma_2) \oplus_{\sigma_2} \hat{\sigma}}\, A(\rho^*(\varepsilon_2^*\varepsilon_1)(a))$$

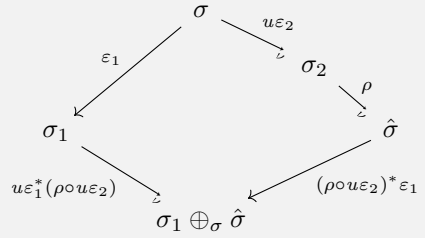
using the diagram



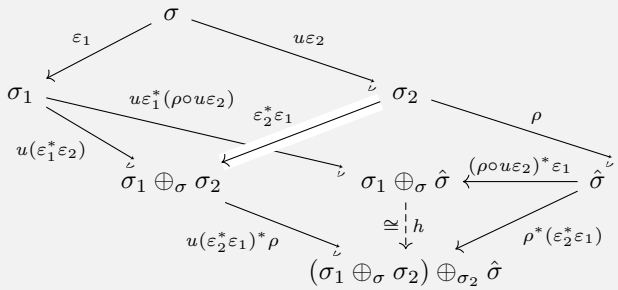
The RHS of the equation gives

$$q_{\sigma_1 \xrightarrow{u\varepsilon_1^*(\rho \circ u\varepsilon_2)} \sigma_1 \oplus_\sigma \hat{\sigma}}\, A((\rho \circ u\varepsilon_2)^*\varepsilon_1)(a)$$

using the diagram



We relate the two resultlets by



where by L.I.C. pasting (Lemma 1.3.18) the two outer L.I.C. squares are isomorphic, witnessed by the iso h. Further, h is an allocation morphism by Lemma 2.3.6. And so, by Lemma 2.3.19, the resultlets are related via h and the identity $1_{(\sigma_1 \oplus_\sigma \sigma_2) \oplus_{\sigma_2} \hat{\sigma}}$. (notice that the data of $\rho^*(\varepsilon_2^*\varepsilon_1)$ and $(\rho \circ u\varepsilon_2)^*\varepsilon_1$ are completely determined by η_{ε_1} and so they match). □

2.3.4 Evaluation and Realisation

An *evaluation* is a collection of resultlets.

Definition 2.3.22 (Evaluation)**.** Let Σ be a finite family of templates and $A : \mathbb{TI} \to \mathsf{Set}$. An A-valued *evaluation* ev over Σ is an indexed family of resultlets $\mathsf{ev} \in \prod_{\sigma_i \in \Sigma} \mathsf{Res}A\sigma_i$.

Let $A : \mathbb{TI} \to \mathsf{Set}$, total cover cov over $\langle w, \Sigma \rangle$, A-valued evaluation ev over Σ, and let ϕ be

$$\begin{array}{ccc} \mathbb{W}(w, u(-)) \times \mathbb{H} & \xrightarrow{\phi} & \mathbb{P}(A \circ 1_-) \\ {\scriptstyle \langle [-], \eta_- \rangle}\uparrow \cong & & \downarrow {\scriptstyle \overset{\mathbb{P}}{\cong}} \\ \mathcal{S}_w 1_- & \dashrightarrow & \mathbb{TP}A1_- \end{array}$$

which induces a morphism from $\mathcal{S}_w 1_- \dashrightarrow \mathbb{TP}A1_-$.

Definition 2.3.23 (Realisation)**.** We say that cov and ev realise ϕ, $\mathsf{cov}, \mathsf{ev} \Vdash \phi$ if for every concrete stencil $\varepsilon : ?_w \to 1_{\hat{w}}$, we have

$$\overset{\mathbb{P}}{\cong}(\phi \langle [\varepsilon], \eta_\varepsilon \rangle) = \mathbb{TP}(A)(\tilde{\varepsilon})(\pi_{|\tilde{\varepsilon}|(\star)}\mathsf{ev})$$

where the morphism $\tilde{\varepsilon} : \sigma_{|\tilde{\varepsilon}|(\star)} \to 1_{\hat{w}}$ is given by totality.

2.4 Finitary FGLS Monad

A $\phi \in \mathcal{T}Xw$ is finitary when it is generated by its actions on a finite number of *stencils*.

Definition 2.4.1 (Finitary FGLS Monad)**.**

$$\begin{aligned} \mathcal{T}^{\mathsf{fin}} :\ & [\mathbb{W}, \mathsf{Set}] \to [\mathbb{W}, \mathsf{Set}] \\ \mathcal{T}^{\mathsf{fin}}Xw := \{\phi \in \mathcal{T}Xw :\ & \textit{exists a finite family of templates } \Sigma, \\ & \textit{a complete cover } \mathsf{cov} \textit{ over } \langle w, \Sigma \rangle, \\ & \textit{an } (X[-] \times \mathbb{TH})\textit{-valued evaluation } \mathsf{ev} \textit{ over } \Sigma, \\ & \textit{such that } \mathsf{cov}, \mathsf{ev} \Vdash \phi\} \end{aligned}$$

We give a functor and monad structure to $\mathcal{T}^{\mathsf{fin}}$ by showing that the structure of $\mathcal{T}$ lifts through the finitary setting.

2.4.1 Action on Morphisms in $\mathbb{W}$

Theorem 1. The factorial action of the $\mathcal{T}^{\mathsf{fin}}$ monad on morphisms lifts through $\mathcal{T}$, that is for every $\sigma : w \to w'$ we have

$$\begin{array}{ccc} \mathcal{T}^{\mathsf{fin}} Xw & \xrightarrow{\exists \mathcal{T}^{\mathsf{fin}} X\sigma} & \mathcal{T}^{\mathsf{fin}} Xw' \\ \downarrow & & \downarrow \\ \mathcal{T} Xw & \xrightarrow{\mathcal{T} X\sigma} & \mathcal{T} Xw' \end{array}$$

Proof. Consider a finitary $\phi \in \mathcal{T}^{\mathsf{fin}} Xw$, that is, we have a complete cover cov over $\langle w, \Sigma \rangle$ and an $(X[-] \times \mathbb{TH})$-valued evaluation ev over Σ s.t. $\mathsf{cov}, \mathsf{ev} \Vdash \phi$. The factorial action on ϕ is given by precomposition expressed as

$$\begin{array}{ccccc} & & \overset{\mathcal{T} X\sigma(\phi)}{\frown} & & \\ \mathbb{W}(w', u(-)) \times \mathbb{H} & \xrightarrow{(-\circ\sigma)\times\mathbb{H}} & \mathbb{W}(w, u(-)) \times \mathbb{H} & \xrightarrow{\phi} & \mathbb{P}(Xu \times \mathbb{H}) \\ \cong \uparrow \langle [-], \eta_- \rangle & & \cong \uparrow \langle [-], \eta_- \rangle & & \downarrow \overset{\mathbb{P}}{\cong} \\ \mathcal{S}_{w'} 1_- & \xrightarrow{\mathcal{S}_\sigma \circ 1_-} & \mathcal{S}_w 1_- & \dashrightarrow & \mathbb{TP}(X[-] \times \mathbb{TH}) 1_- \end{array}$$

We have to find cov^σ, ev^σ s.t. $\mathsf{cov}^\sigma, \mathsf{ev}^\sigma \Vdash \phi \circ (- \circ \sigma) \times \mathbb{H}$, So first rebase cover cov using σ (§2.2.2),

$$\begin{array}{ccc} ?_w & \xrightarrow{?_\sigma} & ?_{w'} \\ {\scriptstyle \mathsf{cov}}\downarrow & \ulcorner & \downarrow{\scriptstyle \mathsf{cov}^\sigma} \\ \Sigma & \xrightarrow[\mathsf{rebase}^\sigma]{} & \Sigma^\sigma \end{array}$$

which gives us a complete cover cov^σ.
Define evaluation ev^σ at $\sigma_k \in \Sigma^\sigma$, via $\pi_k \circ \mathsf{rebase}^\sigma : \sigma_{|\mathsf{rebase}^\sigma|(k)} \to \sigma_k$, by

$$\pi_k \mathsf{ev}^\sigma \mapsto \mathsf{Res}(X[-] \times \mathbb{TH})(\pi_k \circ \mathsf{rebase}^\sigma)(\pi_{|\mathsf{rebase}^\sigma|(k)} \mathsf{ev})$$

That $\mathsf{cov}^\sigma, \mathsf{ev}^\sigma \Vdash \phi \circ (- \circ \sigma) \times \mathbb{H}$, consider $?_{w'} \xrightarrow{\varepsilon} 1_{\hat{w}}$

$$\begin{array}{ccccc} ?_w & \xrightarrow{?_\sigma} & ?_{w'} & \overset{\varepsilon}{\searrow} & \\ {\scriptstyle \mathsf{cov}}\downarrow & \ulcorner & \downarrow{\scriptstyle \mathsf{cov}^\sigma} & & \\ \Sigma & \xrightarrow[\mathsf{rebase}^\sigma]{} & \Sigma^\sigma & \overset{\tilde{\varepsilon}}{\dashrightarrow} & 1_{\hat{w}} \\ & \underset{\widetilde{\varepsilon \circ ?_\sigma}}{\searrow} & & & \end{array}$$

which gives

$$\begin{aligned} \overset{\mathbb{P}}{\cong}(\phi \langle [\varepsilon \circ ?_\sigma], \eta_\varepsilon \rangle) &= \mathbb{TP}(X[-] \times \mathbb{TH})(\widetilde{\varepsilon \circ ?_\sigma})(\pi_{|\widetilde{\varepsilon \circ ?_\sigma}|(\star)} \mathsf{ev}) \\ &= \mathbb{TP}(X[-] \times \mathbb{TH})(\tilde{\varepsilon} \circ \mathsf{rebase}^\sigma)(\pi_{|\mathsf{rebase}^\sigma||\tilde{\varepsilon}|(\star)} \mathsf{ev}) \\ &= \mathbb{TP}(X[-] \times \mathbb{TH})(\tilde{\varepsilon})(\pi_{|\tilde{\varepsilon}|\star} \mathsf{ev}^\sigma) \end{aligned}$$

□

2.4.2 Action on Morphisms in $[\mathbb{W}, \mathsf{Set}]$

Theorem 2. The factorial action of the $\mathcal{T}^{\mathsf{fin}}$ monad on morphisms in $[\mathbb{W}, \mathsf{Set}]$ lifts through $\mathcal{T}$, that is for every $\alpha : X \to Y$ we have

$$\begin{array}{ccc} \mathcal{T}^{\mathsf{fin}} Xw & \xrightarrow{\exists \mathcal{T}^{\mathsf{fin}} \alpha_w} & \mathcal{T}^{\mathsf{fin}} Yw \\ \downarrow & & \downarrow \\ \mathcal{T} Xw & \xrightarrow{\mathcal{T}\alpha_w} & \mathcal{T} Yw \end{array}$$

Proof. Consider a finitary $\phi \in \mathcal{T}^{\mathsf{fin}} Xw$, that is, we have a complete cover cov over $\langle w, \Sigma \rangle$ and an $(X[-] \times \mathbb{TH})$-valued evaluation ev over Σ s.t. $\mathsf{cov}, \mathsf{ev} \Vdash \phi$.
The factorial action on ϕ is given by post-composition

$$\begin{array}{ccccc} \mathbb{W}(w, u(-)) \times \mathbb{H} & & \xrightarrow{\phi} & & \mathbb{P}(Xu \times \mathbb{H}) \xrightarrow{\mathbb{P}(\alpha u \times \mathbb{H})} \mathbb{P}(Yu \times \mathbb{H}) \\ {\scriptstyle \langle [-], \eta_- \rangle}\uparrow & & & & \downarrow \cong^{\mathbb{P}} \qquad \downarrow \cong^{\mathbb{P}} \\ \mathcal{S}_w 1_- & \dashrightarrow & \mathbb{TP}(X[-] \times \mathbb{TH}) 1_- & \xrightarrow[\mathbb{TP}(\alpha[-] \times \mathbb{TH}) \circ 1_-]{} & \mathbb{TP}(Y[-] \times \mathbb{TH}) 1_- \end{array}$$

Define a new evaluation as

$$\pi_{\sigma_i} \mathsf{ev}' := \mathsf{Res}(\alpha[-] \times \mathbb{TH})_{\sigma_i}(\pi_{\sigma_i} \mathsf{ev}) : \prod_{\sigma_i \in \Sigma} \mathsf{Res}(Y[-] \times \mathbb{TH})(\sigma_i)$$

Then for $\varepsilon : ?_w \to 1_{\hat{w}}$ and induced morphism $\Sigma \xrightarrow{\tilde{\varepsilon}} 1_{\hat{w}}$.

$$\begin{aligned} \cong^{\mathbb{P}} (\mathbb{P}(\alpha u \times \mathbb{H})_{\hat{w}}(\phi \langle [\varepsilon], \eta_\varepsilon \rangle)) &= \mathbb{TP}(\alpha[-] \times \mathbb{TH})_{1_{\hat{w}}}(\cong^{\mathbb{P}} (\phi \langle [\varepsilon], \eta_\varepsilon \rangle)) \\ &= \mathbb{TP}(\alpha[-] \times \mathbb{TH})_{1_{\hat{w}}}(\mathbb{TP}(X[-] \times \mathbb{TH})(\tilde{\varepsilon})(\pi_{|\tilde{\varepsilon}|(\star)} \mathsf{ev})) \\ &= \mathbb{TP}(Y[-] \times \mathbb{H})(\tilde{\varepsilon})(\mathbb{TP}(\alpha[-] \times \mathbb{TH})_{\sigma_i}(\pi_{|\tilde{\varepsilon}|(\star)} \mathsf{ev})) \\ &= \mathbb{TP}(Y[-] \times \mathbb{H})(\tilde{\varepsilon})(\pi_{|\tilde{\varepsilon}|(\star)} \mathsf{ev}') \end{aligned}$$

and so $\mathsf{cov}, \mathsf{ev}' \Vdash \mathbb{P}(\alpha u \times \mathbb{H}) \circ \phi$. □

2.4.3 Unit of Monad

Theorem 3. The unit of the $\mathcal{T}^{\mathsf{fin}}$ monad lifts through $\mathcal{T}$, that is we have

$$\begin{array}{ccc} Xw & \xrightarrow{\eta^{\mathsf{fin}}_w} & \mathcal{T}^{\mathsf{fin}} Xw \\ & {\scriptstyle \eta_w} \searrow & \downarrow \\ & & \mathcal{T} Xw \end{array}$$

Proof. $\eta_w(x) \in \mathcal{T} Xw$ is given by

$$\begin{array}{ccccc} \mathbb{W}(w, u(-)) \times \mathbb{H} & & \xrightarrow{X(-)(x) \times 1} & & Xu \times \mathbb{H} \xrightarrow{\eta^{\mathbb{P}}} \mathbb{P}(Xu \times \mathbb{H}) \\ \cong\uparrow & & & & \cong\uparrow \qquad \downarrow \cong^{\mathbb{P}} \\ \mathcal{S}_w 1_- & \xrightarrow{(X[-](x) \times \mathbb{TH}) 1_-} & (X[-] \times \mathbb{TH}) 1_- & \xrightarrow{\eta^{\mathbb{TP}} \circ 1_-} & \mathbb{TP}(X[-] \times \mathbb{TH}) 1_- \end{array}$$

To show that $\eta_w(x)$ is finitary consider the singleton template $\star \mapsto ?_w$, and the singleton cover over $\langle w, \star \mapsto ?_w \rangle$, given by the identity stencil $\mathsf{cov}_\star := ?_w \xrightarrow{1} ?_w$. Notice that cov is an isomorphism, and so it is complete. Define the $(X[-] \times \mathbb{TH})$-valued evaluation at $\star$ to be the resultlet

$$\pi_\star \mathsf{ev} := \ \eta_{?_w}\langle x, \emptyset \rangle : \mathsf{Res}(X[-] \times \mathbb{TP})(?_w)$$

That $\mathsf{cov}, \mathsf{ev} \Vdash \eta_w(x)$, consider $?_w \xrightarrow{\varepsilon} 1_{\hat{w}}$. We have

$$\begin{aligned} \overset{\mathbb{P}}{\cong}(\eta_w(x)\langle [\varepsilon], \eta_\varepsilon \rangle) &= \eta^{\mathbb{P}}_{\hat{w}}\langle X[\varepsilon](x), \eta_\varepsilon \rangle \\ &= \eta^{\mathbb{TP}}_{1_{\hat{w}}}\langle X[\varepsilon](x), \eta_\varepsilon \rangle \\ &= \mathbb{TP}(X[-] \times \mathbb{TH})(\varepsilon)(\eta_{?_w}\langle x, \emptyset \rangle) \\ &= \mathbb{TP}(X[-] \times \mathbb{TH})(\varepsilon)(\pi_\star \mathsf{ev}) \end{aligned}$$

□

2.4.4 Join of Monad

Theorem 4. The join of the $\mathcal{T}^{\mathsf{fin}}$ monad on morphisms lifts through $\mathcal{T}$.

$$\begin{array}{ccc} \mathcal{T}^{\mathsf{fin}}\mathcal{T}^{\mathsf{fin}}Xw & \overset{\exists \mu^{\mathsf{fin}}_w}{\dashrightarrow} & \mathcal{T}^{\mathsf{fin}}Xw \\ \downarrow & & \downarrow \\ \mathcal{T}\mathcal{T}^{\mathsf{fin}}Xw & & \\ \downarrow{\scriptstyle \mathcal{T}i_w} & & \\ \mathcal{T}\mathcal{T}Xw & \xrightarrow{\mu_w} & \mathcal{T}Xw \end{array}$$

Proof. For a finitary $\phi \in \mathcal{T}^{\mathsf{fin}}\mathcal{T}^{\mathsf{fin}}Xw$, $\mu_w(\phi)$ is given by

$$\begin{array}{ccccccc} & & \overset{\mu_w(\phi)}{\frown} & & & & \\ \mathbb{W}(w, u(-)) \times \mathbb{H} & \xrightarrow{\phi} & \mathbb{P}(\mathcal{T}^{\mathsf{fin}}Xu \times \mathbb{H}) & \xrightarrow{\mathbb{P}(\mathsf{eval} \circ (i \times 1))} & \mathbb{P}^2(Xu \times \mathbb{H}) & \xrightarrow{\mu^{\mathbb{P}}} & \mathbb{P}(Xu \times \mathbb{H}) \\ {\scriptstyle \langle [-], \eta_- \rangle}\uparrow \cong & & \overset{\mathbb{P}}{\cong}\downarrow \uparrow \overset{\mathbb{P}}{\cong}{}^{-1} & & & & \downarrow \overset{\mathbb{P}}{\cong} \\ \mathcal{S}_w 1_- & \dashrightarrow & \mathbb{TP}(\mathcal{T}^{\mathsf{fin}}X[-] \times \mathbb{TH})1_- & \dashrightarrow & & & \mathbb{TP}(X[-] \times \mathbb{TH})1_- \end{array}$$

which requires

1. A finite family of templates $\Sigma^{\mu\phi}$
2. A complete cover $\mathsf{cov}^{\mu\phi}$ over $\langle w, \Sigma^{\mu\phi} \rangle$
3. An $(X[-] \times \mathbb{TH})$-valued evaluation $\mathsf{ev}^{\mu\phi}$ over $\Sigma^{\mu\phi}$
4. That $\mathsf{cov}^{\mu\phi}, \mathsf{ev}^{\mu\phi} \Vdash \mu_w(\phi)$

From finitariness of ϕ, we have a complete cover cov over $\langle w, \Sigma \rangle$ and $(\mathcal{T}^{\mathsf{fin}}X[-] \times \mathbb{TH})$-valued evaluation ev s.t. $\mathsf{cov}, \mathsf{ev} \Vdash \phi$. Unpacking this, the resultlet at every $\sigma_i \in \Sigma$, $\pi_i \mathsf{ev}$ gives

- An allocation morphism $\rho_i : \sigma_i \nrightarrow \sigma'_i$.
- A finitary computation $\phi_i : \mathcal{T}^{\mathsf{fin}}X[\sigma'_i]$
  - A finite family of templates Σ_i
  - A complete cover $\mathsf{cov}_i : ?_{[\sigma'_i]} \to \Sigma_i$
  - An evaluation $\mathsf{ev}_i : \prod_{\sigma_j \in \Sigma_i} \mathsf{Res}(X[-] \times \mathbb{TH})(\sigma_j)$

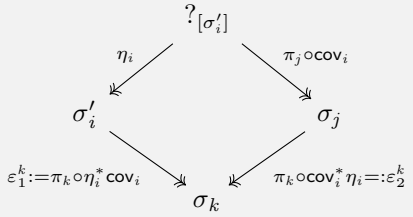
- A template heap $\eta_i : \mathbb{TH}\sigma'_i$ which is a morphism $?_{[\sigma'_i]} \xrightarrow{\eta_i} \sigma'_i \in \mathbb{TI}$.

We lift morphism ρ_i to $\mathbb{TI}$ by selecting the allocation data using $\underline{\mathbb{H}}(\lfloor \rho_i \rfloor, 1)(\eta_i)$, and we denote it by $\varepsilon_i : \sigma_i \nrightarrow \sigma'_i \in \mathbb{TI}$.

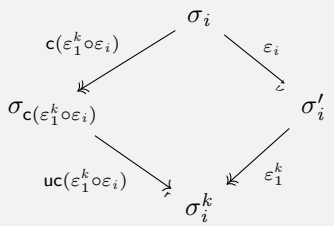
Family of templates $\Sigma^{\mu\phi}$ We first extract all the ways η_i can unify with cover cov_i, that is, we consider the pushout in $\mathsf{Fam}(\mathbb{TI})$

$$\begin{array}{ccc} \prod_{i\in|\Sigma|} ?_{[\sigma'_i]} & \xrightarrow{\prod_{i\in|\Sigma|}\ \mathsf{cov}_i} & \prod_{i\in|\Sigma|}\ \Sigma_i \\ \Big\downarrow{\scriptstyle \prod_{i\in|\Sigma|}\ \eta_i} & & \Big\downarrow{\scriptstyle \prod_{i\in|\Sigma|}\ \mathsf{cov}_i^*\eta_i} \\ \prod_{i\in|\Sigma|}\ \sigma'_i & \xrightarrow[\prod_{i\in|\Sigma|}\ \eta_i^*\mathsf{cov}_i]{} & \prod_{i\in|\Sigma|}\ \sigma'_i \oplus_{?_{[\sigma'_i]}} \Sigma_i) \end{array}$$

Notice that $\mathsf{Fam}(\mathbb{TI})$ is a coextensive category and thus the pushout above is given as the product of the pushouts at every component.
Consider a general unifier for a η_i, that is consider a $\sigma_k \in \sigma'_i \oplus_{?_{[\sigma'_i]}} \Sigma_i$, and let $j := |\mathsf{cov}_i^*\eta_i|(k)$ be the index of the stencil in cov_i that matches. We have a diagram (connectivity follows from Lemma 1.4.42)



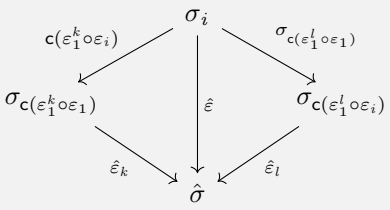
Intuitively, ε_1^k instantiates the variables $\lceil \sigma_i \rceil$ with enough data to match the stencil $\pi_j \circ \mathsf{cov}_i$. But we need to be careful, variables $\lceil \sigma_i \rceil$ represent the extensions on the starting heap before the allocation occurs. As a result, we need to select only the ε_1^k where the instantiated data do not point into the newly allocated data η_{ε_i}. Semantically, this is expressed as ε_1^k should preserve the disconnectivity of ε_i (Definition 1.4.54), which is by Lemmas 2.3.8 and 1.4.58 it is equivalent to requiring that



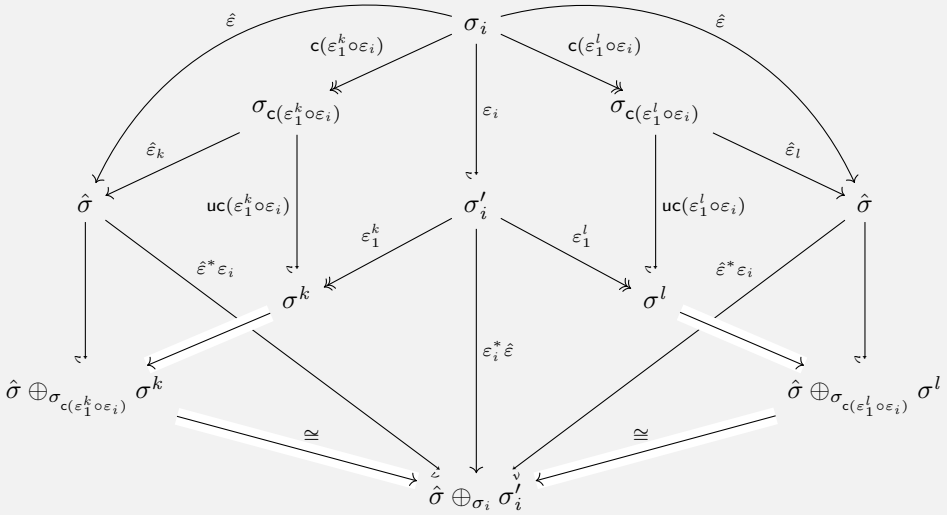
is a L.I.G.U. When this is the case, the reachable component $\sigma_{\mathsf{c}(\varepsilon_1^k \circ \varepsilon_i)}$ represents only instantiations of the variables $\lceil \sigma_i \rceil$ with data that do not point to the allocated locations.
Let $\mathsf{dpres}_{\varepsilon_i}(\sigma'_i \otimes_{?_{[\sigma'_i]}} \Sigma_i) \subset \sigma'_i \otimes_{?_{[\sigma'_i]}} \Sigma_i)$ represent the selection of general unifiers from $\sigma_k \in \sigma'_i \otimes_{?_{[\sigma'_i]}} \Sigma_i$ such that ε_1^k preserves the disconnectivity of ε_i as described.

$$\Sigma \xrightarrow{\overbrace{\prod_{\sigma_i \in \Sigma} \langle \ldots, k \mapsto \mathsf{c}(\varepsilon_1^k \circ \varepsilon_i), \ldots \rangle}^{\text{embed}}} \overbrace{\prod_{\sigma_i \in \Sigma} \prod_{\sigma_k \in \mathsf{dpres}_{\varepsilon_i}(\sigma'_i \otimes_{?_{[\sigma'_i]}} \Sigma_i)} \sigma_{\mathsf{c}(\varepsilon_1^k \circ \varepsilon_i)}}^{\Sigma^{\mu\phi}}$$

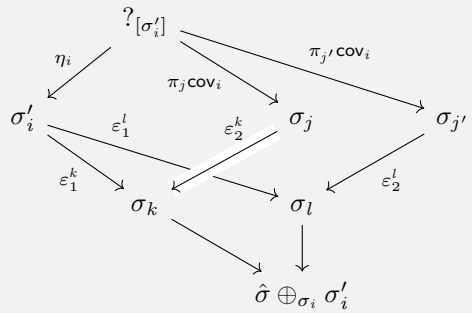
Cover $\mathsf{cov}^{\mu\phi}$ It is enough to show that morphism $\mathsf{embed} : \Sigma \to \Sigma^{\mu\phi}$ is complete, and take the composition $\mathsf{cov}^{\mu\phi} := ?_w \xrightarrow{\mathsf{cov}} \Sigma \xrightarrow{\mathsf{embed}} \Sigma^{\mu\phi}$ as our cover (complete morphisms are closed under composition).
That embed is connected is immediate, so we need to show that is non-overlapping and total. For non-overlappingness, it is enough to show that all diagrams of this form satisfy $k = l$.



Notice that all the inner squares are L.I.G.U, and by Lemmas 1.4.47, 1.4.49 and 1.4.50 we have the given isos

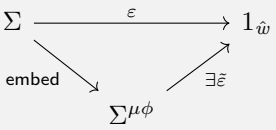


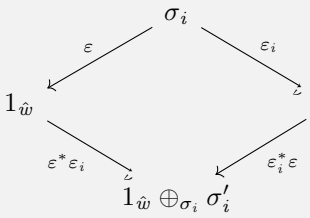
Let $j := |\mathsf{cov}_i^* \eta_i|(k)$ and $j' := |\mathsf{cov}_i^* \eta_i|(l)$ be the indices of their matching stencils. We have a diagram

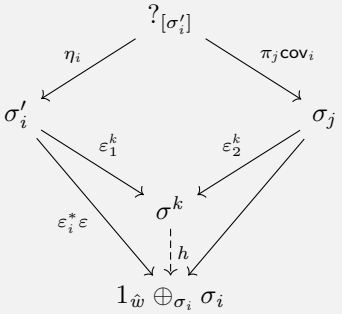


and by cov_i being complete, we have that $j = j'$, and by Lemma 1.4.39, the general unifiers k, l must be iso which means they are equal in our setting since $\sigma_i' \oplus_{?_{[\sigma_i']}} \Sigma_i$ picks representatives for each isomorphism class of general unifiers.

For totality of embed, we need to show that $\tilde{\varepsilon}$ exists for any ε

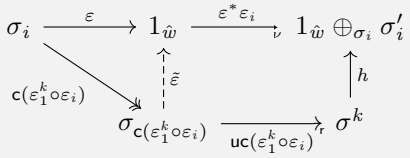


For $i := |\varepsilon|(\star)$, consider the L.I.G.U. given by Lemma 1.4.50  and notice that $1_{\hat{w}} \oplus_{\sigma_i} \sigma'_i$ is a concrete template since it has a variable preserving morphism into it with domain a concrete template.

By totality of cov_i, and further by Lemma 1.4.39 we have  where σ^k is its general unifier.

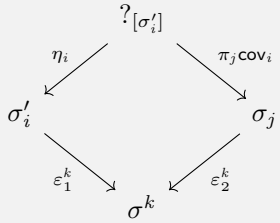
To show that general unifier σ_k with $\varepsilon_1^k, \varepsilon_2^k$ is in $\Sigma^{\mu\phi}$ we only need to show that ε_1^k preserves the disconnectivity of ε_i which follows from Lemmas 1.4.57 and 1.4.58

$$\sigma_i \xrightarrow{\varepsilon_i}_{\mathsf{r}} \sigma'_i \xrightarrow{\varepsilon_1^k} \sigma^k \xrightarrow{h} 1_{\hat{w}} \oplus_{\sigma_i} \sigma'_i \qquad \sigma'_i \xrightarrow{\varepsilon_i^* \varepsilon} 1_{\hat{w}} \oplus_{\sigma_i} \sigma'_i$$

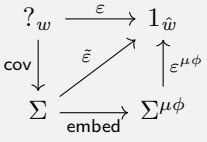
As a result, let the morphism $\tilde{\varepsilon}$ be given by general unifier k and the induced morphism by the factorisation system 

Evaluation $\mathsf{ev}^{\mu\phi}$ We need an evaluation

$$\mathsf{ev}^{\mu\phi} : \prod_{\sigma_{\mathsf{c}(\varepsilon_1^k \circ \varepsilon_i)} \in \Sigma^{\mu\phi}} \mathsf{Res}(X[-] \times \mathbb{TH})(\sigma_{\mathsf{c}(\varepsilon_1^k \circ \varepsilon_i)})$$

We define the resultlet at a $\sigma_{\mathsf{c}(\varepsilon_1^k \circ \varepsilon_i)}$. Let $j = |\mathsf{cov}_i^* \eta_i|(k)$ be the index of its matching stencil, and from the diagram  define the resultlet as

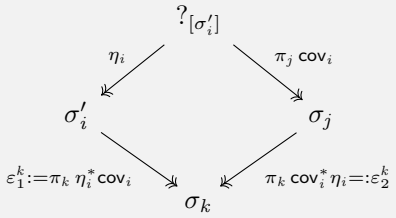
$$\pi_{\sigma_{\mathsf{c}(\varepsilon_1^k \circ \varepsilon_i)}} \mathsf{ev}^{\mu\phi} := \mathsf{hide}_{\mathsf{uc}(\varepsilon_1^k \circ \varepsilon_i)}(\mathsf{Res}(X[-] \times \mathbb{TH})(\varepsilon_2^k)(\pi_j \mathsf{ev}_i))$$

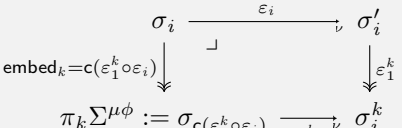
Realisation $\mathsf{cov}^{\mu\phi}, \mathsf{ev}^{\mu\phi} \Vdash \mu_w(\phi)$ For a $\varepsilon : ?_w \to 1_{\hat{w}}$, we have by totality of $\mathsf{cov}^{\mu\phi}$ 

which gives

- An $i := |\tilde{\varepsilon}|(\star)$, and so a $\sigma_i \in \Sigma$, and $\tilde{\varepsilon} : \sigma_i \to 1_{\hat{w}} \in \mathbb{TI}$

- A $k := |\varepsilon^{\mu\phi}|(\star)$, and a $\varepsilon^{\mu\phi} : \sigma_{\mathsf{c}(\varepsilon_1^k \circ \varepsilon_i)} \to 1_{\hat{w}}$ with $\tilde{\varepsilon} = \varepsilon^{\mu\phi} \circ \mathsf{embed}_k$ given by (let $j := |\mathsf{cov}_i^* \eta_i|(k)$)



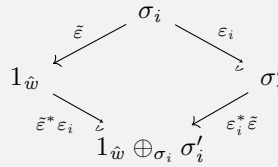


  where the RHS square is a L.I.G.U. by Lemma 2.3.8.

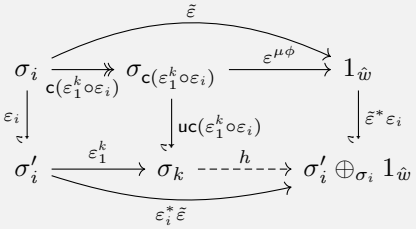
- Notice that the resultlet $\pi_i \mathsf{ev} = inj_{\sigma_i \xrightarrow{\rho_i} \sigma'_i} \langle \phi_i, \eta_i \rangle$ extends to

$$\mathsf{Res}(\mathcal{T}^{\mathsf{fin}} X[-] \times \mathbb{TH})(\tilde{\varepsilon})(\pi_i \mathsf{ev}) = inj_{1_{\hat{w}} \xrightarrow{\tilde{\varepsilon}^* \varepsilon_i} 1_{\hat{w}} \oplus_{\sigma_i} \sigma'_i} \langle \mathcal{T}^{\mathsf{fin}} X[\varepsilon_i^* \tilde{\varepsilon}](\phi_i), \mathbb{TH}(\varepsilon_i^* \tilde{\varepsilon})(\eta_i) \rangle$$

  using the L.I.G.U. (Lemma 1.4.50)



  which further factors as (the outer square is a L.I.S. by Lemma A.1.4, so h is induced by Lemma 1.4.48)



  and notice that the RHS square is itself a L.I.G.U. by Lemma 1.4.49.

  Completeness of cov_i, gives us

$$\overset{\mathbb{P}}{\cong} (\phi_i \langle [\varepsilon_i^* \tilde{\varepsilon}], \mathbb{TH}(\varepsilon_i^* \tilde{\varepsilon})(\eta_i) \rangle) = \mathbb{TP}(X[-] \times \mathbb{TH})(h)(\mathsf{Res}(X[-] \times \mathbb{TH})(\varepsilon_2^k)(\pi_j \mathsf{ev}_i))$$

Lastly we prove $\mathsf{cov}^{\mu\phi}, \mathsf{ev}^{\mu\phi} \Vdash \mu_w(\phi)$ by

$$\begin{aligned}
\overset{\mathbb{P}}{\cong} (\mu_w(\phi) \langle [\varepsilon], \eta_\varepsilon \rangle) &= \overset{\mathbb{P}}{\cong} (\mu^{\mathbb{P}}(\mathbb{P}(\mathsf{eval} \circ (i \times 1))(\phi \langle [\varepsilon], \eta_\varepsilon \rangle))) && (\mathsf{cov}, \mathsf{ev} \Vdash \phi) \\
&= \overset{\mathbb{P}}{\cong} (\mu^{\mathbb{P}}(\mathbb{P}(\mathsf{eval} \circ (i \times 1))(\overset{\mathbb{P}}{\cong}{}^{-1}(\mathbb{TP}(\mathcal{T}^{\mathsf{fin}} X[-] \times \mathbb{TH})(\tilde{\varepsilon})(\pi_i \mathsf{ev}))))) \\
&= \overset{\mathbb{P}}{\cong} (\mu^{\mathbb{P}}(\mathbb{P}(\mathsf{eval} \circ (i \times 1))(\overset{\mathbb{P}}{\cong}{}^{-1}(q_{1_{\hat{w}} \xrightarrow{\tilde{\varepsilon}^* \varepsilon_i} 1_{\hat{w}} \oplus_{\sigma_i} \sigma'_i} \langle \mathcal{T}^{\mathsf{fin}} X[\varepsilon_i^* \tilde{\varepsilon}](\phi_i), \mathbb{TH}(\varepsilon_i^* \tilde{\varepsilon})(\eta_i) \rangle))))) \\
&= \overset{\mathbb{P}}{\cong} (\mu^{\mathbb{P}}(q_{1_{\hat{w}} \xrightarrow{\tilde{\varepsilon}^* \varepsilon_i} 1_{\hat{w}} \oplus_{\sigma_i} \sigma'_i} \phi_i \langle [\varepsilon_i^* \tilde{\varepsilon}], \mathbb{TH}(\varepsilon_i^* \tilde{\varepsilon})(\eta_i) \rangle)) \\
&= \overset{\mathbb{P}}{\cong} (\mathsf{hide}_{\tilde{\varepsilon}^* \varepsilon_i} \phi_i \langle [\varepsilon_i^* \tilde{\varepsilon}], \mathbb{TH}(\varepsilon_i^* \tilde{\varepsilon})(\eta_i) \rangle) \\
&= \mathsf{hide}_{\tilde{\varepsilon}^* \varepsilon_i}(\mathbb{TP}(X[-] \times \mathbb{TH})(h)(\mathsf{Res}(X[-] \times \mathbb{TH})(\varepsilon_2^k)(\pi_j \mathsf{ev}_i))) \\
& && \text{(by Lemma 2.3.21)} \\
&= \mathbb{TP}(X[-] \times \mathbb{TH})(\varepsilon^{\mu\phi})(\mathsf{hide}_{\mathsf{uc}(\varepsilon_1^k \circ \varepsilon_i)}(\mathsf{Res}(X[-] \times \mathbb{TH})(\varepsilon_2^k)(\pi_j \mathsf{ev}_i))) && (\text{def } \mathsf{ev}^{\mu\phi}) \\
&= \mathbb{TP}(X[-] \times \mathbb{TH})(\varepsilon^{\mu\phi})(\pi_k \mathsf{ev}^{\mu\phi})
\end{aligned}$$

□

A Appendix

A.1 Pullback Lemmas

Lemma A.1.1 (Pullbacks and mono post-composition)**.** Let k be mono. Then every square of the form $\begin{array}{ccc} \cdot & \xrightarrow{\pi_1} & \cdot \\ {\scriptstyle \pi_2}\downarrow & & \downarrow{\scriptstyle f} \\ \cdot & \xrightarrow[g]{} & \cdot \end{array}$ is a pullback square iff $\begin{array}{ccc} \cdot & \xrightarrow{\pi_1} & \cdot \\ {\scriptstyle \pi_2}\downarrow & & \downarrow{\scriptstyle f} \\ \cdot & \xrightarrow[g]{} & \cdot \\ & & \downarrow{\scriptstyle k} \\ & \xrightarrow[k]{} & \cdot \end{array}$ is a pullback square.

Proof.
$\rightarrow$) Take $k \circ f \circ \kappa_1 = k \circ g \circ \kappa_2$. From k being mono we have $f \circ \kappa_1 = g \circ \kappa_2$. The induced morphism from the pullback on the left is exactly the one we need.
$\leftarrow$) Take $f \circ \kappa_1 = g \circ \kappa_2$ which gives us $k \circ f \circ \kappa_1 = k \circ g \circ \kappa_2$. The induced morphism from the pullback on the right is exactly the one we need. □

Lemma A.1.2 (Pullbacks and isos)**.** A square $\begin{array}{ccc} \cdot & \xrightarrow{\pi_1} & \cdot \\ {\scriptstyle \pi_2}\downarrow & & \downarrow{\scriptstyle f} \\ \cdot & \xrightarrow[g]{} & \cdot \end{array}$ where π_2 is iso and f is mono, is a pullback square.

Proof. For $g \circ k = f \circ l$, define the unique morphism $h = \pi_2^{-1} \circ k$. Then $k = \pi_2 \circ \pi_2^{-1} \circ k$ and $l = \pi_1 \circ \pi_2^{-1} \circ k$ by post-composition with monomorphism f. h is unique by π_2 being mono. □

Lemma A.1.3 (Pullbacks with Initial Objects)**.** Given a category with a strict initial object and a diagram $\begin{array}{ccccc} 0 & \xrightarrow{?_A} & A & \xrightarrow{h} & D \\ {\scriptstyle ?_B}\downarrow & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle k} \\ B & \xrightarrow{g} & C & \xrightarrow{l} & E \end{array}$ where the outer square is a pullback, we have that $\begin{array}{ccc} 0 & \xrightarrow{?_A} & A \\ {\scriptstyle ?_B}\downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow{g} & C \end{array}$ is also a pullback.

Proof. Consider $f \circ \phi = g \circ \xi$. Then $k \circ h \circ \phi = l \circ f \circ \phi = l \circ g \circ \xi$ and so we have a morphism into 0 which by stictness of the initial object, ϕ, and ξ must be the initial morphisms. □

Lemma A.1.4 (Pullback preserved along factoring)**.** Given a pullback $\begin{array}{ccc} A \otimes_C B & \xrightarrow{\pi_2} & B \\ {\scriptstyle \pi_1}\downarrow & \lrcorner & \downarrow{\scriptstyle g} \\ A & \xrightarrow{f} & C \end{array}$ and a factoring of π_2 as $\begin{array}{ccc} & & B' \\ & \overset{k}{\nearrow} & \downarrow{\scriptstyle l} \\ A \otimes_C B & \xrightarrow{\pi_2} & B \end{array}$ where l is mono, then $\begin{array}{ccc} A \otimes_C B & \xrightarrow{k} & B' \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle g \circ l} \\ A & \xrightarrow{f} & C \end{array}$ is a pullback.

Proof. Consider $g \circ l \circ p_2 = f \circ p_1$. We get a unique morphism h s.t. $p_1 = \pi_1 \circ h$ and $l \circ k \circ h = l \circ p_2$ which by l being mono, we have $k \circ h = p_2$ and uniqueness follows from the pullback. □

Lemma A.1.5 (Monos and Coparallel map in extensive categories)**.** Given an extensive category $\mathcal{C}$, and morphisms $f : A \to B, g : A' \to B' \in \mathcal{C}$, we have that the coparallel map $f + g : A + A' \to B + B'$ is mono iff f, G are.

Proof. For $\to$) direction, extensiveness gives us the pullbacks

$$\begin{array}{ccccc} A & \xrightarrow{inj_A} & A + A' & \xleftarrow{inj_{A'}} & A' \\ \downarrow{\scriptstyle f} & \lrcorner & \vdots{\scriptstyle f+g} & \llcorner & \downarrow{\scriptstyle g} \\ B & \xrightarrow{inj_B} & B + B' & \xleftarrow[inj_{B'}]{} & B' \end{array}$$

which imply f and g are mono.
For $\leftarrow$, extensiveness gives us that for every A', B', we have an equivalence of categories

$$\mathcal{C}/A' \times \mathcal{C}/B' \simeq \mathcal{C}/A' + B'$$

given by the coparallel map. Since equivalences preserve monos, the coparallel map will also be mono. □

A.2 Yoneda Lemmas

Lemma A.2.1 (Restricted Yoneda Embedding preserves Coequalizers)**.** The yoneda embedding $y : \mathcal{C} \to \mathsf{Psh}(\mathcal{C})_{\sqcup}$ preserves coequalizers.

Proof. Given a coequalizing diagram $A \underset{g}{\overset{f}{\rightrightarrows}} B \xrightarrow{e} C$ we show that $yA \underset{yg}{\overset{yf}{\rightrightarrows}} yB \xrightarrow{ye} yC$ is also coequalizing.
Consider a $yB \xrightarrow{\alpha} \Gamma$ s.t. $\alpha \circ yf = \alpha \circ yg$. By yoneda, $\alpha : \Gamma B$ is a morphism $B \xrightarrow{\check{\alpha}} C'$ for some $C' \in \Gamma$ s.t. $\check{\alpha} \circ f = \check{\alpha} \circ g$ and so we have a unique $C \xrightarrow{h} C'$, which by yoneda a morphism $yC \to \Gamma$. Uniqueness follows directly from uniqueness in the base category. □

Lemma A.2.2 (Restricted Yoneda Embedding preserves Pushouts)**.** The yoneda embedding $y : \mathcal{C} \to \mathsf{Psh}(\mathcal{C})_{\sqcup}$ preserves pushouts.

Proof. Given a pushout diagram $\begin{array}{ccc} A & \xrightarrow{g} & C \\ {\scriptstyle f}\downarrow & \ulcorner & \downarrow{\scriptstyle inj_2} \\ B & \xrightarrow[inj_1]{} & B \oplus_A C \end{array}$ we show that $\begin{array}{ccc} yA & \xrightarrow{yg} & yC \\ {\scriptstyle yf}\downarrow & & \downarrow{\scriptstyle y(inj_2)} \\ yB & \xrightarrow[y(inj_1)]{} & y(B \oplus_A C) \end{array}$ is also a pushout.
Consider a $yB \xrightarrow{\alpha} \Gamma, yC \xrightarrow{\beta} \Gamma$ s.t. $\alpha \circ yf = \beta \circ yg$. By yoneda, we have morphisms $B \xrightarrow{\check{\alpha}} D$ and $C \xrightarrow{\check{\beta}} D$ s.t. $\check{\alpha} \circ f = \check{\beta} \circ g$, and so we have a unique $B \oplus_A C \xrightarrow{h} D$, which by yoneda is a morphism $y(B \oplus_A C) \to \Gamma$. Uniqueness follows directly from uniqueness in the base category. □

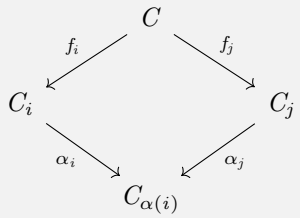
Lemma A.2.3. A $\alpha : \Gamma_1 \to \Gamma_2 \in \mathsf{Psh}(\mathcal{C})_{\sqcup}$ is mono iff it is component-wise injective.

Proof. The $\leftarrow$ direction is clear. For $\to$, consider $inj_i(f_i : C \to C_i) \neq inj_j(f_j : C \to C_j)$. We want

to show that

$$\alpha(inj_i(f_i)) = \Gamma_2(f_i)(\alpha(inj_i 1_{C_i})) \neq \Gamma_2(f_j)(\alpha(inj_j 1_{C_j})) = \alpha(inj_j(f_j))$$

so assume $\Gamma_2(f_i)(\alpha(inj_i 1)) = \Gamma_2(f_j)(\alpha(inj_j 1))$ and we show a contradiction.
We have that $|\alpha|(i) = |\alpha|(j)$ and



We have morphisms $inj_i(f_i), inj_j(f_j) : yC \to \Gamma_1$. And by copairing we have morphisms $[1, inj_i(f_i)], [1, inj_j(f_j)] : \Gamma_1 + yC \to \Gamma_1$.
The assumption $inj_i(f_i) \neq inj_j(f_j)$, gives us exactly that they two copairing morphisms are not equal.
Notice also that $\alpha \circ [1, inj_i(f_i)] = \alpha \circ [1, inj_j(f_j)]$ by construction.
But α being mono gives us that $[1, inj_i(f_i)] = [1, inj_j(f_j)]$ which is a contradiction. □

Corollary A.2.4. If $\alpha : \Sigma_1 \to \Sigma_2 \in \mathsf{Psh}(\mathcal{C})_{\sqcup}$ is mono then every $\alpha(inj_i(1_{C_i}))$ for $C_i \in \Sigma$ is mono.

Proof. Immediate from Lemma A.2.3. □

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